5G RAN

FWA Feature Parameter Description

Issue Draft B

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  (TDD) Added descriptions of FWA Ultra-Large Coverage and Uplink 4 Layers for a Single FWA User in **3 Overview**.

  (TDD) Modified the baseband processing unit requirements for uplink 4-layer transmission for a single FWA user and FWA Ultra-Large Coverage. For details, see **10.3.3 Hardware** and **12.3.3 Hardware**.

  (TDD) Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between user PRB control for FWA and enhanced L4S congestion control based on rate adjustment. For details, see **5.1.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added "resource allocation adjustment for downlink 8-layer transmission" and "MU pairing adjustment for downlink 8-layer transmission". For details, see **9.1 Principles**, **9.4.1.1 Data Preparation**, and **9.4.1.2 Using MML Commands**.	Added the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** and **DL_EIGHT_LAYER_MU_ADJ_SW** options to the **NRDUCellDlRank**.***DlRankSelAlgoExtSw*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between user PRB control for FWA and "NR DU cell resource management between network slices" in high-frequency TDD. For details, see **5.1.3.2.2 Mutually Exclusive Functions**.	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship between downlink resource isolation and time-domain power saving BWP. For details, see **5.6.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Modified the impact relationship between "downlink 8-layer transmission for a single FWA user" and "interference rejection precoding for downlink MU-MIMO in NLOS scenarios". For details, see **9.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the downlink experience satisfaction guarantee function. For details, see **5.8 Downlink Experience Satisfaction Guarantee**.	Added the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter. Added the **DL_THPT_L_SLOT_STATS_POL_SW** option to the **NRDUCellOamParam**.***StatisticsStrategySwitch*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between downlink experience satisfaction guarantee and downlink resource isolation. For details, see **5.6.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between downlink experience satisfaction guarantee and rate adaptation for device-pipe synergy. For details, see **7.2.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added dependency of downlink MU-MIMO QoS guarantee on downlink experience satisfaction guarantee. For details, see **5.4.3.2.1 Prerequisite Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for "downlink 8-layer transmission for a single FWA user" in NSA networking and in CA scenarios with three or more CCs. For details, see **9 Downlink 8-Layer Transmission for a Single FWA User**.	Added the **NRDUCellCarrMgmt**.***DlEightLayerMaxCCNum*** parameter. Added the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option to the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added FOFD-100206 FWA Ultra-Large Coverage, which includes the following functions: • FWA device-pipe-based uplink dense codebooks • FWA device-pipe-based uplink high power For details, see **12 FWA Ultra-Large Coverage**.	Added the **UL_DENSER_CODEBOOK_SW** and **UL_HIGH_POWER_UE_SW** options to the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added uplink 4-layer transmission for a single FWA user. For details, see **10 Uplink 4-Layer Transmission for a Single FWA User**.	Added the **UL_SU_FOUR_LAYER_SW** option to the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

  Added descriptions of the scenario to enable downlink RLC experience-based scheduling. For details, see **5.3.4.1.1 Data Preparation** and **5.3.4.1.2 Using MML Commands**.

  Revised the descriptions of the benefits and impacts of downlink 8-layer transmission for a single FWA user. For details, see **9.2.1 Benefits** and **9.2.2 Impacts**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-051204	Experience-based FWA	**5.1 User PRB Control for FWA** **5.2 Uplink Experience-based Scheduling** **5.3 Downlink Experience-based Scheduling** **5.4 Downlink MU-MIMO QoS Guarantee (Low-Frequency TDD)** **5.5 Multi-Bearer Downlink Target Rate Adaptation** **5.8 Downlink Experience Satisfaction Guarantee**
FOFD-051205	FWA Private Line	**6.1 Basic Service Assurance for FWA Private Line** **6.2 FWA Private Line Service Performance Improvement**

Feature ID	Feature Name	Chapter/Section
FOFD-081220	FWA Service Turbo	7.1 Device-Pipe Synergy Preallocation 7.2 Rate Adaptation for Device-Pipe Synergy
FOFD-081231	8 Layers for a Single FWA User (Trial)	9 Downlink 8-Layer Transmission for a Single FWA User
FOFD-100205	Uplink 4 Layers for a Single FWA User (Trial)	10 Uplink 4-Layer Transmission for a Single FWA User
FOFD-070301	mmWave Coverage Extension	11 FWA mmWave Coverage Extension (High-Frequency TDD)
FOFD-100206	FWA Ultra-Large Coverage (Trial)	12 FWA Ultra-Large Coverage

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
FWA user identification	• In SA networking, 5QIs are used to identify users. • In NSA networking, QCIs are used to identify users.	4.1 FWA User Identification
SRS filtering-based downlink beamforming weight optimization (low-frequency TDD)	None	4.2 SRS Filtering-based Downlink Beamforming Weight Optimization (Low-Frequency TDD)
PDCCH inter-symbol load balancing optimization	None	4.3 PDCCH Inter-Symbol Load Balancing Optimization

Function Name	Difference	Chapter/Section
Power saving BWP deactivation for FWA users (low-frequency TDD)	None	**4.4 Power Saving BWP Deactivation for FWA Users (Low-Frequency TDD)**
SCell activation optimization for FWA users	None	**4.5 SCell Activation Optimization for FWA Users**
Differentiated carrier selection for FWA users	Supported in both NSA and SA networking, with the following differences: ● NSA networking: QCI-based carrier selection ● SA networking: 5QI-based cell reselection	**4.6 Differentiated Carrier Selection for FWA Users**
High-power UE (low-frequency TDD)	Supported only in SA networking	**4.7 High-Power UE (Low-Frequency TDD)**
FWA mmWave GP symbol number adaptation (high-frequency TDD)	None	**4.8 FWA mmWave GP Symbol Number Adaptation (High-Frequency TDD)**
SRS antenna selection for RedCap users in FWA scenarios (low-frequency TDD)	Supported only in SA networking	**8.1 SRS Antenna Selection for RedCap Users in FWA Scenarios (Low-Frequency TDD)**
Uplink CA power enhancement	None	**4.9 Uplink CA Power Enhancement**
User PRB control for FWA	None	**5.1 User PRB Control for FWA**

Function Name	Difference	Chapter/Section
Uplink experience-based scheduling	None	**5.2 Uplink Experience-based Scheduling**
Downlink experience-based scheduling	None	**5.3 Downlink Experience-based Scheduling**
Downlink MU-MIMO QoS guarantee	None	**5.4 Downlink MU-MIMO QoS Guarantee (Low-Frequency TDD)**
Multi-bearer downlink target rate adaptation	None	**5.5 Multi-Bearer Downlink Target Rate Adaptation**
Downlink resource isolation (low-frequency TDD)	None	**5.6 Downlink Resource Isolation (Low-Frequency TDD)**
Downlink MU-MIMO layer isolation (low-frequency TDD)	None	**5.7 Downlink MU-MIMO Layer Isolation (Low-Frequency TDD)**
Downlink experience satisfaction guarantee	Supported in both NSA and SA networking. However, in NSA networking, this function can take effect only in NR cells.	**5.8 Downlink Experience Satisfaction Guarantee**
Basic service assurance for FWA private line	None	**6.1 Basic Service Assurance for FWA Private Line**
FWA private line service performance improvement	None	**6.2 FWA Private Line Service Performance Improvement**
Device-pipe synergy preallocation	Supported only in SA networking	**7.1 Device-Pipe Synergy Preallocation**

Function Name	Difference	Chapter/Section
Rate adaptation for device-pipe synergy	Supported only in SA networking	**7.2 Rate Adaptation for Device-Pipe Synergy**
Downlink 8-layer transmission for a single FWA user	Supported in both NSA and SA networking, with the following differences: In SA networking, this function is controlled by the **DL_SU_EIGHT_LAYER_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter. In NSA networking, this function is controlled by both the **DL_SU_EIGHT_LAYER_SW** and **DL_EIGHT_LAYER_SCENARIO_EXT_SW** options of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter.	**9 Downlink 8-Layer Transmission for a Single FWA User**
Uplink 4-layer transmission for a single FWA user	Supported only in SA networking	**10 Uplink 4-Layer Transmission for a Single FWA User**
FWA mmWave coverage extension (high-frequency TDD)	None	**11 FWA mmWave Coverage Extension (High-Frequency TDD)**
FWA Ultra-Large Coverage	Supported only in SA networking	**12 FWA Ultra-Large Coverage**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
FWA user identification	None	4.1 FWA User Identification
SRS filtering-based downlink beamforming weight optimization (low-frequency TDD)	Supported only in low frequency bands	4.2 SRS Filtering-based Downlink Beamforming Weight Optimization (Low-Frequency TDD)
PDCCH inter-symbol load balancing optimization	Supported only in low frequency bands	4.3 PDCCH Inter-Symbol Load Balancing Optimization
Power saving BWP deactivation for FWA users (low-frequency TDD)	Supported only in low frequency bands	4.4 Power Saving BWP Deactivation for FWA Users (Low-Frequency TDD)
SCell activation optimization for FWA users	Supported only in low frequency bands	4.5 SCell Activation Optimization for FWA Users
Differentiated carrier selection for FWA users	None	4.6 Differentiated Carrier Selection for FWA Users
High-power UE (low-frequency TDD)	Supported only in low frequency bands	4.7 High-Power UE (Low-Frequency TDD)
FWA mmWave GP symbol number adaptation (high-frequency TDD)	Supported only in high frequency bands	4.8 FWA mmWave GP Symbol Number Adaptation (High-Frequency TDD)

Function Name	Difference	Chapter/Section
SRS antenna selection for RedCap users in FWA scenarios (low-frequency TDD)	Supported only in low frequency bands	**8.1 SRS Antenna Selection for RedCap Users in FWA Scenarios (Low-Frequency TDD)**
Uplink CA power enhancement	Supported only in low frequency bands	**4.9 Uplink CA Power Enhancement**
User PRB control for FWA	Supported in both high and low frequency bands, with the following differences: User PRB control for FWA and MU-MIMO can be enabled together only in low frequency bands.	**5.1 User PRB Control for FWA**
Downlink experience-based scheduling	Supported in both high and low frequency bands, with the following differences: Downlink RLC experience-based scheduling is supported only in low frequency bands.	**5.3 Downlink Experience-based Scheduling**
Uplink experience-based scheduling	None	**5.2 Uplink Experience-based Scheduling**
Downlink MU-MIMO QoS guarantee	Supported only in low frequency bands	**5.4 Downlink MU-MIMO QoS Guarantee (Low-Frequency TDD)**
Multi-bearer downlink target rate adaptation	None	**5.5 Multi-Bearer Downlink Target Rate Adaptation**
Downlink resource isolation (low-frequency TDD)	Supported only in low frequency bands	**5.6 Downlink Resource Isolation (Low-Frequency TDD)**

Function Name	Difference	Chapter/Section
Downlink MU-MIMO layer isolation (low-frequency TDD)	Supported only in low frequency bands	5.7 Downlink MU-MIMO Layer Isolation (Low-Frequency TDD)
Downlink experience satisfaction guarantee	None	5.8 Downlink Experience Satisfaction Guarantee
Basic service assurance for FWA private line	None	6.1 Basic Service Assurance for FWA Private Line
FWA private line service performance improvement	Supported only in low frequency bands	6.2 FWA Private Line Service Performance Improvement
Device-pipe synergy preallocation	Supported only in low frequency bands	7.1 Device-Pipe Synergy Preallocation
Rate adaptation for device-pipe synergy	Supported only in low frequency bands	7.2 Rate Adaptation for Device-Pipe Synergy
Downlink 8-layer transmission for a single FWA user	Supported only in low frequency bands	9 Downlink 8-Layer Transmission for a Single FWA User
Uplink 4-layer transmission for a single FWA user	Supported only in low frequency bands	10 Uplink 4-Layer Transmission for a Single FWA User
FWA mmWave coverage extension (high-frequency TDD)	Supported only in high frequency bands	11 FWA mmWave Coverage Extension (High-Frequency TDD)

Function Name	Difference	Chapter/Section
FWA Ultra-Large Coverage	Supported only in low frequency bands	**12 FWA Ultra-Large Coverage**

3 Overview

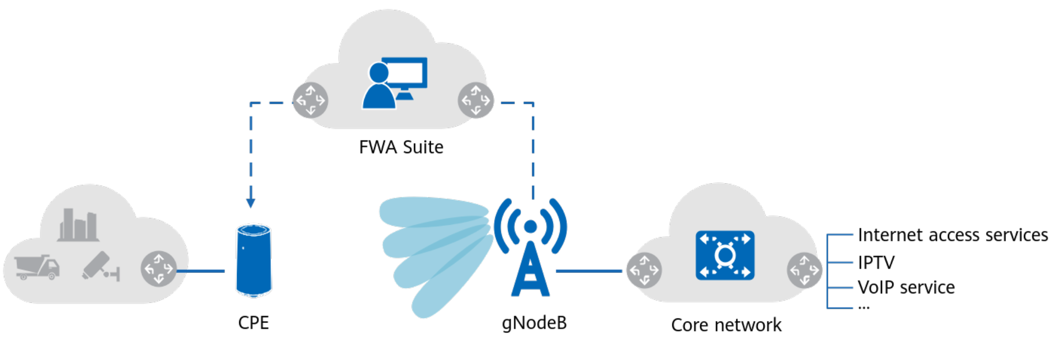
Definition of FWA

Fixed wireless access (FWA) enables various home or enterprise terminals to directly access wireless networks through the customer-premises equipment (CPE). FWA is a cost-efficient wireless broadband solution that offers the same service experience as the optical broadband solution does.

FWA terminals refer to wireless broadband access devices: mainly indoor or outdoor CPEs. A CPE connects to the user equipment through a wired or Wi-Fi connection on one side and to the gNodeB on the other side through the wireless air interface. **Figure 3-1** shows the FWA networking. The FWA Suite is an O&M tool that assists operators in operating FWA services. For details, see descriptions of WTTx Suite in iMaster MAE Product Documentation.

FWA can be applied in business to home (B2H) to provide the home broadband solution or in enterprises to provide the enterprise broadband solution. FWA services are classified into Internet access services, Internet Protocol television (IPTV) services, Voice over Internet Protocol (VoIP) services, gaming services, and so on. All these services can be used in B2H or enterprise scenarios.

Figure 3-1 FWA networking



NOTE

Only unicast IPTV services, but not multicast IPTV services, are available in this release.

Advantages of FWA

- 5G FWA combines large bandwidth and massive MIMO to provide fiber-like user-perceived rates.
- Compared with wired access technologies, FWA, using wireless access, features low construction and maintenance costs and flexible network deployment.
- FWA uses CPEs that have more antennas and higher antenna gains than eMBB mobile phones. In addition, CPEs are stationary with relatively stable radio signals, providing better user experience and higher spectral efficiency.

FWA Solution

Table 3-1 describes the functions in the FWA solution.

Table 3-1 FWA solution

Feature/Function	Principle	Reference
Basic FWA functions	FWA user identification is used together with technologies like downlink beamforming weight optimization/PDCCH inter-symbol load balancing optimization/differentiated power saving policy/differentiated CA activation policy/UE transmit power increase/GP symbol number adaptation/SRS antenna selection/differentiated carrier selection, improving service experience of FWA users.	**4 Basic FWA Functions**
FOFD-051204 Experience-based FWA	Differentiated resource allocation policies can be flexibly customized for different users and uplink and downlink scheduling priorities can be adjusted to improve service experience of FWA users while ensuring service experience of eMBB users.	**5 Experience-based FWA**
FOFD-051205 FWA Private Line	A series of functions are introduced to perform special processing on FWA private line users, ensuring service experience of FWA private line users.	**6 FWA Private Line Service Assurance**
FOFD-081220 FWA Service Turbo	Preallocation, adaptive rate adjustment, and others can be performed for FWA users through device-pipe synergy, ensuring service experience of FWA users.	**7 FWA Service Turbo**

Feature/Function	Principle	Reference
FWA RedCap	To reduce terminal costs, the FWA solution introduces RedCap CPEs defined in 3GPP Release 17. To improve the downlink experience of FWA RedCap users, you can enable "SRS antenna selection for RedCap users in FWA scenarios". You can also enable functions under the feature RedCap Downlink Experience Improvement (NR TDD), which is detailed in chapter "RedCap Downlink Experience Improvement (NR TDD)" in *RedCap*. Other RedCap functions to enhance experience, capacity, and coverage also apply to FWA scenarios. For details about these functions, see chapters "RedCap Introduction" and "Premium RedCap" in *RedCap*.	**8 FWA RedCap**
FOFD-081231 8 Layers for a Single FWA User (Trial)	By using multi-antenna technologies, spatial multiplexing of time-frequency resources can be enabled on the PDSCH for a single user. In this way, a single user supports data transmission over a maximum of eight layers in the downlink.	**9 Downlink 8-Layer Transmission for a Single FWA User**
FOFD-100205 Uplink 4 Layers for a Single FWA User (Trial)	By using multi-antenna technologies, spatial multiplexing of time-frequency resources can be enabled on the PUSCH for a single user. In this way, a single user supports data transmission over a maximum of four layers in the uplink.	**10 Uplink 4-Layer Transmission for a Single FWA User**
FOFD-070301 mmWave Coverage Extension	By changing the demodulation distance and slot structure of the PRACH, the cell coverage is expanded to improve the long-distance coverage capability.	**11 FWA mmWave Coverage Extension (High-Frequency TDD)**
FOFD-100206 FWA Ultra-Large Coverage (Trial)	Coverage for cell-edge UEs is enhanced by instructing UEs to expand their codebook sets for the PUSCH beyond standard codebooks and to use high power.	**12 FWA Ultra-Large Coverage**

Rate-based CPE allocation for FWA

CPE allocation refers to the process in which a CPE user selects a service package and the operator assesses the capability to provide services committed in this package for the CPE user and registers the CPE user with the package.

FWA networks are wireless networks. It is much more difficult to guarantee the average rate of FWA users than fixed-line users. Therefore, traditional FWA CPE allocation is based on the peak rate capability, which can either be the capability committed in the service package or the capability evaluated by the operator to provide services for users. However, with the rise of services such as IPTV that have high requirements on the average rate, CPE allocation based on the peak rate capability becomes uncompetitive against fixed network services. Against this backdrop, rate-based CPE allocation is introduced. With rate-based CPE allocation, a service package is charged based on the average rate and the operator determines the capability to provide the average rate required in the package for users. If the average rate can be provided as required, the operator registers the users with the package. If the average rate cannot be provided as required, the operator does not register the users with the package.

The process of rate-based CPE allocation is as follows:

1. On the CPE allocation window of the MAE, the operator staff enters the address of the user requiring FWA services.
2. The CPE allocation system obtains the primary cell that this address is homed to.
3. The gNodeB provides the CPE allocation system with information such as signal strength and system load of the primary cell.
4. Based on the gNodeB-provided information and other related information, the CPE allocation system estimates the average rate that can be provided for the user. If the average rate meets the package requirements, CPE allocation is allowed. If the requirements are not met, CPE allocation is disallowed.

For details about the rate-based CPE allocation process, see related descriptions in *iMaster MAE Product Documentation*.

4 Basic FWA Functions

During user access, the gNodeB uses QoS class identifiers (QCIs) to identify FWA users and provides service commissioning, rate guarantee, and other performance improvement functions for the FWA users. This meets FWA service requirements while ensuring enhanced Mobile Broadband (eMBB) service experience.

NOTE

QCIs belong to QoS parameters of evolved packet system (EPS) bearers in non-standalone (NSA) networking, and 5G QoS Identifiers (5QIs) belong to QoS parameters of QoS flows in SA networking. However, regardless of whether NSA networking or SA networking is used, services are finally mapped to QCI-specific bearers of a gNodeB. Differentiated service (DiffServ) is implemented by allocating different bearers. For details, see *QoS Management*.

4.1 FWA User Identification

4.1.1 Principles

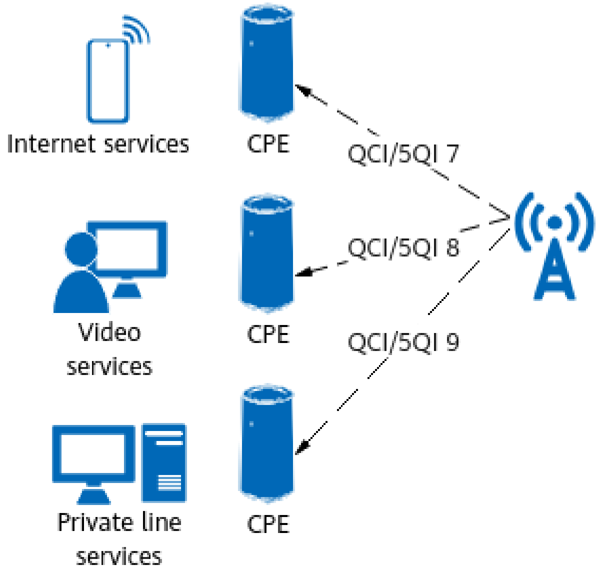
Operators plan for FWA users to use different QCIs (in NSA networking) or 5QIs (in SA networking) from eMBB users to carry services. The FWA user identification process is as follows:

1. The core network assigns QCIs or 5QIs to users.

   Different QCIs or 5QIs are assigned to FWA and eMBB users. The core network notifies the corresponding gNodeB of QCIs or 5QIs assigned for each type of user during RAB setup. In SA networking, after receiving 5QIs assigned for a type of user, the gNodeB maps the 5QIs to the corresponding QCIs.
2. The gNodeB configures the service type for each QCI (by using the **gNBQciBearer**.***ServiceType*** parameter), as listed in **Figure 4-1**.
   - If the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_INTERNET**, this QCI indicates Internet services in FWA scenarios.
   - If the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_VIDEO**, this QCI indicates video services in FWA scenarios.
   - If the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_PRIVATE_LINE**, this QCI indicates private line services in FWA scenarios.

3. If QCI-specific bearers have been set up for a user's Internet services, video services, or private line services, the gNodeB identifies the user as an Internet user, video user, or private line user, respectively. After a user is identified as an Internet user, video user, or private line user, if there is no change in the user bearer but a change in the service type configuration of the corresponding QCI on the base station side (**gNBQciBearer**.***ServiceType*** configuration), the user attribute does not change unless the user accesses the network again.

Figure 4-1 Mapping between QCIs/5QIs and service types



NOTE

- For VoIP services, the default bearer with QCI 5 is always used for Session Initiation Protocol (SIP) signaling, and the dedicated bearer with QCI 1 is always used for voice packets.
- Internet services are assigned the default bearer for Internet PDU sessions, which is a non-guaranteed bit rate (non-GBR) bearer. Video services can be assigned a GBR bearer or a non-GBR bearer.
- Internet users in FWA scenarios are also referred to as B2H users or home broadband users.
- Internet users in FWA scenarios with the **supportOfRedCap-r17** capability defined in 3GPP Release 17 are referred to as RedCap users in FWA scenarios or RedCap FWA users.

4.1.2 Network Analysis

4.1.2.1 Benefits

FWA user identification enables the base station to identify various types of FWA users. The identification is a prerequisite for certain FWA features and functions to take effect. For details, see the section "Prerequisite Functions" for each FWA feature or function.

4.1.2.2 Impacts

The impacts between this function and other functions are described in **4.1.3.2.3 Function Impacts**.

This function has no impact on the network.

4.1.3 Requirements

4.1.3.1 Licenses

None

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

None

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS filtering-based downlink beamforming weight optimization (low-frequency TDD)	**DL_BF_WEIGHTFILTER_OPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	**4.2 SRS Filtering-based Downlink Beamforming Weight Optimization (Low-Frequency TDD)**	FWA-related service types (Internet services, video services, or private line services) must be configured for FWA user identification. Otherwise, SRS filtering-based downlink beamforming weight optimization does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.*FwaAlgoSwitch* parameter	5.1 User PRB Control for FWA	• If the **NRDUCellFwaQos**.*FwaInternetUserResId* parameter is set to a value other than **255**, the Internet service type must be configured for FWA user identification. Otherwise, cell-level PRB proportion control for FWA Internet users does not take effect in user PRB control for FWA. • If the **NRDUCellFwaQos**.*FwaPrivateLineUserResId* parameter is set to a value other than **255**, the private line service type must be configured for FWA user identification. Otherwise, cell-level PRB proportion control for FWA private line users does not take effect in user PRB control for FWA.

RAT	Function Name	Function Switch	Reference	Description
TDD	FWA private line service assurance	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	**6.1 Basic Service Assurance for FWA Private Line**	The private line service type must be configured for FWA user identification. Otherwise, the FWA private line service assurance function does not take effect.
Low-frequency TDD	Flexible spectrum access	**FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter	*Uplink Flexible Spectrum Scheduling*	Flexible spectrum access (FSA) does not take effect for FWA users.
Low-frequency TDD	SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	FWA users do not switch to a sparse search space set group (SSSG).
Low-frequency TDD	RedCap SSSG switching	**REDCAP_SSSG_SWITCHING_SW** or **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*RedCap*	FWA users do not switch to a sparse SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent power control energy saving	**SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Intelligent power control energy saving does not take effect for FWA users.

4.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Service Type	**gNBQciBearer**.***ServiceType***	Set this parameter based on the network plan.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to **4.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Modifying QCI-specific bearer configurations

```
//Changing the service type of QCI-7 bearers to Internet access
MOD GNBQCIBEARER: Qci=7, ServiceType=FWA_INTERNET;
//Changing the service type of QCI-8 bearers to video
MOD GNBQCIBEARER: Qci=8, ServiceType=FWA_VIDEO;
//Changing the service type of QCI-9 bearers to private line
MOD GNBQCIBEARER: Qci=9, ServiceType=FWA_PRIVATE_LINE;
```

- Adding QCI-specific bearer configurations

```
//Adding a QCI-11 bearer with service type being Internet access
ADD GNBQCIBEARER: Qci=11, ServiceType=FWA_INTERNET;
//Adding a QCI-12 bearer with service type being video
ADD GNBQCIBEARER: Qci=12, ServiceType=FWA_VIDEO;
//Adding a QCI-13 bearer with service type being private line
ADD GNBQCIBEARER: Qci=13, ServiceType=FWA_PRIVATE_LINE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Changing the service type of QCI-7 bearers to normal service
MOD GNBQCIBEARER: Qci=7, ServiceType=NORMAL;
//Changing the service type of QCI-8 bearers to normal service
MOD GNBQCIBEARER: Qci=8, ServiceType=NORMAL;
//Changing the service type of QCI-9 bearers to normal service
MOD GNBQCIBEARER: Qci=9, ServiceType=NORMAL;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

Run the **DSP NRCELLUENUMBER** command to query the number of FWA users. If the value of **FWA B2H UE Number** or **FWA Private Line UE Number** is not 0, the FWA user identification function has been enabled.

NOTE

In the **DSP NRCELLUENUMBER** command output, the number of users is measured based on the bearer type. If a user has multiple bearers of different types, it is measured for each bearer type once. If a user has multiple bearers of the same type, repeated measurement is prevented.

4.1.4.3 Network Monitoring

- When there are B2H users on a network, observe the values of the following counters to monitor the performance of these users:
  - **N.User.RRCConn.B2H.Avg**
  - **N.User.RRCConn.Active.B2H.Avg**
  - **N.ThpVol.DL.B2H**
  - **N.ThpVol.UL.B2H**
  - **N.ChMeas.PRB.UL.FWA.B2H.Used**
  - **N.ChMeas.PRB.DL.FWA.B2H.Used**
- When there are IPTV users on a network, observe the values of the following counters or CHRs to monitor the performance of these users:
  - Counters
    - NSA networking
      - **N.User.RRCConn.Active.QCI.Avg**
      - **N.User.RRCConn.Active.QCI.Max**
      - **N.ThpVol.DL.QCI**
      - **N.ThpVol.DL.LastSlot.QCI**
      - **N.ThpTime.DL.RmvLastSlot.QCI**
    - SA networking
      - **N.User.RRCConn.Active.DuCell5QI.Avg**
      - **N.User.RRCConn.Active.DuCell5QI.Max**
      - **N.ThpVol.DL.DuCell5QI**
      - **N.ThpVol.DL.LastSlot.DuCell5QI**
      - **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
  - CHRs

    PERIOD_UE_PACKET_MEASURE_PDCP or **PERIOD_UE_PACKET_MEASURE_RLC** (used to monitor the IPTV transmission quality, including the packet loss rate, packet delay, and packet jitter)

- When there are FWA private line users on a network, observe the values of the following counters to monitor the performance of these users:
  - NSA networking
    - **N.User.RRCConn.Active.QCI.Avg**
    - **N.User.RRCConn.Active.QCI.Max**
    - **N.ThpVol.DL.QCI**
    - **N.ThpVol.DL.LastSlot.QCI**
    - **N.ThpTime.DL.RmvLastSlot.QCI**
    - **N.ChMeas.PRB.UL.FWA.PrivateLine.Used**
    - **N.ChMeas.PRB.DL.FWA.PrivateLine.Used**
  - SA networking
    - **N.User.RRCConn.Active.DuCell5QI.Avg**
    - **N.User.RRCConn.Active.DuCell5QI.Max**
    - **N.ThpVol.DL.DuCell5QI**
    - **N.ThpVol.DL.LastSlot.DuCell5QI**
    - **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
    - **N.ChMeas.PRB.UL.FWA.PrivateLine.Used**
    - **N.ChMeas.PRB.DL.FWA.PrivateLine.Used**

NOTE

- In NSA networking, only QCI 9-related counters are subscribed to by default. To subscribe to the performance counters of another QCI, set the **gNBQciBearer**.***CounterObjSubscriptionInd*** parameter corresponding to this QCI to **YES** before network monitoring.
- In SA networking, only 5QI 9-related counters are subscribed to by default. To subscribe to the performance counters of another 5QI, set the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter corresponding to this 5QI to **YES** before network monitoring.

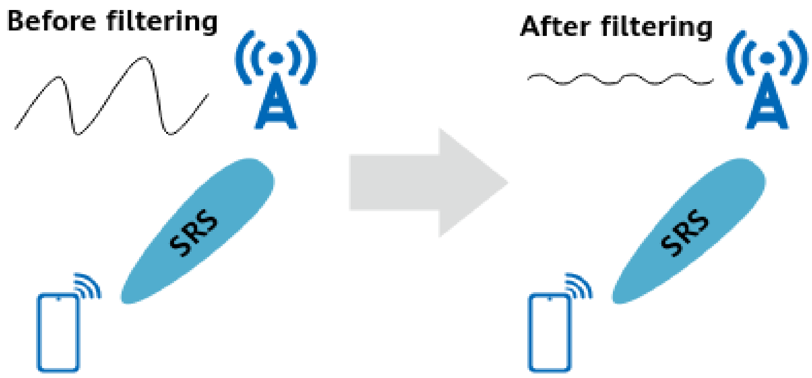
4.2 SRS Filtering-based Downlink Beamforming Weight Optimization (Low-Frequency TDD)

4.2.1 Principles

Generally, FWA users are fixedly located and changes in their channel quality are typically caused by external noise. To resist noise and improve the accuracy of channel quality estimation by the gNodeB for stationary users, the SRS filtering-based downlink beamforming weight optimization function is introduced. After this function is enabled, the gNodeB performs filtering on the SRS and then calculates the downlink beamforming weights for identified FWA users (as detailed in the section on downlink beamforming in *MIMO (TDD)*) to improve the accuracy of downlink beam directions, as shown in **Figure 4-2**.

SRS filtering-based downlink beamforming weight optimization is controlled by the **DL_BF_WEIGHTFILTER_OPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter. This function takes effect only in massive MIMO cells and is recommended in FWA or FWA +eMBB hybrid scenarios.

Figure 4-2 SRS filtering



After the function switch is turned on or off, users need to re-access the network to make this function take effect or terminate this function. Since FWA users stay online for a long time, for immediate effect, run the **STR NRCELLSERVICEPARAMEFFECT** command. For details, see **4.2.4.1 Data Configuration**.

4.2.2 Network Analysis

4.2.2.1 Benefits

This function improves the average downlink user throughput (**User Downlink Average Throughput (DU)**) on the entire network when there is a high proportion of users at the cell edge or at a medium distance from the cell center. For users with a low signal-to-noise ratio (SNR), this function increases the downlink spectral efficiency and MCS indexes, reduces the number of RBs paired for MU beamforming at each layer, and slightly changes the number of paired layers and the ranks used in scheduling. A lower SNR leads to more significant gains. There are no significant gains for users with a high SNR.

4.2.2.2 Impacts

The impacts between this function and other functions are described in **4.2.3.2.3 Function Impacts**.

This function has the following network impacts: For users with a high SNR, negative impacts may be generated in mobility scenarios.

4.2.3 Requirements

4.2.3.1 Licenses

None

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
FWA user identification	**gNBQciBearer**.***ServiceType***	**4.1 FWA User Identification**	FWA-related service types must be configured for FWA user identification. Otherwise, SRS filtering-based downlink beamforming weight optimization does not take effect.

Function Name	Function Switch	Reference	Description
Downlink rank detection	**NRDUCellDlRank**.***DlRankSelAlgoSw***	None	Either of the following conditions must be met before SRS filtering-based downlink beamforming weight optimization is enabled: 1. The **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is selected. 2. When the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is deselected, the **NRDUCellDlRank**.***DlRankDetectCorrThld*** parameter must be set to **100**.
SRS noise reduction for low-SNR channels	**LOW_SNR_CHN_DENOISE_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Massive MIMO AHR (Low-Frequency TDD)*	If both SRS filtering-based downlink beamforming weight optimization and SRS noise reduction for low-SNR channels are enabled, the latter takes effect.

4.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	**NRDUCellAlgoSwitch.*AdaptiveEdgeExpEnhSwitch***	DL_BF_WEIGHTFILTER_OPT_SW	Select the **DL_BF_WEIGHTFILTER_OPT_SW** option.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to **4.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SRS filtering-based downlink beamforming weight optimization
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
AdaptiveEdgeExpEnhSwitch=DL_BF_WEIGHTFILTER_OPT_SW-1;
//Making this function take effect or be terminated immediately for users (using an SA user with a 5QI of 7
as an example)
STR NRCELLSERVICEPARAMEFFECT: NrCellId=1, NrNetworkingOption=SA, First5qi=7;
```

NOTE

STR NRCELLSERVICEPARAMEFFECT is a high-risk command. After this command is executed, intra-cell handovers are automatically triggered for 5QI-or QCI-specific users, causing service interruption for hundreds of milliseconds.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.


```
//Disabling SRS filtering-based downlink beamforming weight optimization
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
AdaptiveEdgeExpEnhSwitch=DL_BF_WEIGHTFILTER_OPT_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

- Observing the switch status

  Run the **LST NRDUCELLALGOSWITCH** command. If the value of **Adaptive Edge Experience Enhance Switch** is **DL BF Weight Filtering Optimize Switch:On** in the command output, this function has been activated.
- Observing performance indicators

  Observe the indicators described in **4.2.4.3 Network Monitoring** to check whether this function has taken effect.

4.2.4.3 Network Monitoring

Observe the average downlink user throughput (**User Downlink Average Throughput (DU)**).

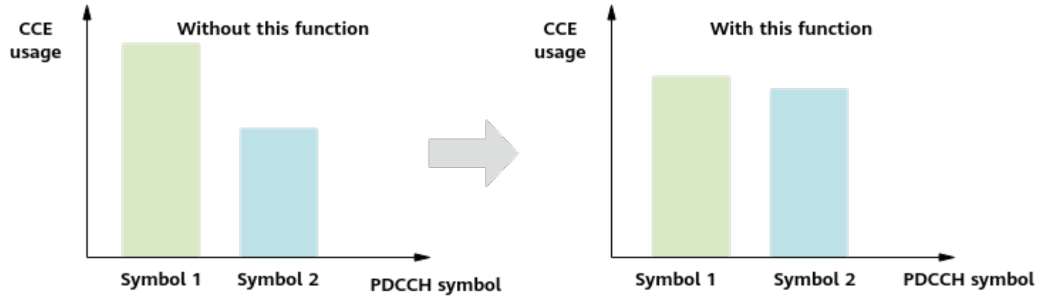
4.3 PDCCH Inter-Symbol Load Balancing Optimization

4.3.1 Principles

PDCCH inter-symbol load balancing optimization is introduced to balance CCE usage among PDCCH symbols. With this function disabled, the gNodeB adjusts the number of users that can access PDCCH symbols for the balance purpose, but the CCE usage is actually imbalanced among symbols; with this function enabled, **the gNodeB adjusts CCE allocation to balance CCE usage among symbols and achieve PDCCH symbol load balancing**, as illustrated in **Figure 4-3**.

PDCCH inter-symbol load balancing optimization is controlled by the **PDCCH_SYMBOL_LOAD_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter.

Figure 4-3 PDCCH inter-symbol load balancing optimization



4.3.2 Network Analysis

4.3.2.1 Benefits

It is recommended that this function be enabled in heavily loaded cells (PDCCH CCE usage > 20%). After this function takes effect, the number of users scheduled in the uplink and downlink, CCE usage, and average uplink and downlink cell throughput increase. If this function is enabled in other scenarios, the expected gains may not be achieved.

NOTE

- PDCCH CCE usage = (**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**) x 100%
- Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**

4.3.2.2 Impacts

The impacts between this function and other functions are described in **4.3.3.2.3 Function Impacts**.

This function has no impact on the network.

4.3.3 Requirements

4.3.3.1 Licenses

None

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

None

4.3.3.2.2 Mutually Exclusive Functions

None

4.3.3.2.3 Function Impacts

None

4.3.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Table 4-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Enhancement Switch	**NRDUCellPdcch**.***PdcchAlgoEnhSwitch***	PDCCH_SYMBOL_LOAD_OPT_SW	Select the **PDCCH_SYMBOL_LOAD_OPT_SW** option.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling PDCCH inter-symbol load balancing optimization
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=PDCCH_SYMBOL_LOAD_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling PDCCH inter-symbol load balancing optimization
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=PDCCH_SYMBOL_LOAD_OPT_SW-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

- Observing the switch status

  Run the **LST NRDUCELLPDCCH** command. If the value of **PDCCH Algorithm Enhancement Switch** is **PDCCH Symbol Load Optimization Sw:On** in the command output, this function has been activated.

- Observing performance indicators

  Observe the indicators described in **4.3.4.3 Network Monitoring** to check whether this function has taken effect.

4.3.4.3 Network Monitoring

Monitor the performance of this function using the following indicators:

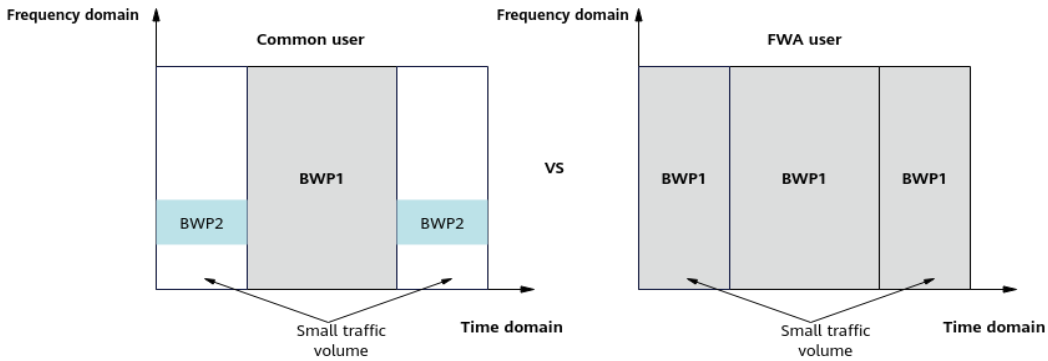
- CCE usage = (**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**) x 100%
- Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**

4.4 Power Saving BWP Deactivation for FWA Users (Low-Frequency TDD)

4.4.1 Principles

Due to the use of large bandwidths in 5G, mobile UEs have high power consumption. After power saving BWP (detailed in *UE Power Saving*) is enabled in a cell, UEs automatically switch to the power saving BWP with a small bandwidth when the traffic volume is low. However, user experience deteriorates after the UEs switch to the power saving BWP. **FWA users have external power supply and do not have power saving requirements. As such, FWA users in a cell with power saving BWP enabled can have such BWP not take effect to enjoy better service experience, as illustrated in Figure 4-4.**

Figure 4-4 Power saving BWP deactivation for FWA users



Power saving BWP deactivation for FWA users is introduced to deactivate power saving BWP for all FWA users. Specifically, after this function is enabled:

- For newly admitted FWA users, power saving BWP is not configured.
- For FWA users with power saving BWP already configured, the configuration will be removed during subsequent RRC reconfiguration.

Power saving BWP deactivation for FWA users is controlled by the **FWA_BWP2_DISABLE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

4.4.2 Network Analysis

4.4.2.1 Benefits

- This function improves the network performance of FWA users in cells with the power saving BWP function enabled. The following performance indicators reflect the main improvements:
  - Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
  - Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
  - Average uplink user throughput = **User Uplink Average Throughput (DU)**
  - Average downlink user throughput = **User Downlink Average Throughput (DU)**
  - Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**

- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

- This function disables the power saving BWP function for FWA users, in turn, alleviating the deterioration of related indicators. For details about the network indicators affected by the power saving BWP function, see the impact analysis section of the power saving BWP function in *UE Power Saving*.

4.4.2.2 Impacts

The impacts between this function and other functions are described in **4.4.3.2.3 Function Impacts**.

This function has the following network impacts: The number of users for which power saving BWP takes effect decreases, and the value of the **N.Bwp2.Dur.Total** counter decreases.

4.4.3 Requirements

4.4.3.1 Licenses

None

4.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Power saving BWP deactivation for FWA users takes effect only when the power saving BWP function is enabled.

4.4.3.2.2 Mutually Exclusive Functions

None

4.4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
FWA user identification	**gNBQciBearer**.***ServiceType***	**4.1 FWA User Identification**	FWA-related service types must be configured for the FWA user identification function. Otherwise, power saving BWP deactivation for FWA users does not take effect.

4.4.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.4.4 Operation and Maintenance

4.4.4.1 Data Configuration

4.4.4.1.1 Data Preparation

Table 4-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	FWA_BWP2_DISABLE_SW	Select this option.

4.4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling power saving BWP deactivation for FWA users
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_BWP2_DISABLE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling power saving BWP deactivation for FWA users
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_BWP2_DISABLE_SW-0;
```

4.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4.2 Activation Verification

- Observing the switch status

  Run the **LST NRDUCELLALGOSWITCH** command. If the value of **FWA Algorithm Switch** is **FWA BWP2 Disabling Switch:On** in the command output, this function has been enabled.

- Observing performance counters

  If the value of **N.Bwp2.Dur.Total** changes, this function has taken effect.

4.4.4.3 Network Monitoring

Monitor the network performance of this function using the following indicators:

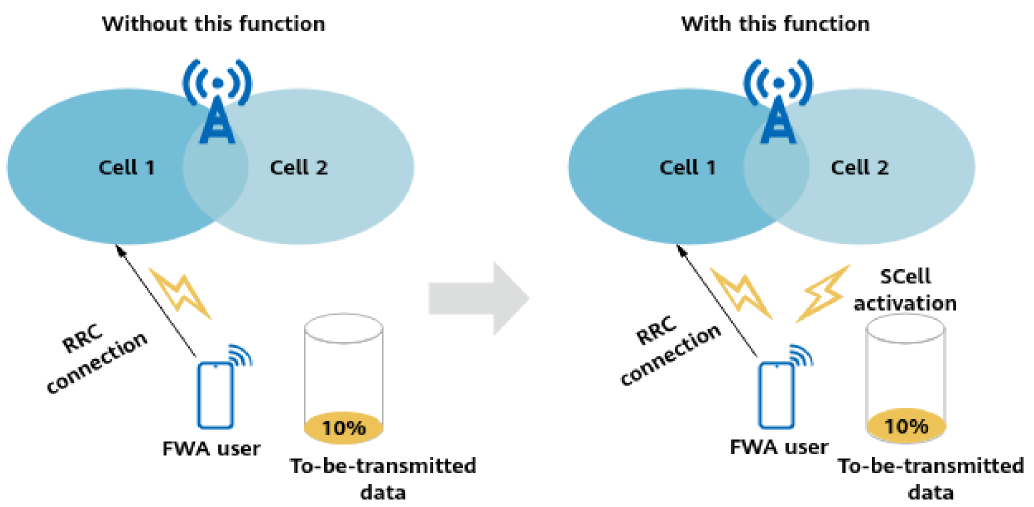
- Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
- Average uplink user throughput = **User Uplink Average Throughput (DU)**
- Average downlink user throughput = **User Downlink Average Throughput (DU)**
- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**

4.5 SCell Activation Optimization for FWA Users

4.5.1 Principles

In carrier aggregation (CA) scenarios, the traffic volume requirements must be met to trigger SCell activation or deactivation for CA users in a cell. For details, see section "SCell Activation" in *Carrier Aggregation*. **To enable FWA users to use SCells for data transmission earlier, SCell activation optimization for FWA users is introduced.** After this function is enabled, SCells can be activated or deactivated for CA-capable FWA Internet users and private line users without requirements on the traffic volume being met, as illustrated in **Figure 4-5**. This function is controlled by the **FWA_CA_FAST_SCC_ACT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

Figure 4-5 SCell activation optimization for FWA users



4.5.2 Network Analysis

4.5.2.1 Benefits

SCells can be activated earlier for FWA Internet users and private line users, improving the user-perceived rate in both the uplink and downlink.

4.5.2.2 Impacts

The impacts between this function and other functions are described in **4.5.3.2.3 Function Impacts**.

This function has the following network impacts: The number of SCell activation or deactivation times decreases. For details, see **4.5.4.3 Network Monitoring**. In addition, SCell activation may cause the values of other performance indicators to fluctuate. For details about the network impacts, see the network analysis section for inter-gNodeB CA in *CA Experience Improvement* and the network analysis sections for the following functions in *Carrier Aggregation*: downlink intra-band CA, uplink intra-band CA, downlink intra-FR inter-band CA, and uplink intra-FR inter-band CA.

4.5.3 Requirements

4.5.3.1 Licenses

None

4.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Carrier Aggregation	**INTRA_BAND_CA_SW**, **INTRA_BAND_UL_CA_SW**, **INTRA_FR_INTER_BAND_CA_SW**, **INTRA_FR_INTER_BAND_UL_CA_SW**, and **INTER_GNODEB_CA_SW** options of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation* *CA Experience Improvement*	SCell activation optimization for FWA users depends on CA. Therefore, at least one CA function needs to be enabled.

4.5.3.2.2 Mutually Exclusive Functions

None

4.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA user identification	**gNBQciBearer**.***ServiceType***	*FWA*	The FWA user identification function requires the configured service type to be FWA Internet service or private line service. If this requirement is not met, SCell activation optimization for FWA users does not take effect.

4.5.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.5.4 Operation and Maintenance

4.5.4.1 Data Configuration

4.5.4.1.1 Data Preparation

Table 4-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	FWA_CA_FAST_SCC_ACT_SW	Select this option.

4.5.4.1.2 Using MML Commands

Before using MML commands, refer to **4.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SCell activation optimization for FWA users
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_CA_FAST_SCC_ACT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling SCell activation optimization for FWA users
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_CA_FAST_SCC_ACT_SW-0;
```

4.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4.2 Activation Verification

Observe the indicators described in **4.5.4.3 Network Monitoring** to check whether this function has taken effect.

4.5.4.3 Network Monitoring

Monitor the network performance of this function using the following indicators:

- **N.CA.SCell.Act.Att**: Check whether its value decreases.
- **N.CA.SCell.Act.Succ**: Check whether its value decreases.
- **N.CA.SCell.Deact.Att**: Check whether its value decreases.
- **N.CA.SCell.Deact.Succ**: Check whether its value decreases.

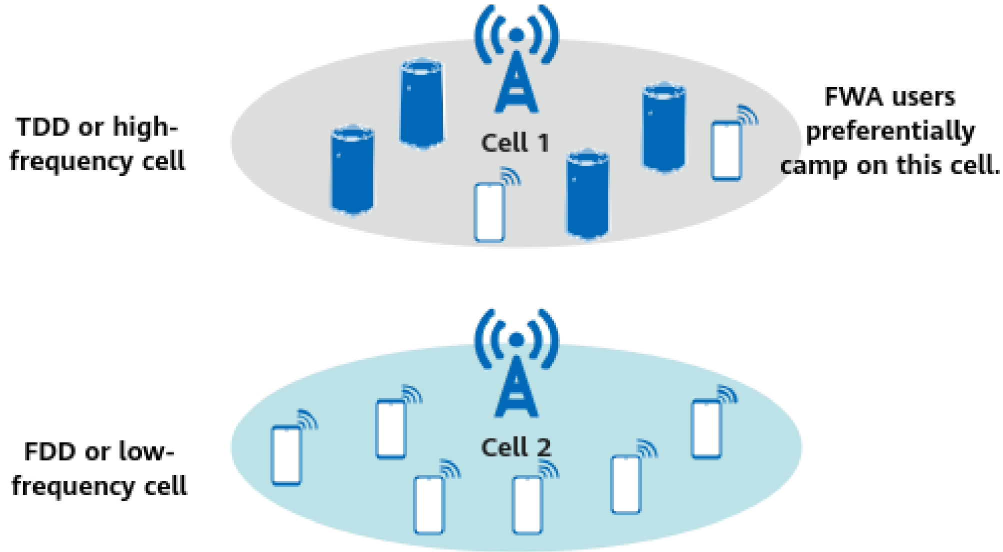
4.6 Differentiated Carrier Selection for FWA Users

4.6.1 Principles

FWA CPEs have higher antenna gains than eMBB mobile phones, and FWA users have better coverage performance than eMBB users. Therefore, FWA users can preferentially camp on carriers in higher frequency bands, rather than compete with eMBB users for carrier resources in low frequency bands.

Differentiated carrier selection for FWA users is introduced to select more suitable carriers for FWA users based on QCIs (in NSA networking) or 5QIs (in SA networking), improving user experience, as illustrated in **Figure 4-6**.

Figure 4-6 Differentiated carrier selection for FWA users



In NSA networking, the working principles of and parameter configurations for this function are as follows on the eNodeB side:

- If the **CellQciPara.*NrScgFreqPriorityGroupId*** parameter is set to a value ranging from **0** to **255**, differentiated carrier selection for FWA users is enabled.
  – Configuring an SCG frequency priority group

    Table 4-6 describes the related parameters.

    Table 4-6 Main related parameters

Parameter Name	Parameter ID	Description
NR SCG Frequency Priority Group ID	**NrScgFreqPriGrp.*NrScgFreqPriorityGroupId***	This parameter specifies the ID of an NR SCG frequency priority group.
PCC DL EARFCN	**NrScgFreqPriGrp.*PccDlEarfcn***	This parameter specifies the downlink frequency of the PCC.
NR SCG DL ARFCN	**NrScgFreqPriGrp.*NrScgDlArfcn***	This parameter specifies the downlink SSB frequency of an NR SCG cell.
NR SCG DL ARFCN Priority	**NrScgFreqPriGrp.*NrScgDlArfcnPriority***	This parameter specifies the downlink SSB frequency priority of an NR SCG cell.

  – Configuring the mapping between QCIs and SCG frequency priority groups

By setting **CellQciPara.*NrScgFreqPriorityGroupId*** to the same value as **NrScgFreqPriGrp**.***NrScgFreqPriorityGroupId***, the SCG carrier selection policy for an FWA user corresponding to **CellQciPara**.***Qci*** can be configured to enable the user to preferentially camp on an SCG carrier on a high-priority frequency during SCG carrier selection. If multiple QCIs are configured for the user, the QCI with the highest priority among those corresponding to valid **CellQciPara**.***NrScgFreqPriorityGroupId*** configurations is used.

After the eNodeB identifies an FWA user by QCI, based on the downlink frequency of the PCC on which the user camps, the eNodeB uses the **NrScgFreqPriGrp**.***PccDlEarfcn*** parameter configuration corresponding to **NrScgFreqPriGrp**.***NrScgFreqPriorityGroupId*** to determine the downlink frequency priority of an SCG.

- If the **CellQciPara.*NrScgFreqPriorityGroupId*** parameter is set to **65535**, differentiated carrier selection for FWA users is disabled.

For details about SCG addition, see descriptions of SCG carrier management in *EPC-based NSA Basics*.

NOTE

Differentiated carrier selection for FWA users in NSA networking requires that carrier management in NSA DC be enabled. This document describes only QCI-based SCG carrier selection for FWA users. For details about carrier management in NSA DC, see *EPC-based NSA Basics*.

In SA networking, the working principles of and parameter configurations for this function are as follows on the gNodeB side:

- If the **NRCellQciBearer**.***gNBFreqPriorityGroupId*** parameter is set to a value ranging from **0** to **254**, differentiated carrier selection for FWA users is enabled.
  – Configuring a gNodeB frequency priority group
    Table 4-7 describes the related parameters.

Table 4-7 Main related parameters

Parameter Name	Parameter ID	Description
gNodeB Frequency Priority Group ID	**gNBFreqPriorityGroup**.***gNBFreqPriority-GroupId***	This parameter specifies the ID of a gNodeB frequency priority group.
Frequency Index	**gNBFreqPriorityGroup**.***FreqIndex***	This parameter specifies the index of a frequency in the gNodeB frequency priority group.
RAT Type	**gNBFreqPriorityGroup**.***RatType***	This parameter specifies the RAT corresponding to the configured frequency.

Parameter Name	Parameter ID	Description
SSB Frequency Position Describe Method	**gNBFreqPriorityGroup**.***SsbDescMethod***	This parameter specifies the frequency-domain position description method of the SSB on a neighboring frequency.
SSB Frequency Position	**gNBFreqPriorityGroup**.***SsbFreqPos***	This parameter specifies the SSB frequency-domain position of an NR frequency. This parameter must be configured when **gNBFreqPriorityGroup**.***RatType*** is set to **NR**.
Downlink EARFCN	**gNBFreqPriorityGroup**.***DlEarfcn***	This parameter specifies the downlink frequency of a neighboring E-UTRAN cell. This parameter must be configured when **gNBFreqPriorityGroup**.***RatType*** is set to **EUTRAN**.
Cell Reselection Priority	**gNBFreqPriorityGroup**.***CellReselPri***	This parameter specifies the cell reselection priority of the configured frequency. A larger value of this parameter indicates a higher priority.
Cell Reselection Sub-Priority	**gNBFreqPriorityGroup**.***CellReselSubPri***	This parameter specifies the cell reselection sub-priority of the configured frequency. A larger value of this parameter indicates a higher priority.

– Configuring the mapping between 5QIs and gNodeB frequency priority groups

By setting **NRCellQciBearer**.***gNBFreqPriorityGroupId*** to the same value as **gNBFreqPriorityGroup**.***gNBFreqPriorityGroupId***, the cell reselection policy can be configured for an FWA user corresponding to **NRCellQciBearer**.***Qci*** to enable the user to preferentially camp on a cell on a high-priority frequency during cell reselection.

After identifying an FWA user by 5QI, the gNodeB sends an RRCRelease message containing the 5QI-specific frequency priority group to the user.

a. The gNodeB generates a frequency priority list based on the configured frequencies (**gNBFreqPriorityGroup**.***SsbFreqPos*** or **gNBFreqPriorityGroup**.***DlEarfcn***), priority (**gNBFreqPriorityGroup**.***CellReselPri***), and sub-priority (**gNBFreqPriorityGroup**.***CellReselSubPri***).

b. The gNodeB filters out NR frequencies not supported by the user.

c. The gNodeB sends the frequencies that remain after filtering to the user through the *cellReselectionPriorities* IE in an RRCRelease message. The user adds up the cell reselection priority and the cell reselection sub-priority and uses the sum as the final cell reselection priority for a configured frequency.

- If the **NRCellQciBearer**.***gNBFreqPriorityGroupId*** parameter is set to **255**, differentiated carrier selection for FWA users is disabled.

NOTE

Differentiated carrier selection for FWA users in SA networking requires that basic cell reselection functions be enabled.

This document describes only 5QI-based cell reselection for non-RedCap FWA users. For details about basic cell reselection functions, see *SA Mobility Management in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*. For details about 5QI-based cell reselection for RedCap users, see the "Frequency selection for RedCap" section in *RedCap*.

4.6.2 Network Analysis

4.6.2.1 Benefits

In NSA networking, in the cell on the highest-priority frequency corresponding to a QCI, this function increases the proportion of users with this QCI (**N.User.RRCConn.Active.QCI.Avg**/**N.User.RRCConn.Active.Avg** x 100%).

In SA networking, in the cell on the highest-priority frequency corresponding to a 5QI, this function increases the proportion of users with this 5QI (**N.User.RRCConn.Active.DuCell5QI.Avg**/**N.User.RRCConn.Active.Avg** x 100%).

4.6.2.2 Impacts

The impacts between this function and other functions are described in **4.6.3.2.3 Function Impacts**.

This function has the following network impacts:

After this function is enabled, the number of online users, average uplink or downlink user throughput, and uplink or downlink PRB usage of a cell may decrease if users in this cell are transferred to a cell on the highest-priority frequency.

If functions that may cause handovers in connected mode are also enabled, users transferred to a cell on the highest-priority frequency may experience handovers in connected mode, and ultimately fail to camp on such a cell. Examples of these functions include connected mode MLB (detailed in *Mobility Load Balancing in Connected Mode*) and SA mobility management in connected mode (detailed in *SA Mobility Management in Connected Mode*).

4.6.3 Requirements

4.6.3.1 Licenses

None

4.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.6.3.2.1 Prerequisite Functions

None

4.6.3.2.2 Mutually Exclusive Functions

None

4.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	SPID-based SCG frequency priority configuration	**SPIDCFG**.***NrScgFreqPriorityGroupId***	*EPC-based NSA Basics*	In NSA networking, if both SPID-based SCG frequency priority configuration and differentiated carrier selection for FWA users take effect for a user, the SCG frequency priority configured by SPID takes precedence for the user.

RAT	Function Name	Function Switch	Reference	Description
TDD	Operator-specific SCG frequency priority configuration	**CnOperator**.***NrScgFreqPriorityGroupId***	*EPC-based NSA Basics*	In NSA networking, if both operator-specific SCG frequency priority configuration and differentiated carrier selection for FWA users take effect for a user, the SCG frequency priority configured by QCI takes precedence for the user.
TDD	NSA carrier combination selection based on user experience	LTE: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	In NSA networking, if both differentiated carrier selection for FWA users and NSA carrier combination selection based on user experience are enabled, the latter function does not take effect for users. In addition, no handover for NSA carrier combination selection based on user experience will be initiated towards a base station with differentiated carrier selection for FWA users enabled. If NSA carrier combination selection based on user experience is required, do not enable differentiated carrier selection for FWA users.

RAT	Function Name	Function Switch	Reference	Description
TDD	QCI-specific PSCell management policy	**QciPara**.***MultiCarrCfgGroupId***	*EPC-based NSA Basics*	In NSA networking, when differentiated carrier selection for FWA users is enabled together with the QCI-specific PSCell management policy function: SCG frequency priority in effect = SCG frequency priority of the QCI with the highest priority configured by **CellQciPara**.***NrScgFreqPriorityGroupId*** + NR SCG frequency priority offset specified by **CellMultiCarrCfgGrp**.***NrScgPriorityOffset*** corresponding to the QCI with the highest priority configured by **NrScgFreqConfig**.***ScgDlArfcnPriority***.
TDD	Low-frequency deprioritisation-based idle mode MLB	**NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**, **NRCellMlb**.***IdleMlbPolicy*** set to **DECREASE_PRIORITY**, and **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected	*Mobility Load Balancing in Idle Mode*	In SA networking, low-frequency deprioritisation-based idle mode MLB does not take effect for users for which differentiated carrier selection for FWA users has taken effect.
TDD	Low-frequency dedicated-priority-based idle mode MLB	**NRCellMlb**.***IdleMlbPolicy*** set to **DEDICATED_PRIORITY** and the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected	*Mobility Load Balancing in Idle Mode*	In SA networking, low-frequency dedicated-priority-based idle mode MLB does not take effect for users for which differentiated carrier selection for FWA users has taken effect.

RAT	Function Name	Function Switch	Reference	Description
TDD	RFSP-based cell reselection policy	**gNBRfspConfig**.***gNBFreqPriorityGroupId***	*Flexible User Steering*	In SA networking, differentiated carrier selection for FWA users does not take effect for users configured with RFSP-based cell reselection policy.
Low-frequency TDD	Operator-specific dedicated cell reselection priority management	**NRCellOpPolicy**.***gNBFreqPriorityGroupId***	*Multi-Operator Sharing*	In SA networking, operator-specific dedicated cell reselection priority management does not take effect on users with differentiated carrier selection for FWA users in effect.

4.6.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.6.4 Operation and Maintenance

4.6.4.1 Data Configuration

4.6.4.1.1 Data Preparation

Table 4-8 and Table 4-9 describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-8 Parameters used for activation in NSA networking (eNodeB)

Parameter Name	Parameter ID	Setting Notes
NR SCG Frequency Priority Group ID	**CellQciPara**.***NrScgFreqPriorityGroupId***	Set this parameter to the same value as the **NrScgFreqPriGrp**.***NrScgFreqPriorityGroupId*** parameter.
NR SCG Frequency Priority Group ID	**NrScgFreqPriGrp**.***NrScgFreqPriorityGroupId***	Set this parameter based on the network plan.
PCC DL EARFCN	**NrScgFreqPriGrp**.***PccDlEarfcn***	Set this parameter based on the network plan.
NR SCG DL ARFCN	**NrScgFreqPriGrp**.***NrScgDlArfcn***	Set this parameter based on the network plan.
NR SCG DL ARFCN Priority	**NrScgFreqPriGrp**.***NrScgDlArfcnPriority***	Set this parameter based on the network plan.
QoS Class Indication	**CellQciPara**.***Qci***	Set this parameter based on the network plan.

Table 4-9 Parameters used for activation in SA networking (gNodeB)

Parameter Name	Parameter ID	Setting Notes
gNodeB Frequency Priority Group ID	**gNBFreqPriorityGroup**.***gNBFreqPriorityGroupId***	Set this parameter based on the network plan.
Frequency Index	**gNBFreqPriorityGroup**.***FreqIndex***	Set this parameter based on the network plan.
RAT Type	**gNBFreqPriorityGroup**.***RatType***	If an E-UTRAN frequency is to be configured, set this parameter to **EUTRAN**. If an NR frequency is to be configured, set this parameter to **NR**.
SSB Frequency Position	**gNBFreqPriorityGroup**.***SsbFreqPos***	Set this parameter based on the network plan if **gNBFreqPriorityGroup**.***RatType*** is set to **NR**.

Parameter Name	Parameter ID	Setting Notes
SSB Frequency Position Describe Method	**gNBFreqPriorityGroup**.***SsbDescMethod***	Set this parameter to **SSB_DESC_TYPE_GSCN**.
Downlink EARFCN	**gNBFreqPriorityGroup**.***DlEarfcn***	Set this parameter based on the network plan if **gNBFreqPriorityGroup**.***RatType*** is set to **EUTRAN**.
Cell Reselection Priority	**gNBFreqPriorityGroup**.***CellReselPri***	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	**gNBFreqPriorityGroup**.***CellReselSubPri***	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	**NRCellQciBearer**.***gNBFreqPriorityGroupId***	The value must be the same as that of **gNBFreqPriorityGroup**.***gNBFreqPriorityGroupId***.
QoS Class Identifier	**NRCellQciBearer**.***Qci***	Set this parameter based on the network plan.

4.6.4.1.2 Using MML Commands

Before using MML commands, refer to **4.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

In contiguous networking, the configurations must be made for all cells.

Activation Command Examples

- NSA networking

```
//Enabling differentiated carrier selection for FWA users (on the eNodeB side)
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=0, PccDlEarfcn=1500, NrScgDlArfcn=636666,
NrScgDlArfcnPriority=7;
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=0, PccDlEarfcn=1500, NrScgDlArfcn=361000,
NrScgDlArfcnPriority=6;
MOD CELLQCIPARA: LocalCellId=0, Qci=9, NrScgFreqPriorityGroupId=0;
```

- SA networking

```
//Enabling differentiated carrier selection for FWA users (on the gNodeB side)
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, CellReselPri=5, CellReselSubPri=0DOT2;
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR,
```

```
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6300, CellReselPri=3, CellReselSubPri=0DOT2;
MOD NRCELLQCIBEARER: NrCellId=3, Qci=9, gNBFreqPriorityGroupId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- NSA networking

```
//Disabling differentiated carrier selection for FWA users (on the eNodeB side)
MOD CELLQCIPARA: LocalCellId=0, Qci=9, NrScgFreqPriorityGroupId=65535;
```

- SA networking

```
//Disabling differentiated carrier selection for FWA users (on the gNodeB side)
MOD NRCELLQCIBEARER: NrCellId=3, Qci=9, gNBFreqPriorityGroupId=255;
```

4.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.6.4.2 Activation Verification

In NSA networking, observe the message tracing result as follows:

Step 1 Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > LTE > Application Layer > Uu Interface Trace**.

Step 3 If the UE sends an RRC_CONN_SETUP_CMP message to the base station, random access of the UE is successful.

Step 4 If the RRC_CONN_RECFG message sent from the base station to the UE contains measurement configurations for the PCC and corresponding SCCs and SCG frequencies, differentiated carrier selection for FWA users has taken effect.

----End

In SA networking, observe the message tracing result as follows:

Step 1 Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.

Step 3 If the *cellReselectionPriorities* IE in the RRCRelease message delivered by the base station contains information about frequencies (of cells that have neighbor relationships with the local cell) configured by running the **ADD GNBFREQPRIORITYGROUP** command, differentiated carrier selection for FWA users has taken effect.

----End

4.6.4.3 Network Monitoring

None

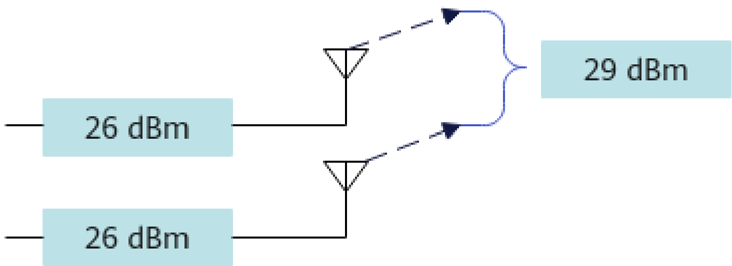
4.7 High-Power UE (Low-Frequency TDD)

4.7.1 Principles

3GPP specifications define power class 1.5 for NR frequency bands. This power class allows UEs to transmit at a maximum power of 29 dBm (as shown in Figure 4-7), which is 3 dB higher than the maximum of 26 dBm for power class 2.0. The high-power UE function is introduced to increase the uplink transmit power of cell edge users, thereby increasing their uplink user-perceived rates. This function is implemented as follows:

1. **NRDUCellSelConfig**.***MaxUeTransmitPower*** is set to **29** so that the base station allows the maximum transmit power of a UE to reach 29 dBm.
2. The base station determines whether a UE supports the power class 1.5 capability based on the **ue-PowerClass-v1610** field in the *BandNR* IE in the UE-reported UECapabilityInformation message.
3. If the UE is a non-CA UE in the uplink, the power class 1.5 capability can take effect on the UE.

Figure 4-7 Transmit power of high-power UEs



NOTE

- For the relevant definition in 3GPP specifications, see Table 6.2.1-1: UE Power Class in section 6.2.1 "UE maximum output power" in 3GPP TS 38.101-1.
- The following frequency band and networking conditions must be met for the high-power UE function to take effect:
  - UE: works in one or more of TDD frequency bands n41, n77, n78, and n79.
  - Cell: works in frequency bands n41, n77, n78, or n79 in SA networking.

4.7.2 Network Analysis

4.7.2.1 Benefits

After this function takes effect, the uplink transmit power of cell edge users increases and the average uplink user throughput (**User Uplink Average Throughput (DU)**) slightly increases.

4.7.2.2 Impacts

The impacts between this function and other functions are described in **4.7.3.2.3 Function Impacts**.

This function has the following network impacts: After this function takes effect, the uplink transmit power of cell edge users increases, slightly increasing the average uplink interference (**N.UL.NI.Avg**).

4.7.3 Requirements

4.7.3.1 Licenses

None

4.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.7.3.2.1 Prerequisite Functions

None

4.7.3.2.2 Mutually Exclusive Functions

None

4.7.3.2.3 Function Impacts

None

4.7.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.7.3.4 Others

This function requires the power class 1.5 capability on UEs, which is indicated by the **ue-PowerClass-v1610** field in the *BandNR* IE in the UECapabilityInformation message. This capability is supported if the field value is **pc1dot5**.

4.7.4 Operation and Maintenance

4.7.4.1 Data Configuration

4.7.4.1.1 Data Preparation

Table 4-10 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-10 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Max UE Transmit Power	**NRDUCellSelConfig**.***MaxUeTransmitPower***	Set this parameter to **29**.

NOTE

The **NRDUCellSelConfig**.***MaxUeTransmitPower*** parameter specifies the maximum transmit power that the base station allows a UE to use and corresponds to the *p-Max* IE in SIB1 as defined in 3GPP TS 38.331. The maximum transmit power of a UE is determined by the UE capability. If this parameter is set to **28**, the maximum transmit power of a UE can reach 28 dBm.

4.7.4.1.2 Using MML Commands

Before using MML commands, refer to 4.7.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the high-power UE function on the base station and allowing the maximum transmit power of UEs to reach 29 dBm
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=29;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the high-power UE function on the base station
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=255;
```

4.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.7.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

Step 3 If the value of **p-Max** in the **frequencyInfoUL** field in SIB1 sent by the base station is **29**, the high-power UE function has been enabled.

Step 4 If the value of the **ue-PowerClass-v1610** field in the *BandNR* IE in the UECapabilityInformation message reported by the UE is **pc1dot5**, the UE supports power class 1.5 capability.

----End

4.7.4.3 Network Monitoring

After this function is enabled, observe the following to monitor the running status of this function:

- Average uplink user throughput (**User Uplink Average Throughput (DU)**)
- Average uplink interference (**N.UL.NI.Avg**)

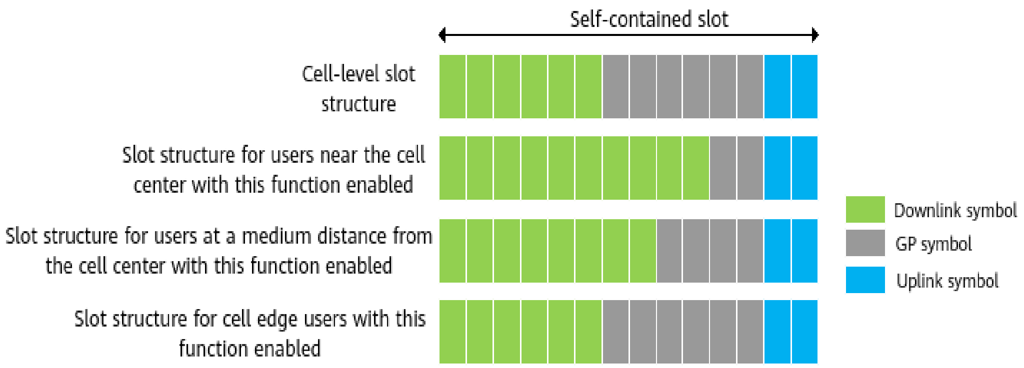
4.8 FWA mmWave GP Symbol Number Adaptation (High-Frequency TDD)

4.8.1 Principles

A farther distance of a user from the base station requires a longer time advance of signal transmission and a larger number of GP symbols. However, users in a cell are located at different locations. If too many GP symbols are configured for users closer to the base station, resources are likely wasted, affecting user experience. As such, FWA mmWave GP symbol number adaptation is introduced. After this function is enabled, **the base station determines the distances away from users based on the user-reported TA measurements. For users with the ue-SpecificUL-DL-Assignment capability, the base station adaptively adjusts the number of PDSCH symbols in self-contained slots based on the distances, reducing unnecessary GP symbol overheads**. The base station uses the **Time domain resource assignment** field in the DCI for downlink scheduling to indicate the start symbol and the length of consecutive symbols allocated to the PDSCH. **Figure 4-8** illustrates this function by using the cell-level slot structure of SS206 as an example.

FWA mmWave GP symbol number adaptation is controlled by the **S_SLOT_GP_SYM_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

Figure 4-8 GP symbol number adaptation



4.8.2 Network Analysis

4.8.2.1 Benefits

This function reduces the GP symbol overheads for users near the cell center or at a medium distance from the cell center and increases downlink data volume, improving the average downlink user throughput (**User Downlink Average Throughput (DU)**) and average downlink cell throughput (**Cell Downlink Average Throughput (DU)**). In addition, the number of uplink response packets increases, and the average uplink user throughput (**User Uplink Average Throughput (DU)**) and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) may increase.

4.8.2.2 Impacts

The impacts between this function and other functions are described in **4.8.3.2.3 Function Impacts**.

This function reduces the GP symbol overheads for users near the cell center or at a medium distance from the cell center and increases the downlink data volume. The impacts are as follows:

- The number of online users in the cell decreases.
- The average downlink MCS index, average downlink rank, and average uplink rank may increase.
- The PDCCH CCE usage (**N.CCE.Used.Avg**/**N.CCE.Avail.Avg** x 100%), uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**), and downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) fluctuate, depending on the cell load and the distributions of users near the cell center, at a medium distance from the cell center, and at the cell edge.

4.8.3 Requirements

4.8.3.1 Licenses

None

4.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.8.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	FWA mmWave GP symbol number adaptation requires that the slot configuration be **4_1_DDDSU**.
Slot structure	**NRDUCell**.***SlotStructure***	*Standards Compliance*	FWA mmWave GP symbol number adaptation requires that the slot structure be **SS203**, **SS204**, **SS205**, or **SS206**.

4.8.3.2.2 Mutually Exclusive Functions

None

4.8.3.2.3 Function Impacts

None

4.8.3.3 Hardware

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

4.8.3.4 Others

This function requires that UEs support the **ue-SpecificUL-DL-Assignment** capability.

4.8.4 Operation and Maintenance

4.8.4.1 Data Configuration

4.8.4.1.1 Data Preparation

Table 4-11 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-11 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	To enable FWA mmWave GP symbol number adaptation, select the **S_SLOT_GP_SYM_ADAPT_SW** option.

4.8.4.1.2 Using MML Commands

Before using MML commands, refer to **4.8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling FWA mmWave GP symbol number adaptation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=S_SLOT_GP_SYM_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling FWA mmWave GP symbol number adaptation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=S_SLOT_GP_SYM_ADAPT_SW-0;
```

4.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.8.4.2 Activation Verification

After this function is enabled, perform the following operations to check whether it has taken effect:

1. Run the **LST NRDUCELLALGOSWITCH** command. If the value of **FWA Algorithm Switch** is **Self-contained Slot GP Symbol Adapt Sw:On** in the command output, this function has been enabled.
2. Run the **DSP NRDUCELL** command to check the cell status and verify that the cell has been activated.

4.8.4.3 Network Monitoring

After this function is enabled, monitor the running status of this function by observing any increase in the following:

- Average downlink user throughput (**User Downlink Average Throughput (DU)**)
- Average downlink cell throughput (**Cell Downlink Average Throughput (DU)**)

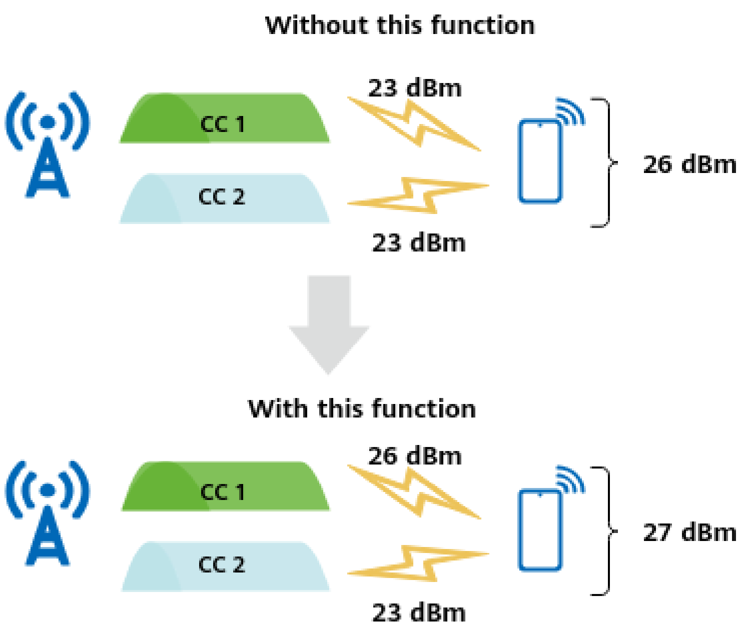
4.9 Uplink CA Power Enhancement

4.9.1 Principles

In versions earlier than 3GPP Release 17, the two carriers of a UE share the rated transmit power in the 2CC case of uplink inter-band CA. That is, the total transmit power cannot exceed the protocol-defined maximum transmit power corresponding to the power class of the UE. According to section 6.2A.1.3 "UE maximum output power for Inter-band CA" in 3GPP TS 38.101 (Release 17), in the 2CC case of uplink inter-band CA, if a UE that reports the **higherPowerLimit-r17** field is of power class 2, with one carrier of power class 3 and the other carrier of power class 2, then the upper limit of the total transmit power of the UE can reach the sum of the transmit power of the two frequency bands to exceed the protocol-defined maximum transmit power corresponding to the power class of the UE.

To adapt to this optimization in 3GPP Release 17, uplink CA power enhancement is introduced. **After this function is enabled, the maximum transmit power of UEs that support the higherPowerLimit-r17 field in the 2CC case of uplink inter-band CA can be increased. That is, this function allows for a high transmit power (27 dBm at most for the moment) of UEs and increases the uplink coverage and the uplink user-perceived rate of UEs**, as illustrated in Figure 4-9. This function is controlled by the **UL_CA_POWER_ENHANCE_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoSwitch*** parameter.

Figure 4-9 Uplink CA power enhancement



This function works only in the 2CC case of uplink intra-FR inter-band CA (detailed in *Carrier Aggregation*).

NOTE

Uplink CA power enhancement is a trial function.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

4.9.2 Network Analysis

4.9.2.1 Benefits

After this function is enabled, the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and average uplink UE throughput (**User Uplink Average Throughput (DU)**) may increase.

4.9.2.2 Impacts

The impacts between this function and other functions are described in **4.9.3.2.3 Function Impacts**.

This function has the following network impacts: After this function takes effect, the uplink transmit power of cell edge users increases, slightly increasing the average uplink interference (**N.UL.NI.Avg**).

4.9.3 Requirements

4.9.3.1 Licenses

None

4.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.9.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	Uplink CA power enhancement takes effect only when uplink intra-FR inter-band CA is enabled.

4.9.3.2.2 Mutually Exclusive Functions

None

4.9.3.2.3 Function Impacts

None

4.9.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.9.3.4 Others

This function requires that UEs have the **higherPowerLimit-r17** capability, that is, the value of this field is **supported**.

4.9.4 Operation and Maintenance

4.9.4.1 Data Configuration

4.9.4.1.1 Data Preparation

Table 4-12 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-12 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Uplink Power Control Algorithm Switch	**NRDUCellUlPcConfig**.*UlPwrCtrlAlgoSwitch*	Select the **UL_CA_POWER_ENHANCE_SW** option.

4.9.4.1.2 Using MML Commands

Before using MML commands, refer to 4.9.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink CA power enhancement
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=UL_CA_POWER_ENHANCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink CA power enhancement
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=UL_CA_POWER_ENHANCE_SW-0;
```

4.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.9.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

Step 3 If the value of the **higherPowerLimit-r17** field in the UE-reported UECapabilityInformation message is **support**, the UE supports uplink CA power enhancement.

Step 4 If the value of the **p-NR-FR1** field sent by the base station is **27**, uplink CA power enhancement has been enabled.

----End

4.9.4.3 Network Monitoring

After this function is enabled, monitor its running status by checking the increases in **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**.

5 Experience-based FWA

5.1 User PRB Control for FWA

5.1.1 Principles

User PRB control for FWA is introduced to control the maximum PRB usage of Internet users, non-Internet users, and private line users separately, thereby preventing a certain type of users from occupying excess PRB resources and impacting experience of other types of users. Different types of users are defined as follows and FWA user identification is detailed in **4.1 FWA User Identification**:

- Internet users refer to users for which the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_INTERNET**.
- Private line users refer to users for which the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_PRIVATE_LINE**.
- Non-Internet users refer to users other than Internet users and include private line users.

User PRB control for FWA is controlled by the **PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. If this option is selected, this function takes effect only when RB resources are configured for Internet users, non-Internet users, or private line users separately by using parameters described in **Table 5-1**.

Table 5-1 Parameters to be configured

Parameter Name	Parameter ID	Description
Resource Usage Object	**NRDUCellRes**.***ResUsageObject***	Set this parameter to **FWA**, indicating that resources are used by FWA.

Parameter Name	Parameter ID	Description
Uplink RB Average Ratio	**NRDUCellRes**.***UlRbAverageRatio***	This parameter specifies the average proportion of uplink RBs that can be used by Internet and non-Internet users.
Downlink RB Average Ratio	**NRDUCellRes**.***DlRbAverageRatio***	This parameter specifies the average proportion of downlink RBs that can be used by Internet and non-Internet users.
Uplink RB Max Ratio	**NRDUCellRes**.***UlRbMaxRatio***	This parameter specifies the maximum proportion of uplink RBs that can be used by Internet users, non-Internet users, and private line users.
Downlink RB Max Ratio	**NRDUCellRes**.***DlRbMaxRatio***	This parameter specifies the maximum proportion of downlink RBs that can be used by Internet users, non-Internet users, and private line users.
FWA Internet User Resource ID	**NRDUCellFwaQos**.***FwaInternetUserResId***	This parameter must be set to the same value as **NRDUCellRes**.***CellResId*** of the corresponding cell.
Non-FWA-Internet User Resource ID	**NRDUCellFwaQos**.***NonFwaInternetUserResId***	This parameter must be set to the same value as **NRDUCellRes**.***CellResId*** of the corresponding cell.
FWA Private Line User Resource ID	**NRDUCellFwaQos**.***FwaPrivateLineUserResId***	This parameter must be set to the same value as **NRDUCellRes**.***CellResId*** of the corresponding cell.
Priority Reduction Extent	**NRDUCellFwaQos**.***PriReduction***	This parameter specifies the decrease in the scheduling priority when the PRB usage exceeds the specified average uplink/downlink RB proportion.

After the preceding parameter configurations are completed and this function is enabled, the gNodeB adjusts the scheduling priority or stops scheduling based on the uplink and downlink PRB usages of users. **Table 5-2** and **Figure 5-1** provide the adjustment process.

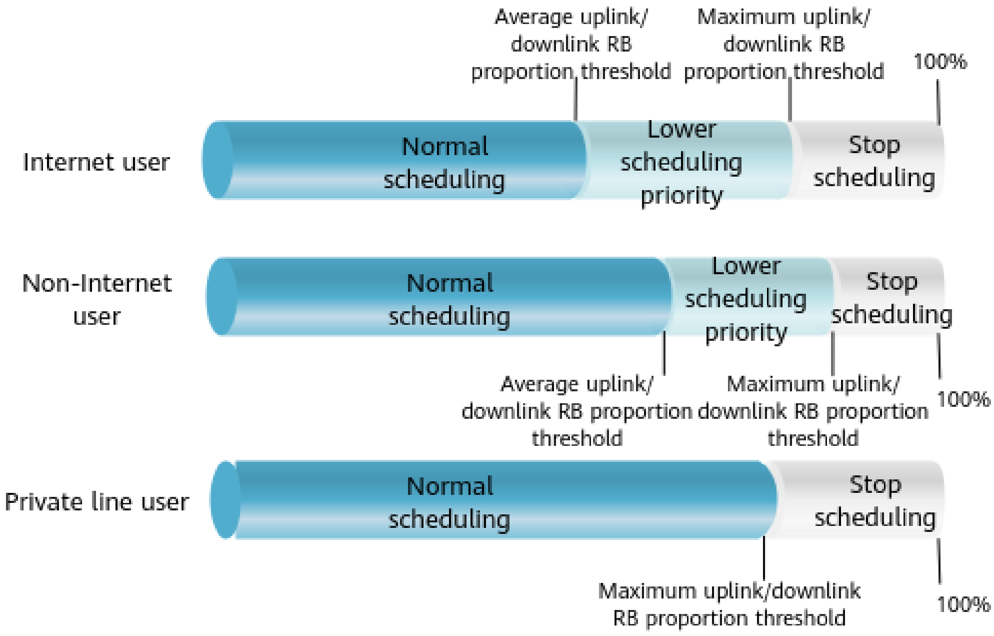
Table 5-2 Adjustment process for the user PRB control for FWA function

Criterion	Internet User or Non-Internet User[a]	FWA Private Line User
The uplink/downlink PRB usage of a user does not exceed the configured average uplink/downlink RB proportion.	Normal scheduling	N/A
The uplink/downlink PRB usage of a user exceeds the configured average uplink/downlink RB proportion.	The gNodeB lowers the uplink/downlink scheduling priority for these users.	N/A
The uplink/downlink PRB usage of a user exceeds the configured maximum uplink/downlink RB proportion.	The gNodeB stops uplink/downlink scheduling for these users.	The gNodeB stops uplink/downlink scheduling for these users.
a: Although non-Internet users include private line users, the uplink or downlink scheduling priority of private line users is not lowered when their registered rates are not fulfilled.		

NOTE

- The preceding restrictions apply only to initial transmissions of data on non-GBR bearers. When the proportion of traffic relevant to retransmissions, special-priority services (such as low latency services, VoNR services, and services carried on GBR bearers), RLC status reporting, TA information, scheduling requests, or signaling is high, the expected proportion of RB resources for each type of user cannot be guaranteed with this function enabled.
- The maximum uplink or downlink RB proportion configured for private line users must be less than that configured for non-Internet users.
- The sum of the maximum uplink or downlink RB proportions of Internet users, non-Internet users, and private line users can exceed 100%, so can the sum of their average uplink or downlink RB proportions. The specific parameter settings depend on the network plan.
- When FWA users participate in MU pairing, the proportion of RBs used by FWA users is the same as that of the paired layers they use. For example, if FWA users use 4 of 6 paired layers, they occupy 4 of 6 total RB resources.
- After a user is identified as an Internet user, non-Internet user, or private line user, if there is no change in the user bearer but a change in the service type configuration of the corresponding QCI on the base station side (**gNBQciBearer**.***ServiceType*** configuration), the user attribute does not change (the PRB usage is measured based on the original user attribute) unless the user accesses the network again.

Figure 5-1 Principle of user PRB control for FWA



In RAN sharing with common carrier scenarios, the uplink or downlink PRB proportions for Internet users, non-Internet users, and private line users can be controlled separately based on the RB proportions planned for each operator. The following parameter configurations are required:

- The **NRDUCellOpFwaQos**.***FwaInternetUserResId*** parameter is configured to specify the cell resource ID for Internet users of a specific operator.
- The **NRDUCellOpFwaQos**.***NonFwaInternetUserResId*** parameter is configured to specify the cell resource ID for non-Internet users of a specific operator.
- The **NRDUCellOpFwaQos**.***FwaPrivateLineUserResId*** parameter is configured to specify the cell resource ID for private line users of a specific operator.
- The preceding IDs must be the same as the corresponding **NRDUCellRes**.***CellResId*** setting to configure cell resources for Internet users, non-Internet users, and private line users.

For example, if the average downlink RB proportion configured for an operator is *A* and the average downlink RB proportion configured for Internet users of the operator is *B*, the effective control threshold is *A* multiplied by *B*.

NOTE

- The total number of **NRDUCellRes** resource groups used by RAN sharing with common carrier and this function cannot exceed six.
- In RAN sharing with common carrier, if multiple operators are configured with the same cell resource ID (**NRDUCellOp**.***CellResId***), they are considered as an operator group for RB resource management. For operators in the same operator group, RB control is available by using the **NRDUCELLFWAQOS** MO to allow these operators to share RB resources, while the uplink/downlink PRB proportions of Internet users, non-Internet users, and private line users cannot be controlled by using the **NRDUCELLOPFWAQOS** MO.

Dynamic User PRB Control for FWA

After user PRB control for FWA is enabled, dynamic user PRB control for FWA is introduced to fully utilize spectrum resources. This function is controlled by the **DYN_PRB_CTRL_SW** option of **NRDUCellAlgoSwitch**.***FwaAlgoSwitch***. After this function is enabled, the gNodeB dynamically adjusts the average and maximum uplink or downlink RB proportions for Internet users based on relevant thresholds. The related decision thresholds are defined as follows:

- Uplink RB proportion threshold = 100% – Value of **NRDUCellRes**.***UlRbMaxRatio*** for Internet users
- Downlink RB proportion threshold = 100% – Value of **NRDUCellRes**.***DlRbMaxRatio*** for Internet users
- Threshold expressed as a ratio of active non-Internet users to active Internet users = **NRDUCellFwaQos**.***ActiveUeNumRatioThld*** value

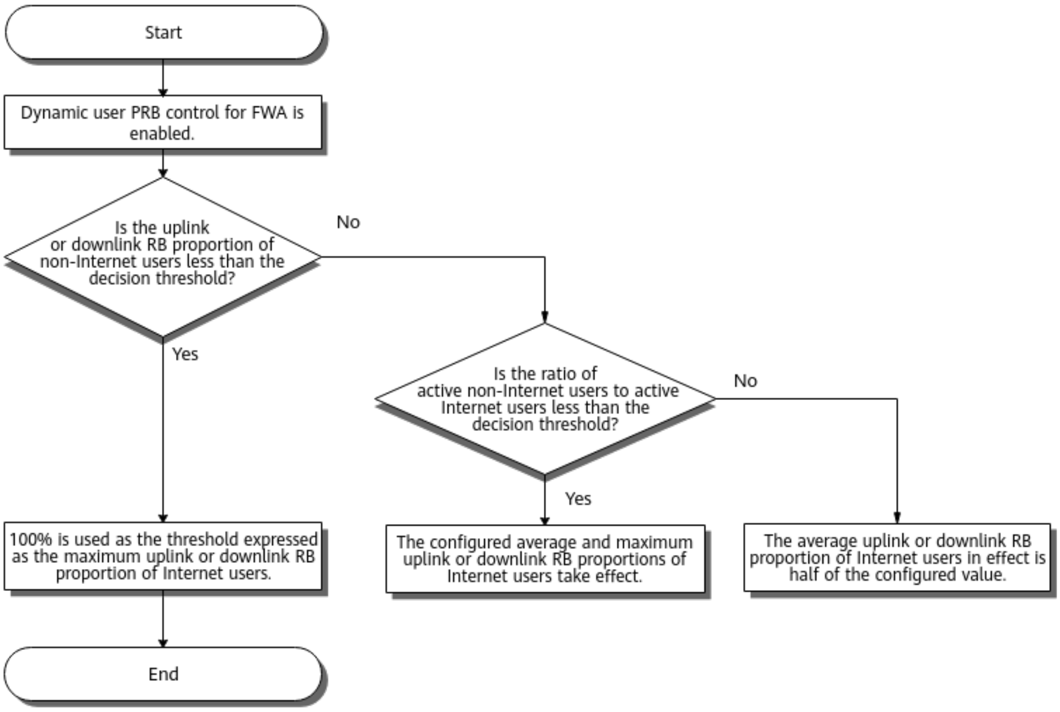
Figure 5-2 shows the dynamic adjustment process for TDD cells.

- If the proportion of uplink or downlink RBs consumed by non-Internet users is less than the corresponding decision threshold in a monitoring period,

  In the next period, the gNodeB adjusts the maximum uplink or downlink RB proportion to 100% for Internet users. That is, the gNodeB no longer restricts the uplink or downlink RBs for Internet users.
- If the proportion of uplink or downlink RBs consumed by non-Internet users is greater than or equal to the corresponding decision threshold in a monitoring period,
  - When the ratio of active non-Internet users to active Internet users is greater than or equal to the corresponding decision threshold in a monitoring period:

    In the next period, the gNodeB automatically decreases the average uplink or downlink RB proportion of Internet users to half of the value of **NRDUCellRes**.***UlRbAverageRatio*** or **NRDUCellRes**.***DlRbAverageRatio***. In this case, the uplink or downlink RBs used by Internet users are more strictly restricted.
  - Otherwise, in the next period, the gNodeB performs control based on the configured values of the average and maximum uplink or downlink RB proportions for Internet users.

Figure 5-2 Working principles of dynamic user PRB control for FWA in TDD cells



For SUL cells, the dynamic adjustment procedure is as follows:

- If the proportion of uplink or downlink RBs consumed by non-Internet users is less than the corresponding decision threshold in a monitoring period, the gNodeB adjusts the maximum uplink or downlink RB proportion to 100% for Internet users in the next monitoring period. That is, the gNodeB no longer restricts the uplink or downlink RBs for Internet users.
- If the proportion of uplink or downlink RBs consumed by non-Internet users is greater than or equal to the corresponding decision threshold in a monitoring period, the gNodeB restores the maximum uplink or downlink RB proportion for Internet users to the configured value in the next monitoring period.

5.1.2 Network Analysis

5.1.2.1 Benefits

User PRB control for FWA prevents specific types of users from occupying excess PRB resources, increasing the average uplink or downlink throughput of other types of users in the cell.

After dynamic user PRB control for FWA is enabled, when non-Internet users occupy a relatively small number of RBs, Internet users will fully utilize cell RB resources, increasing the average uplink or downlink throughput of Internet users and the average uplink or downlink cell throughput; when there is a relatively large number of non-Internet users, RB resource allocation for Internet users is stricter. This increases the average uplink or downlink throughput of non-Internet users.

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
- Average uplink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.UL.QCI – N.ThpVol.UL.SmallPkt.QCI**)/ **N.ThpTime.UL.RmvSmallPkt.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
- Average uplink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.UL.DuCell5QI – N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**

5.1.2.2 Impacts

The impacts between this function and other functions are described in **5.1.3.2.3 Function Impacts**.

This function has the following network impacts:

- After user PRB control for FWA is enabled, if the PRB usage of Internet users, non-Internet users, or private line users reaches the configured maximum uplink or downlink RB proportion, the scheduling of these users is restricted. This reduces the cell PRB usage and the average uplink or downlink cell throughput. Therefore, if there are requirements on peak rates, you are not advised to set the **NRDUCellRes**.***UlRbMaxRatio*** or **NRDUCellRes**.***DlRbMaxRatio*** parameter.
- After dynamic user PRB control for FWA is enabled, if there is a large number of non-Internet users, RB resource allocation for Internet users is stricter. This decreases the average uplink or downlink throughput of Internet users.

NOTE

- Uplink PRB usage in a cell: **Uplink Resource Block Utilizing Rate (DU)**
- Downlink PRB usage in a cell: **Downlink Resource Block Utilizing Rate (DU)**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**

5.1.3 Requirements

5.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
TDD	FWA user identification	**gNBQciBearer**.***ServiceType***	*FWA*	• For the function of user PRB control for FWA to take effect for Internet users, the Internet service type must be configured for FWA user identification. • For the function of user PRB control for FWA to take effect for private line users, the private line service type must be configured for FWA user identification.
TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Dynamic user PRB control for FWA depends on user PRB control for FWA.
TDD	Maximum downlink RB proportion	**NRDUCellRes**.***DlRbMaxRatio***	*FWA*	The maximum downlink RB proportion configured for private line users must be less than that configured for non-Internet users.

RAT	Function Name	Function Switch	Reference	Description
TDD	Maximum uplink RB proportion	**NRDUCellRes**.***UlRbMaxRatio***	*FWA*	The maximum uplink RB proportion configured for private line users must be less than that configured for non-Internet users.

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	User PRB control for FWA cannot work with Hyper Cell.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	User PRB control for FWA cannot work with Cell Combination.
Low-frequency TDD	DACQ	**DACQ_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	None	User PRB control for FWA cannot work with DACQ.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	User PRB control for FWA cannot work with NR DU cell resource management between network slices.
TDD	Operator-specific RB management	**RB_DYNAMIC_SHARING_SW**, **MAX_RB_LIMIT_SW**, and **MIN_RB_GUARANTEE_SW** options of the **NRDUCellAlgoSwitch.*RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	Dynamic user PRB control for FWA can only be enabled when the three options for the operator-specific RB management function are all deselected.
Low-frequency TDD	Resource reservation	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***HighReliability-Switch*** parameter	*URLLC*	User PRB control for FWA cannot be enabled together with resource reservation.
Low-frequency TDD	RFSP-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	User PRB control for FWA and RFSP-based RB resource control cannot be enabled at the same time.
High-frequency TDD	MU-MIMO	**UL_MU_MIMO_SW** or **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	None	User PRB control for FWA cannot be enabled together with MU-MIMO in high-frequency TDD scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink RB restriction for the low-speed cell	**NRDUCellHighSpeedMob**.***DlIntrfLimitRbSchRatio*** set to a non-zero value	*High Speed Mobility*	Downlink RB restriction for the low-speed cell cannot work with user PRB control for FWA.
Low-frequency TDD	Uplink RB restriction for the low-speed cell	**NRDUCellHighSpeedMob**.***UlIntrfLimitRbSchRatio*** set to a non-zero value	*High Speed Mobility*	Uplink RB restriction for the low-speed cell cannot work with user PRB control for FWA.
Low-frequency TDD	Delay-based scheduling for short video services	**SHORT_VIDEO_DELAY_BASED_SCH_SW** option of the **NRDUCellDlSch**.***ShortVideoDlSchSw*** parameter	None	Delay-based scheduling for short video services cannot work with user PRB control for FWA.
Low-frequency TDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffSer-vExpSw*** parameter	*QCI-based Resource Control*	QCI-based resource control cannot work with user PRB control for FWA.

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU-MIMO	**UL_MU_MIMO_SW** or **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	When MU-MIMO is in effect and only one type of users (Internet users or non-Internet users) are paired at a layer, the two user types do not impact RBs of each other. Therefore, if the PRB usage exceeds the average RB threshold, lowering the scheduling priority is not required.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Even uplink RB division optimization	**RB_DIVISION_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter	*Uplink Scheduling*	When there is a small number of users, even uplink RB division optimization affects RB resource allocation, and user PRB control for FWA does not take effect if the number of allocated RBs does not meet the relevant threshold.
TDD	Uplink/Downlink scheduling priority weighting factor	**gNBDUMacParamGroup.*UlSchPriWeightFactor*** **gNBDUMacParamGroup.*DlSchPriWeightFactor***	*QoS Management*	When the uplink or downlink scheduling priority weighting factor is set to a rather small value, if the PRB usage exceeds the average RB proportion threshold after user PRB control for FWA is enabled, lowering the scheduling priority may compromise fairness among users.
TDD	Operator-specific RB management	**RB_DYNAMIC_SHARING_SW**, **MAX_RB_LIMIT_SW**, and **MIN_RB_GUARANTEE_SW** options of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	For the member operators (configured with the same value of **NRDUCellOp**.***CellResId***) in the same operator group, the **NRDUCELLOPFWAQOS** MO cannot be used to specify the uplink or downlink RB proportions for different types of users.
Low-frequency TDD	Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If user PRB control for FWA has taken effect for FWA users, enabling downlink resource isolation may further deteriorate FWA user experience. If user PRB control for FWA has taken effect for eMBB users, enabling downlink resource isolation may deteriorate eMBB user experience.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Device-pipe synergy preallocation	**UL_PREFERENTIAL_SCH_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*FWA*	When both user PRB control for FWA and device-pipe synergy preallocation are enabled, the delay of uplink delay-sensitive services may increase for device-pipe synergy UEs.
Low-frequency TDD	FWA RedCap downlink type 1 scheduling	**FWA_TYPE1_4RB_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	When user PRB control for FWA and FWA RedCap downlink type 1 scheduling are both enabled and the **NRDUCellRes**.***DlRbMaxRatio*** parameter is set to a value other than **255**, downlink PRB resources are more likely to be insufficient because more downlink RBs are allocated at a time to RedCap UEs. As a result, scheduling of RedCap UEs is more likely to be suspended, and the average downlink throughput of RedCap UEs changes.
Low-frequency TDD	Data rate limitation in flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "user PRB control for FWA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of FWA UEs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced L4S congestion control based on rate adjustment	**UL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter and **DL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter	*Adaptive Managed Latency*	If both "user PRB control for FWA" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of FWA UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.

5.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

Table 5-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	To enable user PRB control for FWA, select the **PRB_CTRL_SW** option.
FWA Internet User Resource ID	**NRDUCellFwaQos**.***FwaInternetUserResId***	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, the Internet service type must be configured for FWA user identification. Otherwise, user PRB control for FWA does not take effect for Internet users.
Non-FWA-Internet User Resource ID	**NRDUCellFwaQos**.***NonFwaInternetUserResId***	Set this parameter based on the network plan.
FWA Private Line User Resource ID	**NRDUCellFwaQos**.***FwaPrivateLineUserResId***	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, the private line service type must be configured for FWA user identification. Otherwise, user PRB control for FWA does not take effect for private line users.
Downlink RB Average Ratio	**NRDUCellRes**.***DlRbAverageRatio***	Set this parameter based on the network plan.
Downlink RB Max Ratio	**NRDUCellRes**.***DlRbMaxRatio***	The closer the value of this parameter for private line users to that for non-Internet users, the fewer the downlink resources available for eMBB users.
Uplink RB Average Ratio	**NRDUCellRes**.***UlRbAverageRatio***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Uplink RB Max Ratio	**NRDUCellRes**.***UlRbMaxRatio***	The closer the value of this parameter for private line users to that for non-Internet users, the fewer the uplink resources available for eMBB users.
Resource Usage Object	**NRDUCellRes**.***ResUsageObject***	Set this parameter to **FWA**.
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	To enable dynamic user PRB control for FWA, select the **DYN_PRB_CTRL_SW** option.
Active UE Number Ratio Threshold	**NRDUCellFwaQos**.***ActiveUeNumRatioThld***	Set this parameter based on the network plan.
Priority Reduction Extent	**NRDUCellFwaQos**.***PriReduction***	Set this parameter to **1024**.
FWA Internet User Resource ID	**NRDUCellOpFwaQos**.***FwaInternetUserResId***	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, the Internet service type must be configured for FWA user identification. Otherwise, user PRB control for FWA does not take effect for Internet users.
Non-FWA-Internet User Resource ID	**NRDUCellOpFwaQos**.***NonFwaInternetUserResId***	Set this parameter based on the network plan.
FWA Private Line User Resource ID	**NRDUCellOpFwaQos**.***FwaPrivateLineUserResId***	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, the private line service type must be configured for FWA user identification. Otherwise, user PRB control for FWA does not take effect for private line users.
Cell Resource ID	**NRDUCellRes**.***CellResId***	Set this parameter based on the network plan.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to **5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- User PRB control for FWA
  - Common scenario (PRB control only for FWA Internet users when both FWA Internet users and eMBB users are present)
```
//Enabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-1;
//Adding resource group configurations
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=FWA, DlRbAverageRatio=50,
DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
//Associating FWA Internet users with the resource group
MOD NRDUCELLFWAQOS: NrDuCellId=1, FwaInternetUserResId=1,
NonFwaInternetUserResId=255, FwaPrivateLineUserResId=255, PriReduction=1024;
```
  - Common scenario (PRB control for both FWA Internet users and eMBB users when both types of users are present)
```
//Enabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-1;
//Adding resource group configurations
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=FWA, DlRbAverageRatio=50,
DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=2, ResUsageObject=FWA, DlRbAverageRatio=50,
DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
//Associating Internet users and non-Internet users with the resource groups separately
MOD NRDUCELLFWAQOS: NrDuCellId=1, FwaInternetUserResId=1,
NonFwaInternetUserResId=2, FwaPrivateLineUserResId=255, PriReduction=1024;
```
  - Common scenario (PRB control for FWA Internet users, FWA private line users, and eMBB users when all these types of users are present)
```
//Enabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-1;
//Adding resource group configurations
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=FWA, DlRbAverageRatio=50,
DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=2, ResUsageObject=FWA, DlRbAverageRatio=50,
DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=3, ResUsageObject=FWA, DlRbMaxRatio=60,
UlRbMaxRatio=60;
//Associating Internet users, non-Internet users, and private line users with the resource groups
separately
MOD NRDUCELLFWAQOS: NrDuCellId=1, FwaInternetUserResId=1,
NonFwaInternetUserResId=2, FwaPrivateLineUserResId=3, PriReduction=1024;
```
  - RAN sharing with common carrier scenario (assuming that FWA Internet users, FWA private line users, and eMBB users are all present)

    Before activating this function, refer to section "Using MML Commands" for RAN sharing with common carrier in *Multi-Operator Sharing*.


```
//Enabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-1;
//Adding resource group configurations
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=FWA, DlRbAverageRatio=50, DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=2, ResUsageObject=FWA, DlRbAverageRatio=50, DlRbMaxRatio=80, UlRbAverageRatio=50, UlRbMaxRatio=80;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=3, ResUsageObject=FWA, DlRbMaxRatio=60, UlRbMaxRatio=60;
//Associating Internet users, non-Internet users, and private line users with the resource groups separately for the operator whose OperatorId is 1
MOD NRDUCELLOPFWAQOS: NrDuCellId=1, OperatorId=1, FwaInternetUserResId=1, NonFwaInternetUserResId=2, FwaPrivateLineUserResId=3;
```

- Dynamic user PRB control for FWA

```
//Enabling dynamic user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DYN_PRB_CTRL_SW-1;
//Configuring the threshold expressed as a ratio of active users
MOD NRDUCELLFWAQOS: NrDuCellId=1, ActiveUeNumRatioThld=40;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Dynamic user PRB control for FWA

```
//Disabling dynamic user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DYN_PRB_CTRL_SW-0;
```

- User PRB control for FWA
  - Common scenario

```
//Disabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-0;
```

  - RAN sharing with common carrier scenario

```
//Disabling user PRB control for FWA
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRB_CTRL_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Observe the indicators described in **5.1.4.3 Network Monitoring** to check whether this function has taken effect.

5.1.4.3 Network Monitoring

Observe the following indicators:

- Uplink PRB usage of Internet users = **N.ChMeas.PRB.UL.FWA.B2H.Used**/ **N.PRB.UL.Avail.Avg** x 100%
- Downlink PRB usage of Internet users = **N.ChMeas.PRB.DL.FWA.B2H.Used**/ **N.PRB.DL.Avail.Avg** x 100%

- Uplink PRB usage of non-Internet users = (**N.PRB.UL.Used.Avg – N.ChMeas.PRB.UL.FWA.B2H.Used**)/**N.PRB.UL.Avail.Avg** x 100%
- Downlink PRB usage of non-Internet users = (**N.PRB.DL.Used.Avg – N.ChMeas.PRB.DL.FWA.B2H.Used**)/**N.PRB.DL.Avail.Avg** x 100%
- Uplink PRB usage of private line users = **N.ChMeas.PRB.UL.FWA.PrivateLine.Used**/**N.PRB.UL.Avail.Avg** x 100%
- Downlink PRB usage of private line users = **N.ChMeas.PRB.DL.FWA.PrivateLine.Used**/**N.PRB.DL.Avail.Avg** x 100%
- Proportion of uplink RBs used by Internet users of a specific operator = **N.PRBUsed.UL.FWA.B2H.PLMN**/(**N.PRBUsedOwn.UL.PLMN** + **N.PRBUsedOther.UL.PLMN**) x 100%
- Proportion of downlink RBs used by Internet users of a specific operator = **N.PRBUsed.DL.FWA.B2H.PLMN**/(**N.PRBUsedOwn.DL.PLMN** + **N.PRBUsedOther.DL.PLMN**) x 100%
- Proportion of uplink RBs used by non-Internet users of a specific operator = 100% **– N.PRBUsed.UL.FWA.B2H.PLMN**/(**N.PRBUsedOwn.UL.PLMN** + **N.PRBUsedOther.UL.PLMN**) x 100%
- Proportion of downlink RBs used by non-Internet users of a specific operator = 100% **– N.PRBUsed.DL.FWA.B2H.PLMN**/(**N.PRBUsedOwn.DL.PLMN** + **N.PRBUsedOther.DL.PLMN**) x 100%
- Proportion of uplink RBs used by private line users of a specific operator = **N.PRBUsed.UL.FWA.PrivateLine.PLMN**/(**N.PRBUsedOwn.UL.PLMN** + **N.PRBUsedOther.UL.PLMN**) x 100%
- Proportion of downlink RBs used by private line users of a specific operator = **N.PRBUsed.DL.FWA.PrivateLine.PLMN**/(**N.PRBUsedOwn.DL.PLMN** + **N.PRBUsedOther.DL.PLMN**) x 100%

NOTE

The uplink and downlink PRB usages used during function implementation (as described in **5.1.1 Principles**) are calculated by dividing the number of used RBs by the number of available RBs. The RBs used for common messages and random access signaling are excluded from calculation of these used RBs and available RBs, but are included in measurement of the counters **N.PRB.UL.Used.Avg**, **N.PRB.DL.Used.Avg**, **N.PRB.UL.Avail.Avg**, and **N.PRB.DL.Avail.Avg** in the preceding formulas. In addition, scheduling restrictions are not applied to some services. As such, the PRB usages observed in network monitoring may slightly differ from the expected RB proportions.

5.2 Uplink Experience-based Scheduling

5.2.1 Principles

Operators can plan QCIs, 5QIs, or RFSPs for different types of users on the core network. The gNodeB configures uplink target guaranteed rates for different QCIs, 5QIs, or RFSPs.

- By QCI/5QI

  The **gNBDUMacParamGroup**.***UlExpBasedSchGuarUserRate*** parameter specifies the uplink target guaranteed rate for a QCI/5QI-specific bearer.
- By RFSP

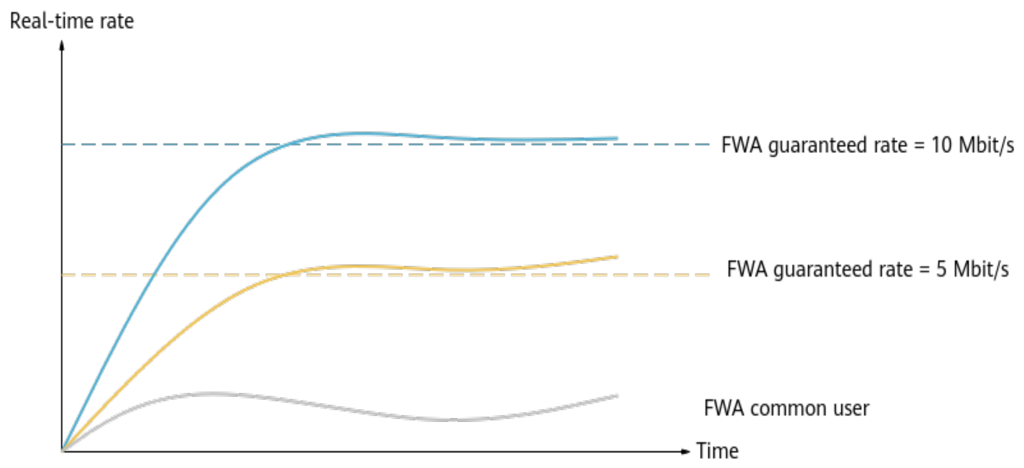
The **gNBDURfspConfig**.***UlExpBasedSchGuarUserRate*** parameter specifies the uplink target guaranteed rate for an RFSP.

This function is controlled by the **UL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. After this function is enabled, the rate of a non-GBR bearer configured with the uplink target guaranteed rate is measured in real time:

- If the rate does not reach the configured target value, the uplink scheduling priority of the bearer is gradually raised by a maximum multiple not higher than the **NRDUCellFwaQos**.***ExpBasedSchMaxWeight*** parameter value.
- If the rate reaches or exceeds the configured target value, the uplink scheduling priority of the bearer is gradually lowered until the initial scheduling priority resumes.

Figure 5-3 shows the rate trend of FWA users configured with different target rates.

Figure 5-3 Uplink experience-based scheduling



With uplink experience-based scheduling enabled, if the uplink target guaranteed rate is changed for a user, the new setting takes effect only when the user accesses the network again.

CA Scenarios

When both uplink experience-based scheduling and CA (detailed in *Carrier Aggregation*) are enabled, the PCC compares the configured target guaranteed rate with the real-time RLC rate, calculates the multiple by which the uplink scheduling priority increases, and sends the multiple to the SCCs. The value of the **NRDUCellFwaQos**.***ExpBasedSchMaxWeight*** parameter on the PCC must be the same as that on the SCCs.

NOTE

- If both RFSP-specific uplink target guaranteed rate and 5QI/QCI-specific uplink target guaranteed rate are set to non-zero values for a user, the gNodeB performs rate guarantee based on the RFSP-specific uplink target guaranteed rate.
- If a user has multiple non-GBR bearers, the uplink scheduling priorities of these bearers are calculated based on the configured uplink target guaranteed rates separately, and the highest uplink scheduling priority among the calculated results takes effect for uplink scheduling of the user.

5.2.2 Network Analysis

5.2.2.1 Benefits

After this function is enabled:

- The average uplink throughput of specific users increases.
- When the network load is light, the benefits for specific users are not significant because scheduling resources are not limited.
- There are also no significant benefits for cell edge users due to limited power.

NOTE

- Observe the average uplink throughput of users based on QCI/5QI-specific indicators:
  - Average uplink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.UL.QCI** – **N.ThpVol.UL.SmallPkt.QCI**)/ **N.ThpTime.UL.RmvSmallPkt.QCI**
  - Average uplink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.UL.DuCell5QI** – **N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- This function adjusts only the scheduling priority. When RB resources are insufficient, the benefits provided by this function decrease.

5.2.2.2 Impacts

The impacts between this function and other functions are described in **5.2.3.2.3 Function Impacts**.

This function has the following network impacts:

- This function preferentially guarantees the user-perceived rate of specific users. Resources allocated to other users without target guaranteed rate configuration decrease, and their uplink user-perceived rate decreases.
- If specific users configured with target guaranteed rates are located at the cell edge, excess resources are consumed to guarantee their services, leading to a decrease in the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and average uplink user throughput (**User Uplink Average Throughput (DU)**).

5.2.3 Requirements

5.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

In TDD in high frequency bands, the number of licenses for a sector must be the same as the number of carriers. Otherwise, this function cannot take effect.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	In TDD in high frequency bands, uplink experience-based scheduling cannot work with inter-FR CA.
Low-frequency TDD	None	None	None	None

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	Uplink guaranteed rate	**gNBDUMacParamGroup**.***UlGuaranteedRate***	*QoS Management*	When the uplink guaranteed rate and uplink experience-based scheduling take effect for different users, resources are preferentially allocated to the user configured with the uplink guaranteed rate. When the uplink guaranteed rate and uplink experience-based scheduling simultaneously take effect for a user, the user rate is increased mainly based on the uplink guaranteed rate.
Low-frequency TDD	Even uplink RB division optimization	**RB_DIVISION_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter	*Uplink Scheduling*	When both even uplink RB division optimization and uplink experience-based scheduling are enabled, the benefits delivered by uplink experience-based scheduling decrease when more users are evenly allocated RBs. This is because even uplink RB division optimization affects RB resource allocation.
Low-frequency TDD	Dynamic adjustment of live streaming resources	**UL_RB_AVG_RATIO_DYN_ADJ_SW** option of the **NRDUCellRes**.***RbRatioDynAdjSwitch*** parameter	*Ultimate Video Experience*	When the uplink guaranteed rate and uplink experience-based scheduling take effect for different users, resources are preferentially allocated to the user configured with the uplink guaranteed rate. When the uplink guaranteed rate and uplink experience-based scheduling simultaneously take effect for a user, the user rate is increased mainly based on the uplink guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Data rate limitation in flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "uplink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the value of the **gNBDUMacParamGroup**.***UlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***UlExpBasedSchGuarUserRate*** parameter is greater than the aforementioned calculated target uplink data rate.

5.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	UL_EXP_BASED_SCH_SW	Select the **UL_EXP_BASED_SCH_SW** option.
UL Exp-based Scheduling Guaranteed User Rate	**gNBDUMacParamGroup**.***UlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.
UL Exp-based Scheduling Guaranteed User Rate	**gNBDURfspConfig**.***UlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.
Exp-based Scheduling Maximum Weight	**NRDUCellFwaQos**.***ExpBasedSchMaxWeight***	None	Set this parameter based on the network plan.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=UL_EXP_BASED_SCH_SW-1;
//Configuring the maximum experience-based scheduling weight
MOD NRDUCELLFWAQOS: NrDuCellId=1, ExpBasedSchMaxWeight=50;
```

Configuring the uplink target guaranteed rate using either of the following methods:

- Method 1: Configuring the uplink target guaranteed rate for a user by QCI/5QI
```
MOD GNBDUMACPARAMGROUP: MacParamGroupId=7, UlExpBasedSchGuarUserRate=50000;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=7, MacParamGroupId=7;
```
- Method 2: Configuring the uplink target guaranteed rate for a user by RFSP
```
ADD GNBDURFSPCONFIG: OperatorId=0, RfspIndex=3, UlExpBasedSchGuarUserRate=50000;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=UL_EXP_BASED_SCH_SW-0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Observe the indicators described in **5.2.4.3 Network Monitoring** to check whether this function has taken effect.

5.2.4.3 Network Monitoring

- When users are differentiated by QCIs/5QIs, the following QCI/5QI-specific indicators can be used to observe the average uplink throughput of users:
  - Average uplink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.UL.QCI** – **N.ThpVol.UL.SmallPkt.QCI**)/ **N.ThpTime.UL.RmvSmallPkt.QCI**
  - Average uplink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.UL.DuCell5QI** – **N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- When users are differentiated by RFSPs, the corresponding RFSPs can be associated with flexible performance counter objects and the uplink user-perceived rates for these objects can be used to observe the average uplink throughput of users. The uplink user-perceived rate for a flexible performance counter object equals (**N.ThpVolForRate.UL.Flex** – **N.ThpVolForRate.UL.SmallPkt.Flex**)/ **N.ThpTimeForRate.UL.RmvSmallPkt.Flex**.

  To associate an RFSP with a flexible performance counter object, configure a **gNBFlexCounterObjCfg** MO with **gNBFlexCounterObjCfg**.***FlexCounterObjType*** set to **NRDUCELL_FLEX** and **gNBFlexCounterObjCfg**.***RfspIndex*** set to this RSRP. For details, see the section "Custom Object Type" in "Object Type" of *Performance Management*.

5.3 Downlink Experience-based Scheduling

5.3.1 Principles

Operators can plan QCIs, 5QIs, or RFSPs for different types of users on the core network. The gNodeB configures downlink target guaranteed rates for different QCIs, 5QIs, or RFSPs.

- By QCI/5QI

  The **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** parameter specifies the downlink target guaranteed rate for a QCI/5QI-specific bearer.
- By RFSP

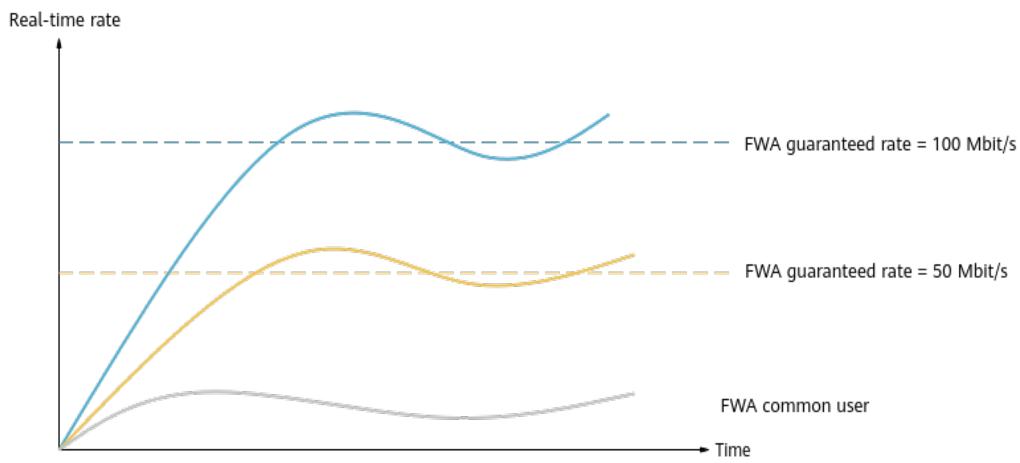
  The **gNBDURfspConfig**.***DlExpBasedSchGuarUserRate*** parameter specifies the downlink target guaranteed rate for an RFSP.

This function is controlled by the **DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. After this function is enabled, the rate of a non-GBR bearer configured with the downlink target guaranteed rate is measured in real time:

- If the rate does not reach the configured target value, the downlink scheduling priority of the bearer is gradually raised by a maximum multiple not higher than the **NRDUCellFwaQos**.***ExpBasedSchMaxWeight*** parameter value.
- If the rate reaches or exceeds the configured target value, the downlink scheduling priority of the bearer is gradually lowered until the initial scheduling priority resumes.

Figure 5-4 shows the rate trend of FWA users configured with different target rates.

Figure 5-4 Downlink experience-based scheduling



With downlink experience-based scheduling enabled, if the downlink target guaranteed rate is changed for a user, the new setting takes effect only when the user accesses the network again.

CA Scenarios

It is recommended that downlink RLC experience-based scheduling be enabled when downlink experience-based scheduling and CA (detailed in *Carrier Aggregation*) are both enabled. The procedure for downlink RLC experience-based scheduling to take effect is as follows:

- When the **DL_RLC_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter is selected, the PCC compares the configured target guaranteed rate with the real-time RLC rate, calculates the multiple by which the scheduling priority increases, and sends the multiple to the SCCs to adjust the downlink scheduling priority.

- When the **DL_RLC_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter is deselected, the PCC and SCCs separately compare the configured target guaranteed rate with the real-time MAC rate, calculate the multiple by which the scheduling priority increases, and adjust the downlink scheduling priority.

In CA scenarios, parameter settings must be consistent between the PCC and SCCs. Any change in parameter settings can take effect only after users access the network again.

Downlink RLC experience-based scheduling applies to low-frequency TDD.

NOTE

- If both RFSP-specific downlink target guaranteed rate and 5QI/QCI-specific downlink target guaranteed rate are set to non-zero values for a user, the gNodeB performs rate guarantee based on the RFSP-specific downlink target guaranteed rate.
- If a user has multiple non-GBR bearers, the downlink scheduling priorities of these bearers are calculated separately based on the configured downlink target guaranteed rates. Therefore, different priorities take effect for different non-GBR bearers of the same user.

5.3.2 Network Analysis

5.3.2.1 Benefits

After this function is enabled, the average downlink throughput of users configured with target guaranteed rates increases.

NOTE

- The average downlink user throughput can be calculated using the following QCI/5QI-specific indicators:
  - Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
  - Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
- When the network load is light, this function does not deliver significant benefits for users configured with target guaranteed rates because scheduling resources are not limited.

5.3.2.2 Impacts

The impacts between this function and other functions are described in **5.3.3.2.3 Function Impacts**.

This function has the following network impacts:

- This function preferentially guarantees users configured with target guaranteed rates. As such, fewer resources can be allocated to users not configured with target guaranteed rates, and the average downlink throughput of these users decreases.

- If users configured with target guaranteed rates are located at the cell edge, excess resources are consumed to guarantee their services, leading to a decrease in the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink user throughput (**User Downlink Average Throughput (DU)**).

5.3.3 Requirements

5.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

NOTE

In TDD in high frequency bands, the number of licenses for a sector must be the same as the number of carriers. Otherwise, this function cannot take effect.

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Downlink RLC experience-based scheduling requires downlink experience-based scheduling.

5.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Downlink experience-based scheduling cannot work with Hyper Cell.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Downlink experience-based scheduling cannot work with Cell Combination.
High-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	In TDD in high frequency bands, downlink experience-based scheduling cannot work with inter-FR CA.
Low-frequency TDD	Delay-based scheduling for short video services	**SHORT_VIDEO_DELAY_BASED_SCH_SW** option of the **NRDUCellDlSch**.***ShortVideoDlSchSw*** parameter	None	Delay-based scheduling for short video services cannot work with downlink experience-based scheduling.

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Load-based adaptive downlink scheduling	**NRDUCellDlSch**.***DlSchPolicy*** set to **ADAPTIVE**	*Massive MIMO AHR (Low-Frequency TDD)*	When downlink experience-based scheduling is in effect, the number of available downlink RBs in a cell decreases. This makes it more likely that the downlink PRB usage falls below the threshold (that determines whether a cell is heavily loaded) for load-based adaptive downlink scheduling to take effect. Consequently, there are fewer opportunities for load-based adaptive downlink scheduling to produce its intended benefits.
Low-frequency TDD	Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If downlink experience-based scheduling has taken effect for FWA users, enabling downlink resource isolation may degrade FWA user experience.
Low-frequency TDD	Rate adaptation for device-pipe synergy	**RATE_ADAPTION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*FWA*	When rate adaptation for device-pipe synergy and downlink experience-based scheduling are both enabled, downlink experience-based scheduling preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Data rate limitation in flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "downlink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***DlExpBasedSchGuarUserRate*** parameter is greater than the aforementioned calculated target downlink data rate.
High-frequency TDD	None	None	None	None

5.3.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	DL_EXP_BASED_SCH_SW	Select the **DL_EXP_BASED_SCH_SW** option.
DL Exp-based Scheduling Guaranteed User Rate	**gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.
DL Exp-based Scheduling Guaranteed User Rate	**gNBDURfspConfig**.***DlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.
Exp-based Scheduling Maximum Weight	**NRDUCellFwaQos**.***ExpBasedSchMaxWeight***	None	Set this parameter based on the network plan.
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	DL_RLC_EXP_BASED_SCH_SW	Select the **DL_RLC_EXP_BASED_SCH_SW** option if downlink experience-based scheduling and CA are both enabled.

5.3.4.1.2 Using MML Commands

NOTICE

Either **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** or **gNBDUMacParamGroup**.***DlGuaranteedRate*** must be set to **0**.

Before using MML commands, refer to 5.3.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_EXP_BASED_SCH_SW-1;
//(Optional) Enabling downlink RLC experience-based scheduling if downlink experience-based scheduling and CA are both enabled
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_RLC_EXP_BASED_SCH_SW-1;
//Configuring the maximum experience-based scheduling weight
MOD NRDUCELLFWAQOS: NrDuCellId=1, ExpBasedSchMaxWeight=50;
```

Configuring the downlink target guaranteed rate using either of the following methods:

- Method 1: Configuring the downlink target guaranteed rate for a user by QCI/5QI
  ```
  MOD GNBDUMACPARAMGROUP: MacParamGroupId=7, DlExpBasedSchGuarUserRate=50000;
  MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=7, MacParamGroupId=7;
  ```
- Method 2: Configuring the downlink target guaranteed rate for a user by RFSP
  ```
  ADD GNBDURFSPCONFIG: OperatorId=0, RfspIndex=3, DlExpBasedSchGuarUserRate=50000;
  ```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling downlink RLC experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_RLC_EXP_BASED_SCH_SW-0;
//Disabling downlink experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_EXP_BASED_SCH_SW-0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Observe the indicators described in **5.3.4.3 Network Monitoring** to check whether this function has taken effect.

5.3.4.3 Network Monitoring

- When users are differentiated by QCIs/5QIs:
  - Observe function activation duration based on QCI/5QI-specific indicators:

  - Proportion of QCI-specific function activation duration in NSA networking = **N.ThpTime.DL.ExpBasedSch.Transfer.RmvLastSlot.QCI**/ **N.ThpTime.DL.Transfer.RmvLastSlot.QCI**
  - Proportion of 5QI-specific function activation duration in SA networking = **N.ThpTime.DL.ExpBasedSch.Transfer.RmvLastSlot.DuCell5QI**/ **N.ThpTime.DL.Transfer.RmvLastSlot.DuCell5QI**
- Observe the average downlink throughput of users based on QCI/5QI-specific indicators:
  - Average downlink throughput of specific QCIs in NSA networking = (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
  - Average downlink throughput of specific 5QIs in SA networking = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

- When users are differentiated by RFSPs, the corresponding RFSPs can be associated with flexible performance counter objects and the downlink user-perceived rates for these objects can be used to observe the average downlink throughput of users. The downlink user-perceived rate for a flexible performance counter object equals (**N.ThpVol.DL.Flex – N.ThpVol.DL.LastSlot.Flex**)/ **N.ThpTime.DL.RmvLastSlot.Flex**.

  To associate an RFSP with a flexible performance counter object, configure a **gNBFlexCounterObjCfg** MO with **gNBFlexCounterObjCfg**.***FlexCounterObjType*** set to **NRDUCELL_FLEX** and **gNBFlexCounterObjCfg**.***RfspIndex*** set to this RSRP. For details, see the section "Custom Object Type" in "Object Type" of *Performance Management*.

NOTE

The proportion of function activation duration is related to multiple factors, such as user distribution, network load, and service model. A lower RSRP, a higher PRB usage in a cell, and a smaller proportion of tail packets result in a higher proportion of function activation duration.

5.4 Downlink MU-MIMO QoS Guarantee (Low-Frequency TDD)

5.4.1 Principles

If home broadband (HBB) users cannot be provided with guaranteed rates by simply using the downlink experience-based scheduling function, **the downlink MU-MIMO QoS guarantee function can be enabled to optimize the correlation threshold for downlink MU-MIMO pairing, further increasing the downlink rates of HBB users**.

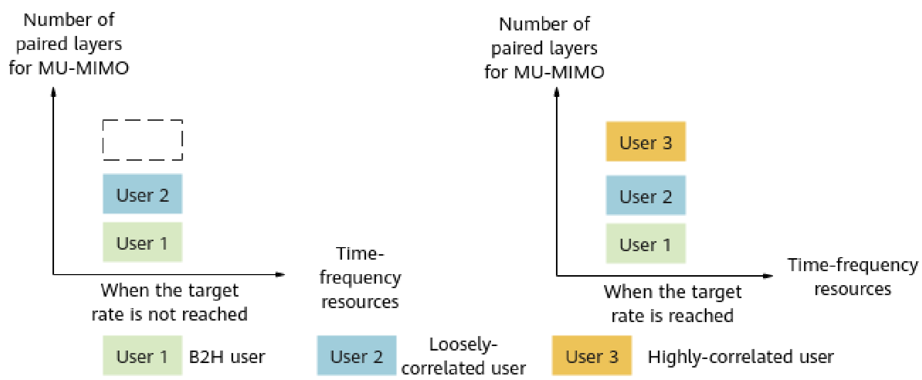
In a massive MIMO cell, two users can be paired for downlink MU-MIMO when their correlation is below the correlation threshold specified by the

NRDUCellPdsch.***DlSrsMuMimoChanIsoThld*** parameter. As illustrated in **Figure 5-5**, after this function is enabled:

- The **NRDUCellPdsch**.***DlSrsMuMimoChanIsoThld*** parameter value is used as the correlation threshold for pairing common users with HBB users whose rates are lower than the guaranteed rates. This correlation threshold is lower than that before the function is enabled. As such, HBB users can be paired with users with looser correlation, reducing interference between paired users and improving downlink experience of HBB users.
- After the guaranteed rates of HBB users are fulfilled, downlink MU-MIMO QoS guarantee no longer takes effect, and the correlation threshold for HBB users is restored to the value used before this function is enabled.

This function is controlled by the **DL_MU_MIMO_QOS_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

Figure 5-5 Downlink MU-MIMO QoS guarantee



This function depends on the downlink experience-based scheduling function and takes effect only for non-GBR bearers configured with guaranteed rates.

5.4.2 Network Analysis

5.4.2.1 Benefits

This function guarantees the rates of HBB users. After this function is enabled, the average downlink throughput of HBB users increases.

NOTE

The average downlink throughput of HBB users of specific QCIs or 5QIs can be calculated as follows:

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.4.2.2 Impacts

The impacts between this function and other functions are described in **5.4.3.2.3 Function Impacts**.

This function has the following network impacts:

- If no guaranteed rate is configured for a user, lowering the correlation threshold further deteriorates downlink experience of the user. In addition, the number of paired layers for downlink MIMO in the cell, number of RBs for downlink MIMO pairing in the cell, and average downlink cell throughput decrease.
- For users configured with a guaranteed rate, the downlink user-perceived rate may fluctuate around the guaranteed rate after the guaranteed rate is reached. The average downlink cell throughput may also fluctuate when there are few users in a cell.

NOTE

- Number of layers paired for downlink MIMO: **N.ChMeas.MIMO.DL.Pair.Layer**
- Number of RBs for downlink MIMO pairing = 1 x **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + 3 x **N.ChMeas.MIMO.DL.MuPairing.3Layers.RB** + 4 x **N.ChMeas.MIMO.DL.MuPairing.4Layers.RB** + 5 x **N.ChMeas.MIMO.DL.MuPairing.5Layers.RB** + 6 x **N.ChMeas.MIMO.DL.MuPairing.6Layers.RB** + 7 x **N.ChMeas.MIMO.DL.MuPairing.7Layers.RB** + 8 x **N.ChMeas.MIMO.DL.MuPairing.8Layers.RB**
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**

5.4.3 Requirements

5.4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Enabling downlink MU-MIMO QoS guarantee requires downlink experience-based scheduling or downlink experience satisfaction guarantee to be enabled.
Downlink experience satisfaction guarantee	**DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter	*FWA*	
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	PDSCH MU-MIMO must be enabled for downlink MU-MIMO QoS guarantee to take effect.
MU pairing enhancement	**NRDUCellAlgoSwitch**.***MuMimoOptSwith***	*Massive MIMO AHR (Low-Frequency TDD)*	MU pairing enhancement must be enabled for downlink MU-MIMO QoS guarantee to take effect.

5.4.3.2.2 Mutually Exclusive Functions

None

5.4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	After downlink MU-MIMO QoS guarantee is enabled, the average downlink cell throughput and number of paired layers for downlink MIMO in a cell may decrease.

Function Name	Function Switch	Reference	Description
Dynamic clustering-based grouping	**NRDUCellPdsch**.***DlMuMimoGroupMode*** set to **DYNAMIC_CLUSTER_GROUP**	*Massive MIMO iBeam (Low-Frequency TDD)*	When dynamic clustering-based grouping takes effect, UEs with low correlations are added to a group first. This affects the downlink experience improvement of downlink MU-MIMO QoS guarantee on HBB users configured with guaranteed rates.
NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	After downlink MU-MIMO QoS guarantee is enabled, if **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY** in NR DU cell resource management between network slices, resources may become limited, thus decreasing the gains yielded by downlink MU-MIMO QoS guarantee.

5.4.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

Table 5-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	DL_MU_MIMO_QOS_SW	Select this option.
DL MU-MIMO Channel Isolation Threshold	**NRDUCellPdsch**.***DlSrsMuMimoChanIsoThld***	None	The default value is recommended. When downlink MU-MIMO QoS guarantee takes effect, a smaller value of this parameter results in a lower pairing probability of HBB users configured with guaranteed rates and a higher probability that the user-perceived rate is close to the target rate, but a larger capacity loss.

5.4.4.1.2 Using MML Commands

Before using MML commands, refer to **5.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the downlink MU-MIMO QoS guarantee switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_MU_MIMO_QOS_SW-1;
//Configuring the channel isolation threshold for downlink MU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=1, DlSrsMuMimoChanIsoThld=30;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the downlink MU-MIMO QoS guarantee switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_MU_MIMO_QOS_SW-0;
```

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

Observe the indicators described in **5.4.4.3 Network Monitoring** to check whether this function has taken effect.

5.4.4.3 Network Monitoring

Observe the average downlink throughput of HBB users of specific QCIs or 5QIs:

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.5 Multi-Bearer Downlink Target Rate Adaptation

5.5.1 Principles

Operators usually offer HBB users various FWA packages with different guaranteed rates. However, a dedicated bearer (for example, with a QCI of 7) is often set up for video services through the core network. In this case, video services from different packages use the same bearer, making it impossible to provide differentiated service guarantee. **Table 5-7** provides an example. User A uses a high-end package with a downlink target rate of 50 Mbit/s and a bearer with a QCI of 128 for Internet services. User B uses a low-end package with a downlink target rate of 30 Mbit/s and a bearer with a QCI of 129 for Internet services. If video services with a QCI of 7 and downlink target rate of 20 Mbit/s are added for the two users, the actual downlink target rate becomes 70 Mbit/s for user A and 50 Mbit/s for user B, exceeding the respective purchased package rates.

Table 5-7 Target user/bearer rates in different service packages before the function is enabled

User	Package Type (Rate)	Single Bearer	Multi-Bearer
User A	High-end package (50 Mbit/s)	QCI 128: Downlink target rate = 50 Mbit/s User downlink target rate = 50 Mbit/s	QCI 128: Downlink target rate = 50 Mbit/s QCI 7: Downlink target rate = 20 Mbit/s User downlink target rate = 70 Mbit/s
User B	Low-end package (30 Mbit/s)	QCI 129: Downlink target rate = 30 Mbit/s User downlink target rate = 30 Mbit/s	QCI 129: Downlink target rate = 30 Mbit/s QCI 7: Downlink target rate = 20 Mbit/s User downlink target rate = 50 Mbit/s

To solve this problem, **Huawei provides the multi-bearer downlink target rate adaptation function, which adaptively allocates proper downlink target rates to different bearers**. If an HBB user has multiple non-GBR bearers configured with downlink target rates, operators can use the multi-bearer downlink target rate adaptation function to guarantee the user-perceived rate.

This function is controlled by the **DL_TARGET_RATE_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. After this function is enabled, the larger one of the downlink target rates configured for multiple non-GBR bearers is used as the downlink target rate of the user, and the actual target rate of each bearer is calculated based on this target rate and the corresponding ratio determined by the weighting factor **gNBDUMacParamGroup**.***DlSchPriWeightFactor***. For example, **Table 5-8** lists the actual target rates of users A and B and each bearer after this function is enabled for the scenarios listed in **Table 5-9**.

Table 5-8 Configurations for different QCIs

QCI	Configured Downlink Target Rate	Weighting Factor
7	20 Mbit/s	600
128	50 Mbit/s	400
129	30 Mbit/s	300

Table 5-9 Target user/bearer rates in different service packages after the function is enabled

User	Package Type (Rate)	Single Bearer	Multi-Bearer
User A	High-end package (50 Mbit/s)	QCI 128: Downlink target rate = 50 Mbit/s User downlink target rate = 50 Mbit/s	QCI 128: Actual downlink target rate = 20 Mbit/s QCI 7: Actual downlink target rate = 30 Mbit/s User downlink target rate = 50 Mbit/s
User B	Low-end package (30 Mbit/s)	QCI 129: Downlink target rate = 30 Mbit/s User downlink target rate = 30 Mbit/s	QCI 129: Actual downlink target rate = 10 Mbit/s QCI 7: Actual downlink target rate = 20 Mbit/s User downlink target rate = 30 Mbit/s

5.5.2 Network Analysis

5.5.2.1 Benefits

This function applies to scenarios with multiple non-GBR bearers. It provides differentiated guarantee for services in different packages and guarantees the rates of HBB users in multi-bearer scenarios.

5.5.2.2 Impacts

The impacts between this function and other functions are described in **5.5.3.2.3 Function Impacts**.

This function has the following network impacts: When a user has multiple bearers configured with target rates, the average downlink throughput of non-GBR bearers may decrease because the target rates of HBB users and non-GBR bearers may decrease.

NOTE

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.5.3 Requirements

5.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

NOTE

In TDD in high frequency bands, the number of licenses for a sector must be the same as the number of carriers. Otherwise, this function cannot take effect.

5.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
TDD	Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Downlink experience-based scheduling must be enabled.

5.5.3.2.2 Mutually Exclusive Functions

None

5.5.3.2.3 Function Impacts

None

5.5.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

Table 5-10 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	DL_TARGET_RATE_ADAPT_SW	Select the **DL_TARGET_RATE_ADAPT_SW** option.
Downlink Scheduling Priority Weight Factor	**gNBDUMacParamGroup**.***DlSchPriWeightFactor***	None	Operators can configure this parameter based on the actual service requirements. A smaller value of this parameter indicates a lower target rate of the bearer after dynamic adjustment.

Parameter Name	Parameter ID	Option	Setting Notes
DL Exp-based Scheduling Guaranteed User Rate	**gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.

5.5.4.1.2 Using MML Commands

NOTICE

You are advised to configure the **gNBDUMacParamGroup**.***DlSchPriWeightFactor*** parameter for each QCI added for a cell.

Before using MML commands, refer to **5.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//The scenario described in the principles section is used as an example.
//Configuring the downlink target rate for each QCI
MOD GNBDUMACPARAMGROUP: MacParamGroupId=1, DlExpBasedSchGuarUserRate=50000;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=128, MacParamGroupId=1;
MOD GNBDUMACPARAMGROUP: MacParamGroupId=2, DlExpBasedSchGuarUserRate=30000;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=129, MacParamGroupId=2;
MOD GNBDUMACPARAMGROUP: MacParamGroupId=7, DlExpBasedSchGuarUserRate=20000;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=7, MacParamGroupId=7;
//Turning on the multi-bearer downlink target rate adaptation switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_TARGET_RATE_ADAPT_SW-1;
//Configuring the downlink scheduling priority weighting factor for each QCI
MOD GNBDUMACPARAMGROUP: MacParamGroupId=1, DlSchPriWeightFactor=400;
MOD GNBDUMACPARAMGROUP: MacParamGroupId=2, DlSchPriWeightFactor=300;
MOD GNBDUMACPARAMGROUP: MacParamGroupId=7, DlSchPriWeightFactor=600;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the multi-bearer downlink target rate adaptation switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_TARGET_RATE_ADAPT_SW-0;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

Observe the indicators described in **5.5.4.3 Network Monitoring** to check whether this function has taken effect.

5.5.4.3 Network Monitoring

Observe the following indicators to obtain the average downlink throughput of QCI- or 5QI-specific services:

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.6 Downlink Resource Isolation (Low-Frequency TDD)

5.6.1 Principles

Downlink resource isolation is introduced to further reduce the impact of FWA users on eMBB users. After this function is enabled, the base station periodically measures the number of active eMBB users and makes decisions as follows:

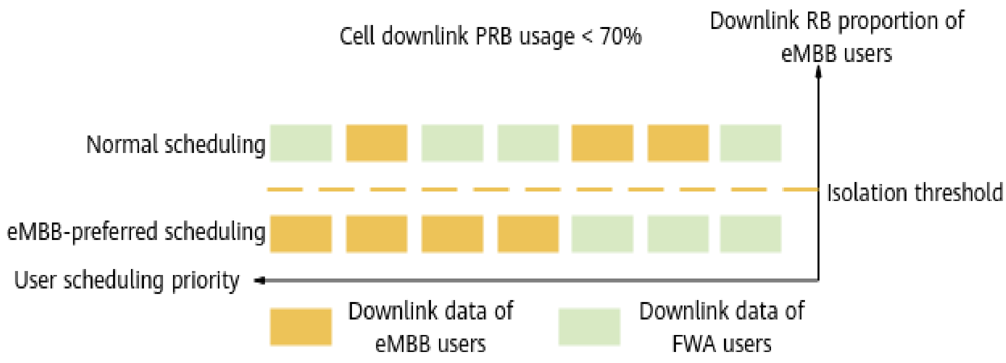
- **When the downlink PRB usage of a cell is less than 70% and the proportion of downlink RBs used by eMBB users is less than the resource isolation threshold, the base station preferentially schedules downlink data for eMBB users**, as illustrated in **Figure 5-6**.

  Resource isolation threshold = **NRDUCellFwaQos**.***EmbbPrioritizeDynPrbThld*** value x Number of active eMBB users
- **Otherwise, the scheduling of eMBB users does not take precedence over that of FWA users.**

FWA users refer to Internet users, for which the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_INTERNET**. eMBB users refer to users other than Internet users.

This function is controlled by the **DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

Figure 5-6 Downlink resource isolation



This function is recommended when FWA services are not developed on the network. In this case, enabling this function and then deploying FWA services can reduce the impact of FWA users on eMBB users. This function is not recommended when FWA services are well developed or the rate of FWA services needs to be guaranteed.

NOTE

- Downlink resource isolation does not take effect for retransmission and signaling, RLC status reports, and special-priority services (such as private line services, low-latency services, VoNR services, and services carried on GBR bearers) of FWA users.
- Downlink resource isolation does not take effect for retransmitted data. When there is a large number of retransmissions, FWA users affect downlink experience of eMBB users. In this case, the **gNBDUSchParamGroup**.***DlMaxHarqRetransNum*** parameter can be configured for FWA users to reduce the maximum number of downlink HARQ retransmissions and reduce the impact on downlink experience of eMBB users.
- When the **NRDUCellFwaQos**.***EmbbPrioritizeDynPrbThld*** parameter value multiplied by the number of active eMBB users exceeds 100%, the resource isolation threshold is 100%.

5.6.2 Network Analysis

5.6.2.1 Benefits

When the downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) of a cell is less than 70%, this function increases the average downlink throughput of eMBB users. For details about how to calculate the average downlink throughput of eMBB users, see **5.6.4.3 Network Monitoring**.

5.6.2.2 Impacts

The impacts between this function and other functions are described in **5.6.3.2.3 Function Impacts**.

This function has the following network impacts: When the downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) of a cell is less than 70%, this function allows preferential scheduling of downlink data of eMBB users in the event of cell resource insufficiency. As such, the average downlink throughput of

FWA users decreases. For details about how to calculate the average downlink throughput of FWA users, see **5.6.4.3 Network Monitoring**.

5.6.3 Requirements

5.6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.6.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
FWA user identification	**gNBQciBearer.*ServiceType***	*FWA*	The service type must be configured for FWA user identification. Otherwise, downlink resource isolation does not take effect.

5.6.3.2.2 Mutually Exclusive Functions

None

5.6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	After the downlink resource isolation function is enabled, if the power saving BWP function takes effect for FWA users, the downlink experience of FWA users decreases more severely. In this case, it is good practice to select the **FWA_BWP2_DISABLE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter so that power saving BWP is disabled for FWA users.
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	After downlink resource isolation is enabled, the number of downlink scheduling opportunities for RedCap FWA users further decreases. As such, the average downlink throughput of FWA users decreases more severely.
Downlink Scheduling Priority Weight Factor	**gNBDUMacParamGroup**.***DlSchPriWeightFactor***	None	After downlink resource isolation is enabled, if the downlink scheduling priority weight factors for FWA users are adjusted due to other functions, the adjustments affect whether this function takes effect.

Function Name	Function Switch	Reference	Description
NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	With downlink resource isolation enabled: If the minimum percentage of downlink RBs in the "NR DU cell resource management between network slices" function is used for FWA users, then downlink resource isolation will not take effect for these FWA users; if this minimum percentage of downlink RBs is used for non-FWA users, the reduction in average downlink throughput of FWA users will be amplified.
Operator-specific RB management	**RB_DYNAMIC_SHARING_SW**, **MAX_RB_LIMIT_SW**, and **MIN_RB_GUARANTEE_SW** options of the **NRDUCellAlgoSwitch.*RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	With downlink resource isolation enabled: If the minimum percentage of downlink RBs in operator-specific RB management is used for FWA users, then downlink resource isolation will not take effect for these FWA users; if this minimum percentage of downlink RBs is used for non-FWA users, the reduction in average downlink throughput of FWA users will be amplified.

Function Name	Function Switch	Reference	Description
Dynamic RB control for UEs requiring low latency and high reliability	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	With downlink resource isolation enabled: If the minimum percentage of downlink RBs in the "dynamic RB control for UEs requiring low latency and high reliability" function is used for FWA users, then downlink resource isolation will not take effect for these FWA users; if this minimum percentage of downlink RBs is used for non-FWA users, the reduction in average downlink throughput of FWA users will be amplified.
RFSP-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	With downlink resource isolation enabled: If the minimum percentage of downlink RBs in RFSP-based RB resource control is used for FWA users, then downlink resource isolation will not take effect for these FWA users; if this minimum percentage of downlink RBs is used for non-FWA users, the reduction in average downlink throughput of FWA users will be amplified.

Function Name	Function Switch	Reference	Description
User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If user PRB control for FWA has taken effect for FWA users, enabling downlink resource isolation may further deteriorate FWA user experience. If user PRB control for FWA has taken effect for eMBB users, enabling downlink resource isolation may deteriorate eMBB user experience.
Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If downlink experience-based scheduling has taken effect for FWA users, enabling downlink resource isolation may degrade FWA user experience.
Downlink experience satisfaction guarantee	**DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter	*FWA*	If downlink experience satisfaction guarantee has taken effect for FWA users, enabling downlink resource isolation may degrade FWA user experience.

Function Name	Function Switch	Reference	Description
QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When QCI-based resource control (controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter) is enabled together with downlink resource isolation and the **NRDUCellRes**.***DlRbAverageRatio*** or **NRDUCellRes**.***DlRbMinRatio*** parameter takes effect for FWA users in a QCI group, the downlink throughput of eMBB users decreases. If the **NRDUCellRes**.***DlRbAverageRatio*** or **NRDUCellRes**.***DlRbMinRatio*** parameter takes effect for non-FWA users in a QCI group, the average downlink throughput of FWA users will decrease more significantly.
Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	After the downlink resource isolation function is enabled, if the time-domain power saving BWP function takes effect on FWA UEs, the downlink experience of FWA UEs deteriorates more severely. In this case, it is good practice to select the **FWA_BWP2_DISABLE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter so that BWP2 will not be configured for FWA UEs.

5.6.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.6.4 Operation and Maintenance

5.6.4.1 Data Configuration

5.6.4.1.1 Data Preparation

Table 5-11 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-11 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	To enable downlink resource isolation, select the **DL_EMBB_PREFERENTIAL_SCH_SW** option of this parameter.
eMBB Prioritization Dynamic PRB Threshold	**NRDUCellFwaQos**.***EmbbPrioritizeDynPrbThld***	Set this parameter based on the network plan.

5.6.4.1.2 Using MML Commands

Before using MML commands, refer to **5.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink resource isolation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_EMBB_PREFERENTIAL_SCH_SW-1;
//Configuring the resource isolation threshold
MOD NRDUCELLFWAQOS: NrDuCellId=1, EmbbPrioritizeDynPrbThld=70;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink resource isolation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_EMBB_PREFERENTIAL_SCH_SW-0;
```

5.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.6.4.2 Activation Verification

If the average downlink throughput of eMBB users increases after this function is enabled, this function has taken effect. For details about related counters, see **5.6.4.3 Network Monitoring**.

5.6.4.3 Network Monitoring

After this function is enabled, you can monitor the running status of this function by observing the decrease in the average downlink throughput of FWA users and the increase in the average downlink throughput of eMBB users.

The average downlink user throughput of FWA users or eMBB users can be calculated using the following QCI/5QI-specific counters:

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.7 Downlink MU-MIMO Layer Isolation (Low-Frequency TDD)

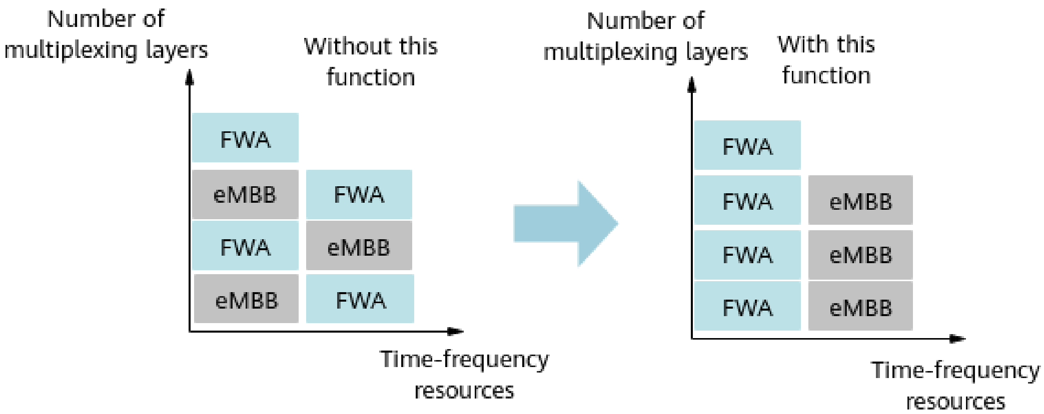
5.7.1 Principles

Downlink MU-MIMO layer isolation is introduced to further reduce the impact of FWA users on eMBB users. After this function is enabled, **user pairing for downlink MU-MIMO is allowed only between eMBB users or between FWA users**, as shown in **Figure 5-7**. This reduces the impact of FWA users on eMBB users and improves downlink experience of eMBB users. In this function:

- FWA users refer to Internet users for which the **gNBQciBearer**.***ServiceType*** parameter is set to **FWA_INTERNET**.
- eMBB users refer to users other than Internet users.

This function is controlled by the **DL_MU_MIMO_ISOLATION_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

Figure 5-7 Downlink MU-MIMO layer isolation



This function is recommended when FWA services are not developed on the network. In this case, enabling this function and then deploying FWA services can reduce the impact of FWA users on eMBB users. This function is not recommended when FWA services are well developed or the rate of FWA services needs to be guaranteed.

5.7.2 Network Analysis

5.7.2.1 Benefits

When the downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) of a cell is less than 70%, this function increases the average downlink throughput of eMBB users. For details about how to calculate the average downlink throughput of eMBB users, see **5.7.4.3 Network Monitoring**.

5.7.2.2 Impacts

The impacts between this function and other functions are described in **5.7.3.2.3 Function Impacts**.

This function has the following network impacts: When the downlink cell PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) is less than 70%, enabling this function causes the downlink MU-MIMO pairing rate to decrease. As such, the total number of layers paired for downlink MIMO (**N.ChMeas.MIMO.DL.Pair.Layer**), average number of layers paired for downlink MIMO (**N.ChMeas.MIMO.DL.Pair.Layer.Avg**), and average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) will decrease. In addition, after this function is enabled, if eMBB users with higher priorities occupy more RBs, the average downlink throughput of FWA users may decrease because FWA and eMBB users cannot be paired with each other. For details about how to calculate the average downlink throughput of FWA users, see **5.7.4.3 Network Monitoring**.

5.7.3 Requirements

5.7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.7.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
FWA user identification	**gNBQciBearer**.***ServiceType***	*FWA*	The service type must be configured for FWA user identification. Otherwise, downlink MU-MIMO layer isolation does not take effect.

Function Name	Function Switch	Reference	Description
Downlink small-packet group merging	**DL_SMALL_PACKET_MERGE_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter	None	Downlink MU-MIMO layer isolation requires that the **DL_SMALL_PACKET_MERGE_SW** option be selected.
Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Downlink MU-MIMO layer isolation requires that downlink resource isolation be enabled.
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Downlink MU-MIMO layer isolation requires that PDSCH MU-MIMO be enabled.

5.7.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Clustering-based MU grouping	**NRDUCellPdsch**.***DlMuMimoGroupMode*** set to **CLUSTER_GROUP**	None	Downlink MU-MIMO layer isolation cannot work with clustering-based MU grouping.
Dynamic clustering-based grouping	**NRDUCellPdsch**.***DlMuMimoGroupMode*** set to **DYNAMIC_CLUSTER_GROUP**	*Massive MIMO iBeam (Low-Frequency TDD)*	Downlink MU-MIMO layer isolation cannot be enabled together with dynamic clustering-based grouping.

5.7.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	If downlink MU-MIMO layer isolation takes effect after PDSCH MU-MIMO is enabled, eMBB users cannot be paired with FWA users for MU-MIMO. In this case, the benefits of PDSCH MU-MIMO decrease.
DL MU-MIMO Group Mode	**NRDUCellPdsch**.***DlMuMimoGroupMode*** set to **ONE_USER_ONE_GROUP**	None	When **NRDUCellPdsch**.***DlMuMimoGroupMode*** is set to **ONE_USER_ONE_GROUP**, downlink MU-MIMO layer isolation does not take effect.
Precise MU-MIMO evaluation	**PRECISE_MUMIMO_EVAL_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	When downlink MU-MIMO layer isolation is enabled, the benefits of precise MU-MIMO evaluation may change due to changes in UE pairing.

5.7.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.7.4 Operation and Maintenance

5.7.4.1 Data Configuration

5.7.4.1.1 Data Preparation

Table 5-12 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-12 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	To enable downlink MU-MIMO layer isolation, select the **DL_MU_MIMO_ISOLATION_SW** option.

5.7.4.1.2 Using MML Commands

Before using MML commands, refer to **5.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink MU-MIMO layer isolation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_MU_MIMO_ISOLATION_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink MU-MIMO layer isolation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=DL_MU_MIMO_ISOLATION_SW-0;
```

5.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.7.4.2 Activation Verification

If the average downlink throughput of eMBB users increases after this function is enabled, this function has taken effect. For details about how to calculate the average downlink throughput of eMBB users, see **5.7.4.3 Network Monitoring**.

5.7.4.3 Network Monitoring

After this function is enabled, monitor the running status of this function by observing value changes of the indicators and counters listed in **5.7.2.1 Benefits** and **5.7.2.2 Impacts**.

The average downlink user throughput of FWA users or eMBB users can be calculated using the following QCI/5QI-specific counters:

- Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
- Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**

5.8 Downlink Experience Satisfaction Guarantee

5.8.1 Principles

FWA operators are launching data plans that guarantee a certain level of downlink experience satisfaction. However, downlink experience satisfaction is at risk of declining as both the committed downlink target rates in the plans and cell loads are rising. The downlink experience satisfaction guarantee function can improve downlink experience satisfaction in cells.

This function is controlled by the **DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter. It is applicable to UEs having non-GBR bearers with target guaranteed rates configured through the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** parameter. If the rate of such a UE is close to the target, the base station preferentially increases the UE's priority to maximize downlink experience satisfaction in the serving cell. Downlink experience satisfaction is calculated using this formula: Number of samples with downlink rates greater than or equal to the target downlink rate/Total number of samples. The numbers of these samples can be calculated based on the values of the following counters (detailed in **5.8.4.2 Activation Verification**).

Application Scenario	Counter Set	Counter Name
NSA networking	QCI-specific RLC-Layer Downlink UE Throughput Distribution	**N.Thp.DL.Samp.QCI.Index0** to **N.Thp.DL.Samp.QCI.Index14**
SA networking	5QI-specific Downlink Throughput Distribution at the RLC Layer	**N.Thp.DL.Samp.DuCell5QI.Index0** to **N.Thp.DL.Samp.DuCell5QI.Index14**

To ensure more accurate calculation of downlink experience satisfaction, you are advised to select the **DL_THPT_L_SLOT_STATS_POL_SW** option of the **NRDUCellOamParam**.***StatisticsStrategySwitch*** parameter. When this option is selected, the duration and traffic of downlink tail data are not excluded from the measurement of the preceding counters. Tail data is the data transmitted in the last slot during which the downlink buffer becomes empty.

NOTE

- In NSA DC data split scenarios, this function can take effect only in NR cells. That is, only NR cells can see a higher downlink experience satisfaction.
- In CA (detailed in *Carrier Aggregation* and *CA Experience Improvement*) scenarios, this function can take effect for both the PCell and SCells serving a UE. To achieve this, the PCell will transfer the updated scheduling priority of the UE to the SCells.

5.8.2 Network Analysis

5.8.2.1 Benefits

After this function is enabled, downlink experience satisfaction increases for UEs configured with target guaranteed rates. For details about how to observe downlink experience satisfaction, see **5.8.4.2 Activation Verification**.

5.8.2.2 Impacts

The impacts between this function and other functions are described in **5.8.3.2.3 Function Impacts**.

The impacts on the network are as follows:

- When downlink experience satisfaction guarantee is enabled, the first packet delay increases, and the average downlink cell throughput, downlink spectral efficiency, and average downlink user-perceived rate may slightly change.
- Downlink experience satisfaction guarantee preferentially assures the priorities of UEs whose rates are close to the target. As a result, service experience will deteriorate for UEs that have already reached the downlink target rates (for example, cell-center UEs) and cell-edge UEs that struggle to reach the downlink target rates.

- When downlink experience satisfaction guarantee is enabled, the service experience will deteriorate for UEs without the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** parameter configured. The deterioration degree depends on factors such as the cell load and traffic model.

NOTE

First packet delay = **N.Traffic.DL.RlcFirstPktDelay.Time**/**N.Qos.DL.RlcFirstPktDelay.Num**

5.8.3 Requirements

5.8.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051204	Experience-based FWA	NR0S00EFWA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.8.3.2.1 Prerequisite Functions

None

5.8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Downlink experience satisfaction guarantee cannot work with Hyper Cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Downlink experience satisfaction guarantee cannot work with Cell Combination.

5.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When downlink experience satisfaction guarantee and downlink experience-based scheduling are both enabled, both functions can take effect, but downlink experience satisfaction guarantee may produce fewer benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA private line	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When downlink experience satisfaction guarantee and basic service assurance for FWA private line are both enabled, and the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** and **gNBDUMacParamGroup**.***DlGuaranteedRate*** parameters are both configured for FWA private line users, the following mechanism applies: ● When the rate of a private line user is lower than the value of **gNBDUMacParamGroup**.***DlGuaranteedRate***, absolute priority scheduling is used for the private line user. Downlink experience satisfaction guarantee does not take effect for the user. ● When the rate of a private line user is greater than or equal to the value of **gNBDUMacParamGroup**.***DlGuaranteedRate***, downlink experience satisfaction guarantee can take effect for the user.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink throughput transmission duration	**DL_THPT_TRANS_TIME_SW** option of the **NRDUCellOamParam**.***StatisticsStrategySwitch*** parameter	None	Downlink experience satisfaction guarantee cannot take effect when the **DL_THPT_TRANS_TIME_SW** option is selected.
Low-frequency TDD	Load-based adaptive downlink scheduling	**NRDUCellDlSch**.***DlSchPolicy*** set to **ADAPTIVE**	*Massive MIMO AHR (Low-Frequency TDD)*	When downlink experience satisfaction guarantee is in effect, the number of available downlink RBs in a cell decreases. This makes it more likely that the downlink PRB usage falls below the threshold (that determines whether a cell is heavily loaded) for load-based adaptive downlink scheduling to take effect. Consequently, there are fewer opportunities for load-based adaptive downlink scheduling to produce its intended benefits. If the downlink PRB usage reaches the threshold for load-based adaptive downlink scheduling to take effect, load-based adaptive downlink scheduling can take effect, but the benefits of downlink experience satisfaction guarantee may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If downlink experience satisfaction guarantee has taken effect for FWA users, enabling downlink resource isolation may degrade FWA user experience.
Low-frequency TDD	Rate adaptation for device-pipe synergy	**RATE_ADAPTION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*FWA*	Downlink experience satisfaction guarantee preferentially takes effect when it is enabled together with rate adaptation for device-pipe synergy.
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	Assume that both downlink experience satisfaction guarantee and flexible experience differentiation are in effect for a UE. If the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** or **gNBDURfspConfig**.***DlExpBasedSchGuarUserRate*** parameter exceeds the downlink target rate determined by the flexible experience differentiation function, the UE's downlink peak rate will be capped at this downlink target rate determined by the flexible experience differentiation function.

5.8.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.8.4 Operation and Maintenance

5.8.4.1 Data Configuration

5.8.4.1.1 Data Preparation

Table 5-13 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Scheduling Algorithm Switch	**NRDUCellFwaQos**.***SchAlgoSwitch***	DL_EXP_SATIS_GUAR_SW	Select this option if downlink experience satisfaction guarantee is required. In addition, to prevent the impacts on the benefits of downlink experience satisfaction guarantee, you are advised to deselect the **DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.
Statistics Strategy Switch	**NRDUCellOamParam**.***StatisticsStrategySwitch***	DL_THPT_L_SLOT_STATS_POL_SW	Select this option.
DL Exp-based Scheduling Guaranteed User Rate	**gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
gNodeB Range Start Value	**gNBCounterParam.*gNBRangeStartValue***	None	To facilitate data collection for calculating downlink experience satisfaction of UEs with a specific QCI/5QI, you can set the start value of the range corresponding to a performance counter based on the configured downlink target guaranteed rate. For example, if the downlink target guaranteed rate is 50 Mbit/s, you can set the start value of range 10 to 50000.

5.8.4.1.2 Using MML Commands

Before using MML commands, refer to **5.8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink experience satisfaction guarantee
MOD NRDUCELLFWAQOS: NrDuCellId=0, SchAlgoSwitch=DL_EXP_SATIS_GUAR_SW-1;
//(Optional) Disabling downlink experience-based scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FwaAlgoSwitch=DL_EXP_BASED_SCH_SW-0;
//Selecting the DL_THPT_L_SLOT_STATS_POL_SW option
MOD NRDUCELLOAMPARAM: NrDuCellId=0, StatisticsStrategySwitch=DL_THPT_L_SLOT_STATS_POL_SW-1;
//Configuring the downlink target guaranteed rate (for example, 50 Mbit/s) for a UE with a specific QCI/5QI
MOD GNBDUMACPARAMGROUP: MacParamGroupId=7, DlExpBasedSchGuarUserRate=50000;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=7, MacParamGroupId=7;
//(Optional) Setting the start value of the range corresponding to a performance counter to the downlink
target guaranteed rate (for example, 50 Mbit/s), to facilitate data collection for calculating downlink
experience satisfaction of UEs with a specific QCI/5QI
MOD GNBCOUNTERPARAM: gNBCounterType=DL_THP_SAMP, gNBCounterIndex=10,
gNBRangeStartValue=50000;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink experience satisfaction guarantee
MOD NRDUCELLFWAQOS: NrDuCellId=0, SchAlgoSwitch=DL_EXP_SATIS_GUAR_SW-0;
//Deselecting the DL_THPT_L_SLOT_STATS_POL_SW option
MOD NRDUCELLOAMPARAM: NrDuCellId=0, StatisticsStrategySwitch=DL_THPT_L_SLOT_STATS_POL_SW-0;
```

5.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.8.4.2 Activation Verification

After this function is enabled, check whether downlink experience satisfaction increases for UEs with a specific QCI/5QI. If it increases, this function has taken effect.

For UEs with a specific QCI/5QI, downlink experience satisfaction is calculated using the following formula: Number of samples with downlink rates greater than or equal to the target downlink rate/Total number of samples. The numbers of these samples can be calculated based on the values of the following counters.

Application Scenario	Counter Set	Counter Name
NSA networking	QCI-specific RLC-Layer Downlink UE Throughput Distribution	**N.Thp.DL.Samp.QCI.Index0** to **N.Thp.DL.Samp.QCI.Index14**
SA networking	5QI-specific Downlink Throughput Distribution at the RLC Layer	**N.Thp.DL.Samp.DuCell5QI.Index0** to **N.Thp.DL.Samp.DuCell5QI.Index14**

For example, if the lower rate limit of range 6 is equal to the downlink target rate, then the downlink rates that fall in range 6 and higher ranges are all greater than or equal to the downlink target rate. In this case, downlink experience satisfaction is calculated as follows:

- SA networking: Downlink experience satisfaction of UEs with a specific 5QI = (**N.Thp.DL.Samp.DuCell5QI.Index6** + **N.Thp.DL.Samp.DuCell5QI.Index7** + ... + **N.Thp.DL.Samp.DuCell5QI.Index14**)/(**N.Thp.DL.Samp.DuCell5QI.Index0** + **N.Thp.DL.Samp.DuCell5QI.Index1** + ... + **N.Thp.DL.Samp.DuCell5QI.Index14**)

- NSA networking: Downlink experience satisfaction of UEs with a specific QCI = (**N.Thp.DL.Samp.QCI.Index6** + **N.Thp.DL.Samp.QCI.Index7** + ... + **N.Thp.DL.Samp.QCI.Index14**)/(**N.Thp.DL.Samp.QCI.Index0** + **N.Thp.DL.Samp.QCI.Index1** + ... + **N.Thp.DL.Samp.QCI.Index14**)

5.8.4.3 Network Monitoring

After this function is enabled, monitor the running status of this function by observing value changes of the indicators and counters listed in **5.8.2.1 Benefits** and **5.8.2.2 Impacts**.

6 FWA Private Line Service Assurance

FWA private line offers the FWA broadband solution for enterprises. FWA private line users are configured with QCI-specific bearers for which **gNBQciBearer**.***ServiceType*** is set to **FWA_Private_Line**.

6.1 Basic Service Assurance for FWA Private Line

6.1.1 Principles

FWA private line requires high QoS and fulfilled subscription rates. **This function reserves some admission resources for FWA private line users to improve access performance. When the guaranteed rates of FWA private line services are not fulfilled, the user scheduling priority is raised to guarantee the service QoS.**

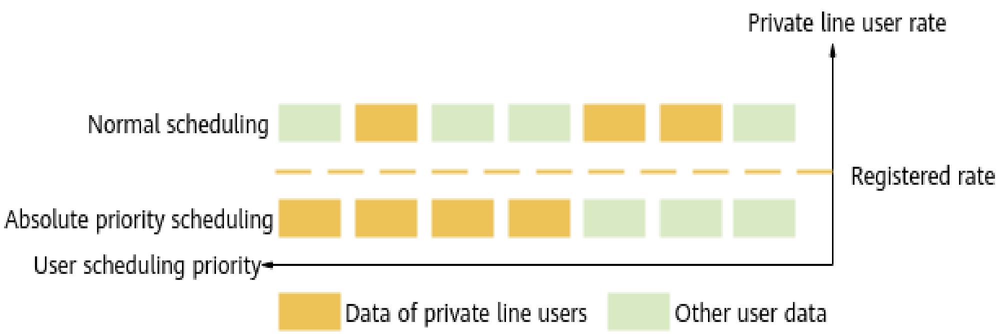
The **PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter specifies whether to enable this function. After this function is enabled, the gNodeB performs QoS rate guarantee and differentiated admission for identified FWA private line users.

QoS Rate Guarantee

For FWA private line, rate guarantee for GBR services is implemented based on the GBR configured on the core network, and that for non-GBR services is implemented by setting the uplink guaranteed rate (**gNBDUMacParamGroup**.***UlGuaranteedRate***) and downlink guaranteed rate (**gNBDUMacParamGroup**.***DlGuaranteedRate***) for the corresponding QCI.

For a private line user whose rate is less than the registered rate, absolute priority scheduling is used for rate guarantee. Scheduling resources can be allocated to other users only when the rate of the user reaches the registered rate. **Figure 6-1** illustrates such rate guarantee.

Figure 6-1 QoS rate guarantee for private line users



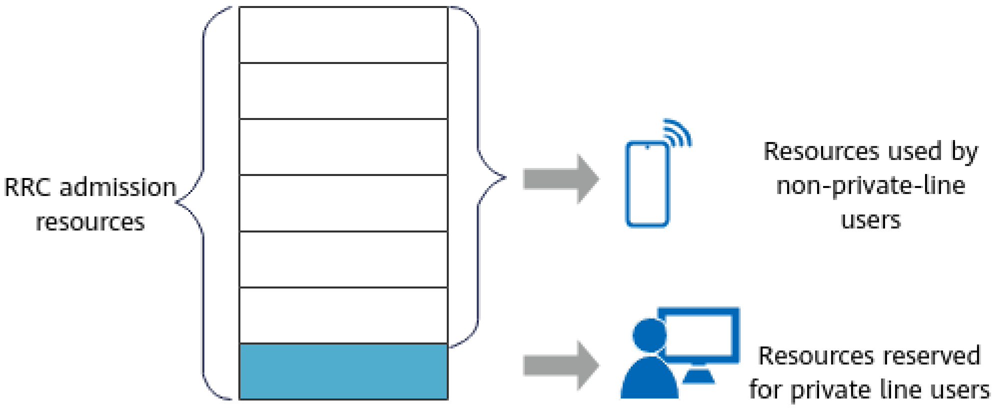
NOTE

After this function is enabled, if the uplink or downlink guaranteed rate is modified, the modification takes effect only after the user accesses the network again.

Differentiated Admission

If the **RRC_SETUP_REASON_ADMISSION_SW** option of the **NRCellAlgoSwitch**.***FwaAlgoSwitch*** parameter is selected, the gNodeB sets up temporary RRC connections only with the cause value "highPriorityAccess" for users, and triggers bearer setup when the number of RRC connections in a cell exceeds the licensed number or the upper limit. Then, the gNodeB performs admission control based on QCIs or 5QIs. **Figure 6-2** illustrates such differentiated admission.

Figure 6-2 Differentiated admission for private line services



6.1.2 Network Analysis

6.1.2.1 Benefits

QoS rate guarantee can improve the average uplink and downlink throughput of FWA private line users, while differentiated admission improves their admission success rate.

NOTE

Average downlink user throughput: **User Downlink Average Throughput (DU)**

Average uplink user throughput: **User Uplink Average Throughput (DU)**

6.1.2.2 Impacts

The impacts between this function and other functions are described in **6.1.3.2.3 Function Impacts**.

This function has the following network impacts: After this function is enabled, the rates of non-private-line users may decrease. If the **RRC_SETUP_REASON_ADMISSION_SW** option of the **NRCellAlgoSwitch**.***FwaAlgoSwitch*** parameter is deselected, the PDU session setup success rate of the cell may decrease.

NOTE

Cell PDU session setup success rate = (**N.PDUSession.Est.Succ**/**N.PDUSession.Est.Att**) x 100%

6.1.3 Requirements

6.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051205	FWA Private Line	NR0S0FWAPL00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

NOTE

In TDD in high frequency bands, the number of licenses for a sector must be the same as the number of carriers. Otherwise, this function cannot take effect.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
TDD	FWA user identification	**gNBQciBearer**.***ServiceType***	*FWA*	The private line service type must be configured for FWA user identification. Otherwise, the FWA private line service assurance function does not take effect.

6.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	FWA private line service assurance cannot work with Hyper Cell.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	FWA private line service assurance cannot work with Cell Combination.

6.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Admission control	**QOS_FLOW_ADMISSION_SW** and **QOS_FLOW_PREEMPTION_SW** options of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter	*Admission Control*	After the differentiated admission switch for FWA private line service assurance (**RRC_SETUP_REASON_ADMISSION_SW** option of the **NRCellAlgoSwitch**.***FwaAlgoSwitch*** parameter) and the user-number-based preemption switch are both turned on, the preempting users for which the cause value of RRC connection setup is not "Emergency", "highPriorityAccess", or "mo-VoiceCall" and private line users can share temporary user resources.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When user PRB control for FWA is enabled together with FWA private line service assurance, if the uplink or downlink PRB usage of private line users exceeds the configured maximum uplink or downlink RB proportion, the uplink or downlink guaranteed rate will not be fulfilled for private line users.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Special Service PVI Mode	**NRDUCellRac**.***SpecServPviMode*** set to **PVI_FOLLOW_ARP**	*Admission Control*	When **NRDUCellRac**.***SpecServPviMode*** is set to **PVI_FOLLOW_ARP** and the bearer for an FWA private line user has a relatively low ARP priority and is preemptable, even if the differentiated admission switch is turned on, resources of such users may be preempted by bearers of other users with a relatively high ARP, leading to admission failures. As such, you are advised to configure a relatively high ARP priority and an attribute indicating not preemptable for such users during registration on the core network.

6.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.1.3.4 Cells

FWA private line service assurance is not supported for SUL cells with **NRDUCell**.***DuplexMode*** set to **CELL_SUL**.

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	PRIVATE_LINE_SW	Select the **PRIVATE_LINE_SW** option.
FWA Algorithm Switch	**NRCellAlgoSwitch**.***FwaAlgoSwitch***	RRC_SETUP_REASON_ADMISSION_SW	Select the **RRC_SETUP_REASON_ADMISSION_SW** option.
Downlink Guaranteed Rate	**gNBDUMacParamGroup**.***DlGuaranteedRate***	None	Set this parameter based on the network plan.
Uplink Guaranteed Rate	**gNBDUMacParamGroup**.***UlGuaranteedRate***	None	Set this parameter based on the network plan.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to **6.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the private line service assurance function
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_SW-1;
//Enabling differentiated admission for private line services
MOD NRCELLALGOSWITCH: NrCellId=1, FwaAlgoSwitch=RRC_SETUP_REASON_ADMISSION_SW-1;
//Configuring the uplink/downlink guaranteed rate for FWA private line services with a QCI of 9 in a cell
//1. Configuring the uplink/downlink guaranteed rate for the gNodeB DU MAC parameter group with the ID of 9
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, UlGuaranteedRate=3000, DlGuaranteedRate=50000;
//2. Configuring the gNodeB DU MAC parameter group ID as 9 for services with a QCI of 9 in a cell
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, MacParamGroupId=9;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling differentiated admission for private line services
MOD NRCELLALGOSWITCH: NrCellId=1, FwaAlgoSwitch=RRC_SETUP_REASON_ADMISSION_SW-0;
//Restoring the uplink/downlink guaranteed rate
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, UlGuaranteedRate=0, DlGuaranteedRate=0;
//Disabling the private line service assurance function
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_SW-0;
```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

1. Run the following MML command to subscribe to the private line-related performance counter objects.
   - SA networking (using 5QI 9 as an example)
     ```
     MOD GNB5QICONFIG: Nr5qi=9, CounterObjSubscriptionInd=YES;
     ```
   - NSA networking (using QCI 9 as an example)
     ```
     MOD GNBQCIBEARER: Qci=9, CounterObjSubscriptionInd=YES;
     ```
2. If the QCI of a service in the cell indicates FWA private line service and the value of the corresponding **N.User.RRCConn.Active.QCI.Max** or **N.User.RRCConn.Active.DuCell5QI.Max** counter is not zero, the private line service assurance function has been enabled.

6.1.4.3 Network Monitoring

Observe the following indicators:

- Average downlink user throughput: **User Downlink Average Throughput (DU)**

- Average uplink user throughput: **User Uplink Average Throughput (DU)**
- **N.User.RRCConn.Active.QCI.Max** of private line services
- **N.User.RRCConn.Active.DuCell5QI.Max** of private line services

6.2 FWA Private Line Service Performance Improvement

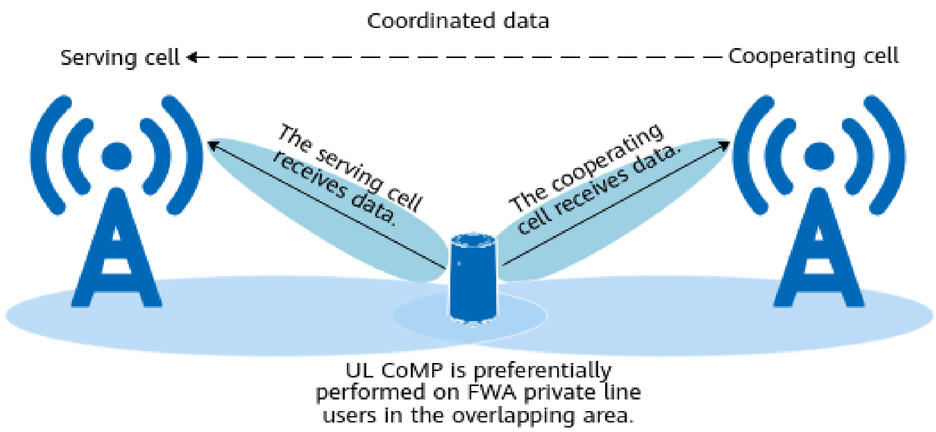
6.2.1 Principles

To further ensure the user-perceived rate of FWA private line users, a series of functions are introduced to perform special processing on these users when other features have taken effect.

UL CoMP Optimization for FWA Private Line

UL CoMP enables the serving and cooperating cells to jointly receive traffic channel data from users in the overlapping area at the cell edge, improving the user-perceived rate of such users. For cells where UL CoMP (detailed in *CoMP*) has taken effect, UL CoMP optimization for FWA private line is introduced to reserve some CoMP resources for FWA private line users so that these users can preferentially obtain CoMP resources in the overlapping area. This function is controlled by the **PRIVATE_LINE_UL_COMP_OPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

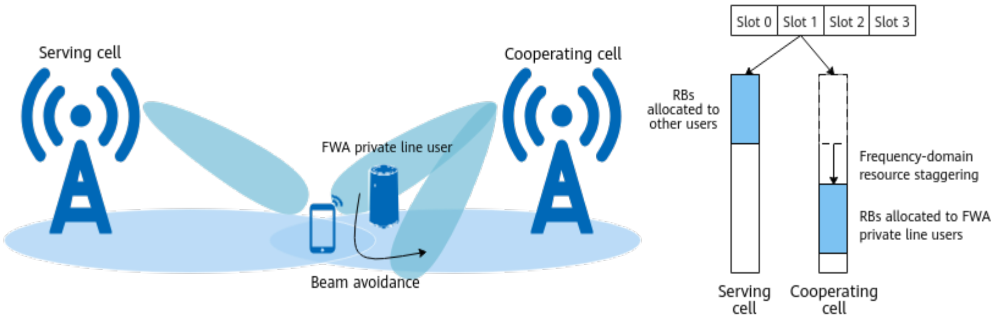
Figure 6-3 Principles of UL CoMP optimization for FWA private line



Downlink CS/CBF Optimization for FWA Private Line (Low-Frequency TDD)

With downlink CS/CBF, frequency-domain resources or beam-domain resources are coordinated through cooperating cells to prevent strong interference to users in overlapping areas and implement beam avoidance for such users. This improves the user-perceived rates of cell-edge users in overlapping areas but affects user experience in cooperating cells. For cells where CS/CBF between intra- or inter-base-station cells (detailed in *Coordinated Interference Management (Low-Frequency TDD)*) has taken effect, downlink CS/CBF optimization for FWA private line is introduced to allow an FWA private line user in a cooperating cell to participate in interference avoidance for beneficiary users in the serving cell only when the downlink rate of this FWA private line user is higher than the registered rate, as illustrated in **Figure 6-4**. This prevents resource loss of FWA private line users in the cooperating cell. In addition, if fast CBF is enabled, FWA private line users in the overlapping area in the serving cell have more opportunities to obtain downlink CS/CBF scheduling resources. This function is controlled by the **PRIVATE_LINE_DL_CS_CBF_OPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. This function applies only to TDD in low frequency bands.

Figure 6-4 Downlink CS/CBF optimization for FWA private line

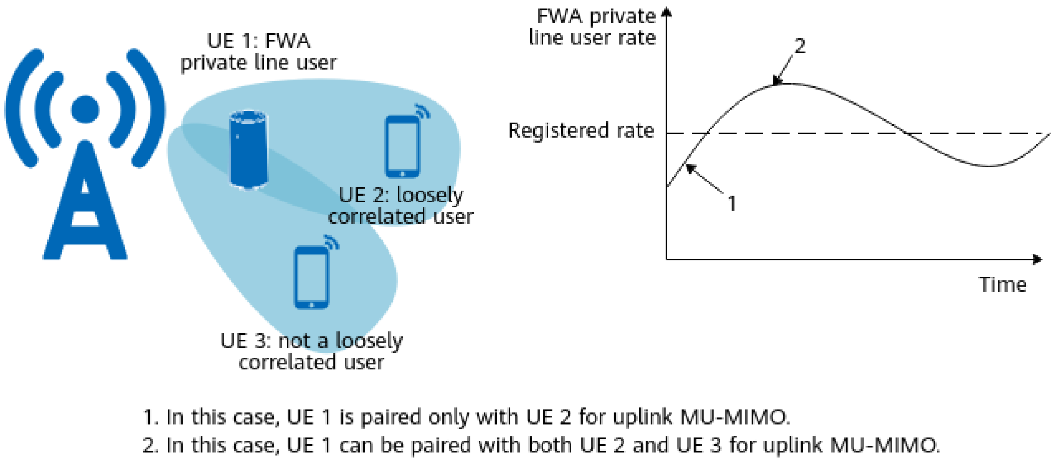


Uplink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

Uplink MU-MIMO implements the spatial multiplexing of time-frequency resources for multiple users during uplink data transmission, increasing uplink and downlink cell capacity and improving spectral efficiency. However, UE pairing for uplink MU-MIMO intensifies inter-user interference, affecting user experience. For cells where PUSCH MU-MIMO (detailed in *MIMO (TDD)*) has taken effect, uplink MU-MIMO optimization for FWA private line is introduced to allow an FWA private line user whose uplink rate is lower than the registered rate to be paired with other loosely correlated users for uplink MU-MIMO, as shown in **Figure 6-5**. This reduces inter-stream interference and improve the uplink throughput of the FWA private line user. This function is controlled by the **PRIVATE_LINE_UL_MUMIMO_OPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. This function applies only to TDD in low frequency bands.

It is recommended that this function be enabled when FWA private line users and common users are densely distributed and the number of FWA private line users is less than half the maximum number of layers that can be paired for uplink MU-MIMO in a cell. For details about the maximum number of paired layers for uplink MU-MIMO, see **6.2.4.3 Network Monitoring**.

Figure 6-5 Uplink MU-MIMO optimization for FWA private line



Downlink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

This function is controlled by the **PRIVATE_LINE_DL_MUMIMO_OPT_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter. After this function is enabled in cells where PDSCH MU-MIMO (detailed in *MIMO (TDD)*) has taken effect, if the downlink rate of an FWA private line user is lower than the registered rate, the FWA private line user is paired with other loosely correlated users for downlink MU-MIMO to reduce inter-stream interference and improve the downlink throughput of the FWA private line user. This function applies only to TDD in low frequency bands.

6.2.2 Network Analysis

6.2.2.1 Benefits

- UL CoMP optimization for FWA private line

  This function improves the average uplink throughput of FWA private line users in overlapping areas. For details, see **6.2.4.3 Network Monitoring**.
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

  This function improves the average downlink throughput of FWA private line users in overlapping areas. For details, see **6.2.4.3 Network Monitoring**.
- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

  This function improves the average uplink throughput of FWA private line users in heavy-load scenarios. For details, see **6.2.4.3 Network Monitoring**.
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

  This function improves the average downlink throughput of FWA private line users in heavy-load scenarios. For details, see **6.2.4.3 Network Monitoring**.

6.2.2.2 Impacts

The impacts between this function and other functions are described in **6.2.3.2.3 Function Impacts**.

This function has the following network impacts:

- UL CoMP optimization for FWA private line

  The number of non-FWA private line users for which UL CoMP takes effect decreases in the overlapping area, and the uplink throughput of non-FWA private line users may decrease.

- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

  The downlink throughput of non-FWA private line users in the overlapping area may decrease.

- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

  The following indicators may decrease: uplink throughput of non-FWA private line users, average number of scheduled users in the uplink in a cell, average uplink cell throughput, number of RBs paired for uplink MU-MIMO, and uplink pairing ratio.

- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

  The following indicators may decrease: downlink throughput of non-FWA private line users, average number of scheduled users in the downlink in a cell, average downlink cell throughput, number of RBs paired for downlink MU-MIMO, and downlink pairing ratio.

NOTE

- Observe the average uplink throughput of users with specific QCIs or 5QIs:
  - Average uplink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.UL.QCI** – **N.ThpVol.UL.SmallPkt.QCI**)/ **N.ThpTime.UL.RmvSmallPkt.QCI**
  - Average uplink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.UL.DuCell5QI** – **N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- Observe the average downlink throughput of users with specific QCIs or 5QIs:
  - Average downlink throughput of QCI-specific bearers in NSA networking = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
  - Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**
- Average number of scheduled users in the uplink = **N.User.Schedule.Ul.Sum**/ (**N.ThpTime.UL.Cell** x 1000)
- Average number of scheduled users in the downlink = **N.User.Schedule.Dl.Sum**/ (**N.ThpTime.DL.Cell** x 1000)
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- Number of RBs paired for uplink MU-MIMO = 1 x **N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + 3 x **N.ChMeas.MIMO.UL.MuPairing.3Layers.RB** + 4 x **N.ChMeas.MIMO.UL.MuPairing.4Layers.RB** + 5 x **N.ChMeas.MIMO.UL.MuPairing.5Layers.RB** + 6 x **N.ChMeas.MIMO.UL.MuPairing.6Layers.RB** + 7 x **N.ChMeas.MIMO.UL.MuPairing.7Layers.RB** + 8 x **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**
- Number of RBs paired for downlink MU-MIMO = 1 x **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + 3 x **N.ChMeas.MIMO.DL.MuPairing.3Layers.RB** + 4 x **N.ChMeas.MIMO.DL.MuPairing.4Layers.RB** + 5 x **N.ChMeas.MIMO.DL.MuPairing.5Layers.RB** + 6 x **N.ChMeas.MIMO.DL.MuPairing.6Layers.RB** + 7 x **N.ChMeas.MIMO.DL.MuPairing.7Layers.RB** + 8 x **N.ChMeas.MIMO.DL.MuPairing.8Layers.RB** + 9 x **N.ChMeas.MIMO.DL.MuPairing.9Layers.RB** + 10 x **N.ChMeas.MIMO.DL.MuPairing.10Layers.RB** + 11 x **N.ChMeas.MIMO.DL.MuPairing.11Layers.RB** + 12 x **N.ChMeas.MIMO.DL.MuPairing.12Layers.RB** + 13 x **N.ChMeas.MIMO.DL.MuPairing.13Layers.RB** + 14 x **N.ChMeas.MIMO.DL.MuPairing.14Layers.RB** + 15 x **N.ChMeas.MIMO.DL.MuPairing.15Layers.RB** + 16 x **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**
- Uplink pairing ratio: **N.ChMeas.MIMO.UL.Pair.Ratio**
- Downlink pairing ratio: **N.ChMeas.MIMO.DL.Pair.Ratio**

6.2.3 Requirements

6.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051205	FWA Private Line	NR0S0FWAPL00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

- UL CoMP optimization for FWA private line

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*
Low-frequency TDD	Basic service assurance for FWA private line	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*

- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** or **INTER_GNB_DL_CS_CBF_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Basic service assurance for FWA private line	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*

- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*
Low-frequency TDD	Basic service assurance for FWA private line	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*

- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*
Low-frequency TDD	Basic service assurance for FWA private line	**PRIVATE_LINE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*

6.2.3.2.2 Mutually Exclusive Functions

None

6.2.3.2.3 Function Impacts

- UL CoMP optimization for FWA private line

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	After UL CoMP optimization for FWA private line is enabled, the number of eMBB users for which UL CoMP takes effect decreases.

- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** or **INTER_GNB_DL_CS_CBF_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	After downlink CS/CBF optimization for FWA private line is enabled, interference avoidance does not take effect when the rates of FWA private line users in the cooperating cell are lower than their guaranteed rates. In this case, the benefits of coordinated interference management decrease.
Low-frequency TDD	Fast CBF	**QUICK_CBF_SW** option of the **NRDUCellComp**.***CsCbfAlgoSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	

- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	After uplink MU-MIMO optimization for FWA private line is enabled, the uplink throughput of eMBB users in the cell and the number of paired layers for uplink MU-MIMO in the cell may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	After uplink MU-MIMO optimization for FWA private line is enabled, if **NRDUCellRes**.***ResStrategyType*** is set to **MEAN_ONLY** in NR DU cell resource management between network slices, resources may become limited, thus decreasing the data rate of FWA private line users.

- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	After downlink MU-MIMO optimization for FWA private line is enabled, the downlink throughput of eMBB users in the cell and the number of paired layers for downlink MU-MIMO in the cell may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	After downlink MU-MIMO optimization for FWA private line is enabled, if **NRDUCellRes**.***ResStrategyType*** is set to **MEAN_ONLY** in NR DU cell resource management between network slices, resources may become limited, thus decreasing the user-perceived rate of FWA private line users.

6.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

- UL CoMP optimization for FWA private line

  In NR TDD, all NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

  All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

  All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

  All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- UL CoMP optimization for FWA private line

  All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

  All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)

  All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)

  All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

UL CoMP Optimization for FWA Private Line

Table 6-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	PRIVATE_LINE_UL_COMP_OPT_SW	Select this option.

Downlink CS/CBF Optimization for FWA Private Line (Low-Frequency TDD)

Table 6-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	PRIVATE_LINE_DL_CS_CBF_OPT_SW	Select this option.

Uplink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

Table 6-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	PRIVATE_LINE_UL_MUMIMO_OPT_SW	Select this option.

Downlink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

Table 6-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	PRIVATE_LINE_DL_MUMIMO_OPT_SW	Select this option.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to **6.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- UL CoMP optimization for FWA private line
```
//Enabling UL CoMP optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_UL_COMP_OPT_SW-1;
```
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)
```
//Enabling downlink CS/CBF optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_DL_CS_CBF_OPT_SW-1;
```
- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)
```
//Enabling uplink MU-MIMO optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
FwaAlgoSwitch=PRIVATE_LINE_UL_MUMIMO_OPT_SW-1;
```
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)
```
//Enabling downlink MU-MIMO optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
FwaAlgoSwitch=PRIVATE_LINE_DL_MUMIMO_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- UL CoMP optimization for FWA private line
```
//Disabling UL CoMP optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_UL_COMP_OPT_SW-0;
```
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)
```
//Disabling downlink CS/CBF optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=PRIVATE_LINE_DL_CS_CBF_OPT_SW-0;
```
- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)
```
//Disabling uplink MU-MIMO optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
FwaAlgoSwitch=PRIVATE_LINE_UL_MUMIMO_OPT_SW-0;
```
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)
```
//Disabling downlink MU-MIMO optimization for FWA private line
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
FwaAlgoSwitch=PRIVATE_LINE_DL_MUMIMO_OPT_SW-0;
```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

UL CoMP Optimization for FWA Private Line

Observe the CHRs and performance counters described in **6.2.4.3 Network Monitoring** to check whether this function has taken effect.

Downlink CS/CBF Optimization for FWA Private Line (Low-Frequency TDD)

Observe the CHRs described in **6.2.4.3 Network Monitoring** to check whether this function has taken effect.

Uplink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

Observe the CHRs and performance counters described in **6.2.4.3 Network Monitoring** to check whether this function has taken effect.

Downlink MU-MIMO Optimization for FWA Private Line (Low-Frequency TDD)

Observe the performance counters described in **6.2.4.3 Network Monitoring** to check whether this function has taken effect.

6.2.4.3 Network Monitoring

- UL CoMP optimization for FWA private line
  - Use external CHRs to monitor the average uplink throughput of FWA private line users in overlapping areas:

    Subscribe to both the **PERIOD_PRIVATE_UE_MEASUREMENT** and **PERIOD_PRIVATE_THROUGHPUT_MEASUREMENT** events, which are associated by **ChrHeader** > **fixed header** > **CallId**. Filter the following statistical items with the QCI-specific service type being **FWA Private Line** and the **OverlapProperty** field being **1** or **2** (indicating that a user is in the overlapping area): **ULThpVol**, **ULSmallPktThpVol**, and **ULThpTimeRmvSmallPkt**.

    Average uplink throughput of FWA private line users = (**ULThpVol** – **ULSmallPktThpVol**)/**ULThpTimeRmvSmallPkt**
  - Use relevant performance counters to monitor the average uplink throughput of FWA private line users:
    - NSA networking:

      Average uplink throughput of QCI-specific bearers for FWA private line services = (**N.ThpVol.UL.QCI – N.ThpVol.UL.SmallPkt.QCI**)/ **N.ThpTime.UL.RmvSmallPkt.QCI**
    - SA networking:

      Average uplink throughput of 5QI-specific bearers for FWA private line services = (**N.ThpVol.UL.DuCell5QI – N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- Downlink CS/CBF optimization for FWA private line (low-frequency TDD)

  Use external CHRs to monitor the average downlink throughput of FWA private line users in overlapping areas:

  Subscribe to both the **PERIOD_PRIVATE_UE_MEASUREMENT** and **PERIOD_PRIVATE_THROUGHPUT_MEASUREMENT** events, which are associated by **ChrHeader** > **fixed header** > **CallId**. Filter the following statistical items with the QCI-specific service type being **FWA Private Line** and the **OverlapProperty** field being **1** or **2** (indicating that a user is in the overlapping area): **DLThpVol**, **DLThpVolLastSlot**, and **DLThpTimeRmvLastSlot**.

Average downlink throughput of FWA private line users = (**DLThpVol** – **DLThpVolLastSlot**)/**DLThpTimeRmvLastSlot**

- Uplink MU-MIMO optimization for FWA private line (low-frequency TDD)
  - On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **Cell Performance Monitoring** to observe **Max Number of UL Transmission Pair Layers**.
  - Observe the maximum number of uplink layers on PRBs in a cell, which is measured by **N.ChMeas.MIMO.UL.Trans.Layer.Max**.
  - Use external CHRs to monitor the average uplink throughput of FWA private line users:

    Subscribe to both the **PERIOD_PRIVATE_UE_MEASUREMENT** and **PERIOD_PRIVATE_THROUGHPUT_MEASUREMENT** events, which are associated by **ChrHeader** > **fixed header** > **CallId**. Filter the following statistical items with the QCI-specific service type being **FWA Private Line**: **ULThpVol**, **ULSmallPktThpVol**, and **ULThpTimeRmvSmallPkt**.

    Average uplink throughput of FWA private line users = (**ULThpVol** – **ULSmallPktThpVol**)/**ULThpTimeRmvSmallPkt**
  - Use relevant performance counters to monitor the average uplink throughput of FWA private line users:
    - NSA networking:

      Average uplink throughput of QCI-specific bearers for FWA private line services = (**N.ThpVol.UL.QCI** – **N.ThpVol.UL.SmallPkt.QCI**)/**N.ThpTime.UL.RmvSmallPkt.QCI**
    - SA networking:

      Average uplink throughput of 5QI-specific bearers for FWA private line services = (**N.ThpVol.UL.DuCell5QI** – **N.ThpVol.UL.SmallPkt.DuCell5QI**)/**N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- Downlink MU-MIMO optimization for FWA private line (low-frequency TDD)
  - On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **Cell Performance Monitoring** to observe **Max Number of DL Transmission Pair Layers**.
  - Observe the maximum number of downlink layers on PRBs in a cell, which is measured by **N.ChMeas.MIMO.DL.Transmission.Layer.Max**.
  - Use relevant performance counters to monitor the average downlink throughput of FWA private line users:
    - NSA networking:

      Average downlink throughput of QCI-specific bearers for FWA private line services = (**N.ThpVol.DL.QCI** – **N.ThpVol.DL.LastSlot.QCI**)/**N.ThpTime.DL.RmvLastSlot.QCI**
    - SA networking:

      Average downlink throughput of 5QI-specific bearers for FWA private line services = (**N.ThpVol.DL.DuCell5QI** –

N.ThpVol.DL.LastSlot.DuCell5QI)/
N.ThpTime.DL.RmvLastSlot.DuCell5QI

7 FWA Service Turbo

FWA Service Turbo includes device-pipe synergy preallocation and rate adaptation for device-pipe synergy. FWA Service Turbo is controlled by the **FWA_SERVICE_TURBO_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter.

7.1 Device-Pipe Synergy Preallocation

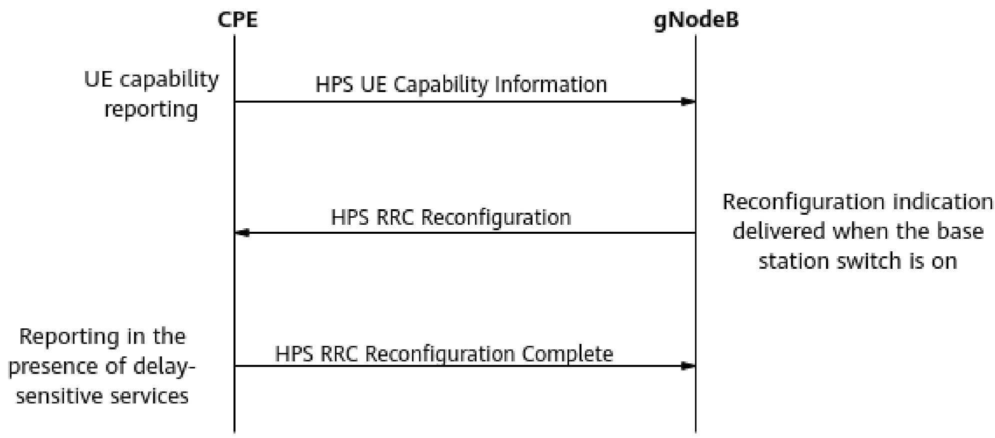
7.1.1 Principles

To improve user experience of delay-sensitive services, the FWA solution introduces the device-pipe synergy preallocation function.

This function requires that the base station and UEs complete the device-pipe identification procedure. For details about this procedure, see *Device-Pipe Synergy*. This document describes only processing exclusive to device-pipe synergy preallocation in comparison to basic device-pipe identification. **Figure 7-1** illustrates the related message exchange.

1. UE capability reporting: A UE reports its device-pipe synergy preallocation capability to the gNodeB through the **hps-smartPreSchedule** field in an HPS UE Capability Information message.
2. Function status notification: The gNodeB sends an HPS RRC Reconfiguration message to notify the UE of the device-pipe synergy preallocation function status and related configuration information. If device-pipe synergy preallocation is enabled, this function takes effect after UEs supporting this function receive this message.
3. Function activation response: If device-pipe synergy preallocation successfully takes effect on the UE, the UE sends an HPS RRC Reconfiguration Complete message as a response to the gNodeB.

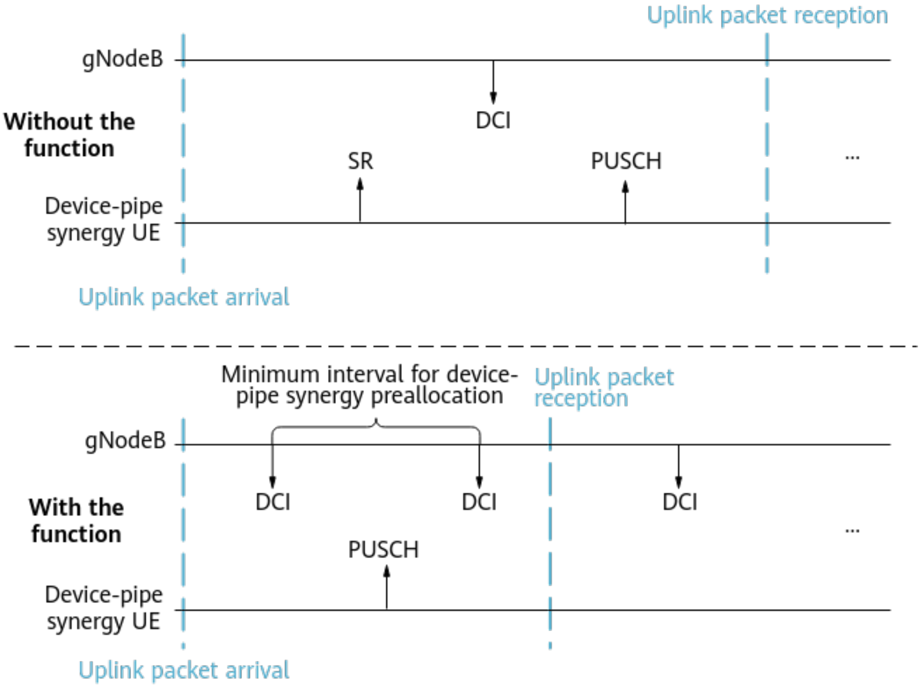
Figure 7-1 Message exchange for device-pipe synergy



The principles for device-pipe synergy preallocation to take effect are as follows:

- When a device-pipe synergy UE performs delay-sensitive services, the UE proactively sends a device-pipe synergy preallocation request to the gNodeB. **The gNodeB increases the uplink scheduling priority of the UE so that uplink preallocation can be preferentially performed for the UE**, as shown in **Figure 7-2**. Regardless of whether a UE has service requests and whether there are remaining resources, the gNodeB proactively schedules the UE at a specified interval to reduce the duration from the time when the UE sends an SRI to the time when the UE obtains the uplink scheduling grant.
- When a device-pipe synergy UE does not perform delay-sensitive services, the UE proactively sends a request for stopping device-pipe synergy preallocation to the gNodeB, and the gNodeB no longer makes this function take effect on the UE.

Figure 7-2 Principles of device-pipe synergy preallocation



This function is controlled by the **UL_PREFERENTIAL_SCH_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. **Table 7-1** describes the parameters related to this function.

Table 7-1 Parameters related to device-pipe synergy preallocation

Parameter Name	Parameter ID	Description
UE Cooperation Min Preallocation Period	**NRDUCellFwaQos**.***UeCoopMinPreallocPeriod***	This parameter specifies the minimum period for device-pipe synergy preallocation.
UE Cooperation Max Preallocation RB Ratio	**NRDUCellFwaQos**.***UeCoopMaxPreallocRbRatio***	This parameter specifies the maximum percentage of RBs that can be used for device-pipe synergy preallocation. After the percentage of RBs used by device-pipe synergy UEs reaches the value of this parameter, the base station no longer preferentially schedules these UEs.
Preallocation Size	**NRDUCellPusch**.***PreallocationSize***	This parameter specifies the amount of data to be scheduled in device-pipe synergy preallocation.

This function applies only to SA networking.

NOTE

- For details about the device-pipe identification procedure, see *Device-Pipe Synergy*.
- If a device-pipe synergy UE does not perform any uplink service for 60 ms, device-pipe synergy preallocation does not take effect for the UE in the next 60 ms. Otherwise, device-pipe synergy preallocation takes effect for the UE.
- Preferential preallocation can be performed for a maximum of two device-pipe synergy UEs in each slot.
- It is recommended that this function be enabled on the default bearer.

7.1.2 Network Analysis

7.1.2.1 Benefits

After this function is enabled, the uplink service delay of device-pipe synergy UEs decreases when the uplink CCE allocation failure rate is less than 30% and the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is less than 90%.

The benefits offered by device-pipe synergy preallocation decrease if this function is enabled together with common preallocation functions (uplink basic preallocation, uplink smart preallocation, and QCI-specific uplink preallocation) for cells with light load. Therefore, it is recommended that this function be enabled for cells with medium or heavy load.

NOTE

Uplink CCE allocation failure rate = 100 x (1 – (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**)

7.1.2.2 Impacts

The impacts between this function and other functions are described in **7.1.3.2.3 Function Impacts**.

This function has the following network impacts: After this function takes effect, uplink preallocation preferentially takes effect for device-pipe synergy UEs. As a result, the uplink data rates of other UEs cannot reach the peak value. When such UEs account for a large proportion or have heavy uplink traffic, the average uplink throughput of other UEs decreases.

7.1.3 Requirements

7.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081220	FWA Service Turbo	NR0S0DPSFSA0	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	**SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter	*Device-Pipe Synergy*	Device-pipe synergy preallocation requires that device-pipe identification be enabled.
FWA Service Turbo	**FWA_SERVICE_TURBO_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Device-pipe synergy preallocation requires that the switch for FWA Service Turbo be turned on.

7.1.3.2.2 Mutually Exclusive Functions

None

7.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	UEs with device-pipe synergy preallocation in effect may fail to enter the sleep time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink preallocation	**UL_PREALLOCATION_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter	*Service-differentiated Experience Guarantee*	QCI-based uplink preallocation does not take effect on a UE for which device-pipe synergy preallocation is in effect.
Low-frequency TDD	Uplink basic preallocation	**UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	Uplink basic preallocation does not take effect on a UE for which device-pipe synergy preallocation is in effect.
Low-frequency TDD	Uplink smart preallocation	**UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	Uplink smart preallocation does not take effect on a UE for which device-pipe synergy preallocation is in effect.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs, device-pipe synergy preallocation takes effect only on the PCC but not on the SCCs.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs, device-pipe synergy preallocation takes effect only on the PCC but not on the SCCs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When both user PRB control for FWA and device-pipe synergy preallocation are enabled, the delay of uplink delay-sensitive services may increase for device-pipe synergy UEs.

7.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.4 Others

This function requires that UEs support device-pipe synergy preallocation. The **hps-smartPreSchedule** field indicates whether a UE supports this function. If the value is **supported**, the UE supports this function.

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	If device-pipe synergy preallocation is required, select the **FWA_SERVICE_TURBO_SW** option.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	If device-pipe synergy preallocation is required and there are UEs capable of device-pipe synergy on the network, select the **UL_PREFERENTIAL_SCH_SW** option.
UE Cooperation Min Preallocation Period	**NRDUCellFwaQos**.***UeCoopMinPreallocPeriod***	The default value is recommended.
UE Cooperation Max Preallocation RB Ratio	**NRDUCellFwaQos**.***UeCoopMaxPreallocRbRatio***	The default value is recommended.
Preallocation Size	**NRDUCellPusch**.***PreallocationSize***	Set this parameter based on the network plan.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Switch for FWA Service Turbo

```
//Turning on the switch for FWA Service Turbo
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_SERVICE_TURBO_SW-1;
```

- Device-pipe synergy preallocation

```
//Enabling device-pipe synergy preallocation for users with a QCI of 9 as an example
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=UL_PREFERENTIAL_SCH_SW-1;
```

```
//Modifying the minimum preallocation period for UE cooperation and the maximum percentage of RBs that can be preallocated
MOD NRDUCELLFWAQOS: NrDuCellId=1, UeCoopMinPreallocPeriod=5,
UeCoopMaxPreallocRbRatio=10;
//Modifying the amount of data to be scheduled in preallocation
MOD NRDUCELLPUSCH: NrDuCellId=1, PreallocationSize=100;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Device-pipe synergy preallocation
```
//Disabling device-pipe synergy preallocation for UEs with a QCI of 9
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9,
ServiceDiffAlgoSwitch=UL_PREFERENTIAL_SCH_SW-0;
```
- Switch for FWA Service Turbo
```
//Turning off the switch for FWA Service Turbo
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_SERVICE_TURBO_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

If the value of the **N.PRB.UL.Prealloc.UeCoop.Used.Avg** counter changes from 0 to a non-zero value after this function is enabled, this function has taken effect.

7.1.4.3 Network Monitoring

After this function is enabled, you can monitor the running status of this function by observing the change in the average number of PRBs used by UEs involved in device-pipe synergy preallocation (indicated by the **N.PRB.UL.Prealloc.UeCoop.Used.Avg** counter).

7.2 Rate Adaptation for Device-Pipe Synergy

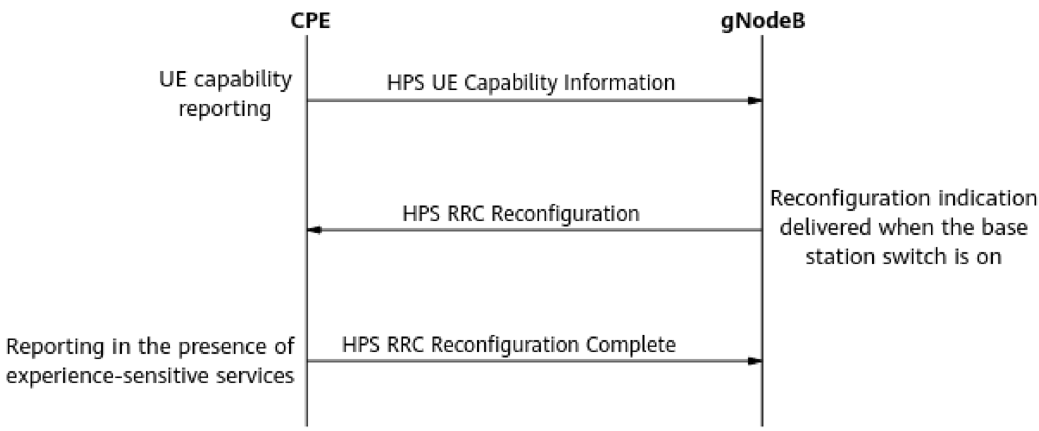
7.2.1 Principles

To improve user experience of specific services, such as video services and other experience-sensitive services, the FWA solution introduces rate adaptation for device-pipe synergy. This function is configured based on QCIs and supported only in SA networking.

This function requires that the base station and UEs complete the device-pipe identification procedure. For details about this procedure, see *Device-Pipe Synergy*. This document describes only processing exclusive to rate adaptation for device-pipe synergy and basic device-pipe identification. **Figure 7-3** illustrates the related message exchange.

1. UE capability reporting: A UE reports its rate adaptation for device-pipe synergy capability to the gNodeB through the **hps-support-RateAdaption** field in an HPS UE Capability Information message.
2. Function status notification: The base station sends an HPS RRC Reconfiguration message to notify the UE of the function status and related configuration information. If rate adaptation for device-pipe synergy is enabled, this function takes effect after UEs supporting this function receive this message.
3. Function activation response: If rate adaptation for device-pipe synergy successfully takes effect on the UE, the UE sends an HPS RRC Reconfiguration Complete message as a response to the gNodeB.

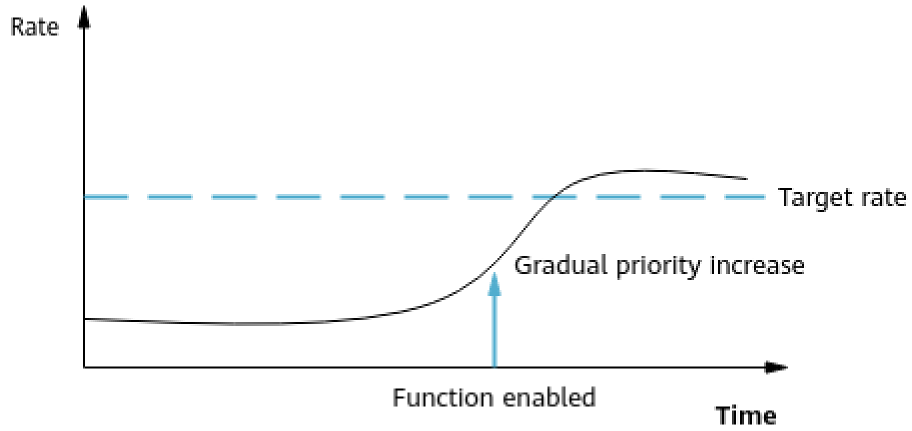
Figure 7-3 Message exchange for device-pipe synergy



This function is controlled by the **RATE_ADAPTION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. The principles for this function to take effect are as follows:

- When a UE is performing a specific service, the UE proactively sends a request to the gNodeB. Then the gNodeB enables rate adaptation for device-pipe synergy for the UE. For QCI-specific bearers with this function enabled, the base station measures the downlink rate of these bearers in real time. This function includes the following optimizations:
  - **Rate guarantee for specific services**, as illustrated in **Figure 7-4**.
    - If the data rate of specific services on a bearer does not reach the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** parameter, the downlink scheduling priority is gradually raised by a maximum multiple not higher than the **NRDUCellFwaQos**.***ExpBasedSchMaxWeight*** parameter value.
    - When the data rate of specific services on a bearer reaches or exceeds the value of the **gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate*** parameter, the downlink scheduling priority is gradually lowered until the initial scheduling priority resumes.

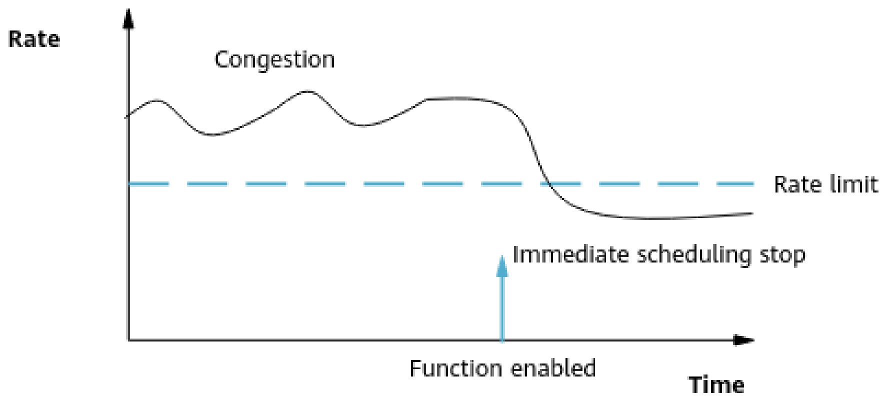
Figure 7-4 Rate guarantee for specific services



– **Rate limitation for specific services during cell congestion**, as illustrated in **Figure 7-5**.

  When the data rate of specific services on a bearer exceeds the **NRDUCellQciBearer**.***DlServiceDiffRateLimit*** parameter value in a cell meeting the congestion conditions, downlink scheduling for these specific services stops and rate limitation is performed. (Congestion conditions: Average downlink PRB usage of the cell within a monitoring period ≥ **NRDUCellFwaQos**.***DlPrbUsageThreshold*** parameter value; and Average number of active UEs in the downlink within a monitoring period ≥ **NRDUCellFwaQos**.***UserNumThreshold*** parameter value)

Figure 7-5 Rate limitation for specific services during cell congestion



- When a UE does not perform any specific service, the UE proactively reports a request to the gNodeB, and the gNodeB no longer makes this function take effect on the UE.

Table 7-3 lists the adaptation procedure of rate guarantee for specific services when rate adaptation for device-pipe synergy is enabled in CA scenarios. For details about CA, see *Carrier Aggregation*.

Table 7-3 Adaptation procedure for rate guarantee for specific services in CA scenarios

Scenario	Adaptation Procedure
Rate adaptation for device-pipe synergy is enabled and downlink experience-based scheduling (detailed in **5.3 Downlink Experience-based Scheduling**) is disabled.	The PCC compares the configured target guaranteed rate with the real-time RLC rate, calculates the multiple by which the scheduling priority increases, and sends the multiple to the SCCs to adjust the downlink scheduling priority.
Rate adaptation for device-pipe synergy and downlink experience-based scheduling are both enabled.	The adaptation procedure is the same as that for downlink experience-based scheduling in CA scenarios. For details, see **CA Scenarios**.

7.2.2 Network Analysis

7.2.2.1 Benefits

- Rate guarantee for specific services: After this function is enabled, the downlink user-perceived rate of device-pipe synergy UEs increases when the UEs are configured with downlink guaranteed rates.
- Rate limitation for specific services during cell congestion: After this function is enabled, when a cell meets the congestion conditions (detailed in **7.2.1 Principles**) and the downlink rates of specific service exceeds the rate limit, rate limitation is performed on the specific services to spare more resources for other services with poor user experience. As such, the downlink throughput of specific user bearers is less likely to fall into low ranges.

Downlink user-perceived rate of device-pipe synergy UEs = (**N.ThpVol.DL.UeCoop** – **N.ThpVol.DL.LastSlot.UeCoop**)/**N.ThpTime.DL.RmvLastSlot.UeCoop**

Counters for measuring the downlink throughput ranges of 5QI-specific bearers in SA networking: **N.Thp.DL.Samp.DuCell5QI.Index0** to **N.Thp.DL.Samp.DuCell5QI.Index14**

7.2.2.2 Impacts

The impacts between this function and other functions are described in **7.2.3.2.3 Function Impacts**.

This function has the following network impacts:

- Rate guarantee for specific services: After this function is enabled, when the cell load is heavy, this function preferentially guarantees users configured with target guaranteed rates. As such, fewer resources can be allocated to users not configured with target guaranteed rates, and the average downlink throughput of these users decreases.
- Rate limitation for specific services during cell congestion: After this function is enabled, when a cell meets the congestion conditions (detailed in **7.2.1**

Principles) and the downlink rate of specific services exceeds the rate limit, rate limitation is performed on the specific services. As such, the average downlink throughput of users configured with downlink rate limit decreases.

NOTE

Average downlink throughput of 5QI-specific bearers in SA networking = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

7.2.3 Requirements

7.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081220	FWA Service Turbo	NR0S0DPSFSA0	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Switch for FWA Service Turbo	**FWA_SERVICE_TURBO_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	Rate adaptation for device-pipe synergy requires that the switch for FWA Service Turbo be turned on.

Function Name	Function Switch	Reference	Description
Device-pipe identification	**SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter	*Device-Pipe Synergy*	Rate adaptation for device-pipe synergy requires that device-pipe identification be enabled.

7.2.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
DACQ Switch	**DACQ_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	None	Rate adaptation for device-pipe synergy cannot work with DACQ.

7.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	**DL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When rate adaptation for device-pipe synergy and downlink experience-based scheduling are both enabled, downlink experience-based scheduling preferentially takes effect.
Low-frequency TDD	Downlink experience satisfaction guarantee	**DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos**.***SchAlgoSwitch*** parameter	*FWA*	Downlink experience satisfaction guarantee preferentially takes effect when it is enabled together with rate adaptation for device-pipe synergy.

7.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.2.3.4 Others

This function requires that UEs be capable of rate adaptation for device-pipe synergy. The **hps-support-RateAdaption** field in the RRC_HPS_UE_CAP_INFO message indicates whether a UE supports the rate adaptation capability. If the value of this field is **supported**, the UE supports the rate adaptation capability.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FWA Algorithm Switch	**NRDUCellAlgoSwitch**.***FwaAlgoSwitch***	Select the **FWA_SERVICE_TURBO_SW** option if rate adaptation for device-pipe synergy is required.

Parameter Name	Parameter ID	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	You are advised to select the **RATE_ADAPTION_SW** option when there are UEs capable of device-pipe synergy on the network.
QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	Set this parameter based on the network plan.
DL Exp-based Scheduling Guaranteed User Rate	**gNBDUMacParamGroup**.***DlExpBasedSchGuarUserRate***	Set this parameter based on the network plan.
Exp-based Scheduling Maximum Weight	**NRDUCellFwaQos**.***ExpBasedSchMaxWeight***	The default value is recommended.
Downlink PRB Usage Threshold	**NRDUCellFwaQos**.***DlPrbUsageThreshold***	Set this parameter based on the network plan.
Downlink Service Differentiation Rate Limit	**NRDUCellQciBearer**.***DlServiceDiffRateLimit***	Set this parameter based on the network plan.
User Number Threshold	**NRDUCellFwaQos**.***UserNumThreshold***	The default value is recommended.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to **7.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Switch for FWA Service Turbo

```
//Turning on the switch for FWA Service Turbo
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_SERVICE_TURBO_SW-1;
```

- Rate adaptation for device-pipe synergy

```
//Enabling rate adaptation for device-pipe synergy for users with a QCI of 9 as an example
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=RATE_ADAPTION_SW-1;
```

```
//Configuring the downlink guaranteed rate
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, DlExpBasedSchGuarUserRate=50000;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, MacParamGroupId=9;
//Configuring the maximum experience-based scheduling weight
MOD NRDUCELLFWAQOS: NrDuCellId=1, ExpBasedSchMaxWeight=50;
//Configuring the downlink PRB usage threshold and user number threshold
MOD NRDUCELLFWAQOS: NrDuCellId=1, DlPrbUsageThreshold=60, UserNumThreshold=2;
//Configuring the rate limit for downlink service differentiation
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, DlServiceDiffRateLimit=4;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Rate adaptation for device-pipe synergy

```
//Disabling rate adaptation for device-pipe synergy for users with a QCI of 9
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=9, ServiceDiffAlgoSwitch=RATE_ADAPTION_SW-0;
```

- Switch for FWA Service Turbo

```
//Turning off the switch for FWA Service Turbo
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, FwaAlgoSwitch=FWA_SERVICE_TURBO_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

Step 3 If the RRC_HPS_CONN_RECFG message contains the **hps-RateAdaption** field, this function has taken effect.

----End

7.2.4.3 Network Monitoring

After this function is enabled, monitor its running status by checking the increase in the downlink user-perceived rate of device-pipe synergy UEs.

Downlink user-perceived rate of device-pipe synergy UEs = (**N.ThpVol.DL.UeCoop** – **N.ThpVol.DL.LastSlot.UeCoop**)/**N.ThpTime.DL.RmvLastSlot.UeCoop**

8 FWA RedCap

To reduce terminal costs, the FWA solution introduces RedCap CPEs defined in 3GPP Release 17. Compared with traditional CPEs, RedCap CPEs have the following characteristics:

- They can work only in SA networking.
- They only support up to 20 MHz bandwidth, which is smaller.
- They are equipped with fewer antennas (1T2R).
- They do not support CA.

FWA users mainly perform large-packet services in the downlink. This may cause the following issues when there are both eMBB users and FWA RedCap users:

- FWA RedCap users have insufficient RB resources when the network is congested because they share and compete for time-frequency resources with eMBB users.
- RedCap users are more vulnerable to PDCCH CCE allocation failures caused by CCE insufficiency because they support a smaller bandwidth.

To improve the downlink experience of FWA RedCap users, you can enable "SRS antenna selection for RedCap users in FWA scenarios". You can also enable functions under the feature RedCap Downlink Experience Improvement (NR TDD), which is detailed in chapter "RedCap Downlink Experience Improvement (NR TDD)" in *RedCap*.

Other RedCap functions to enhance experience, capacity, and coverage also apply to FWA scenarios. For details about these functions, see chapters "RedCap Introduction" and "Premium RedCap" in *RedCap*.

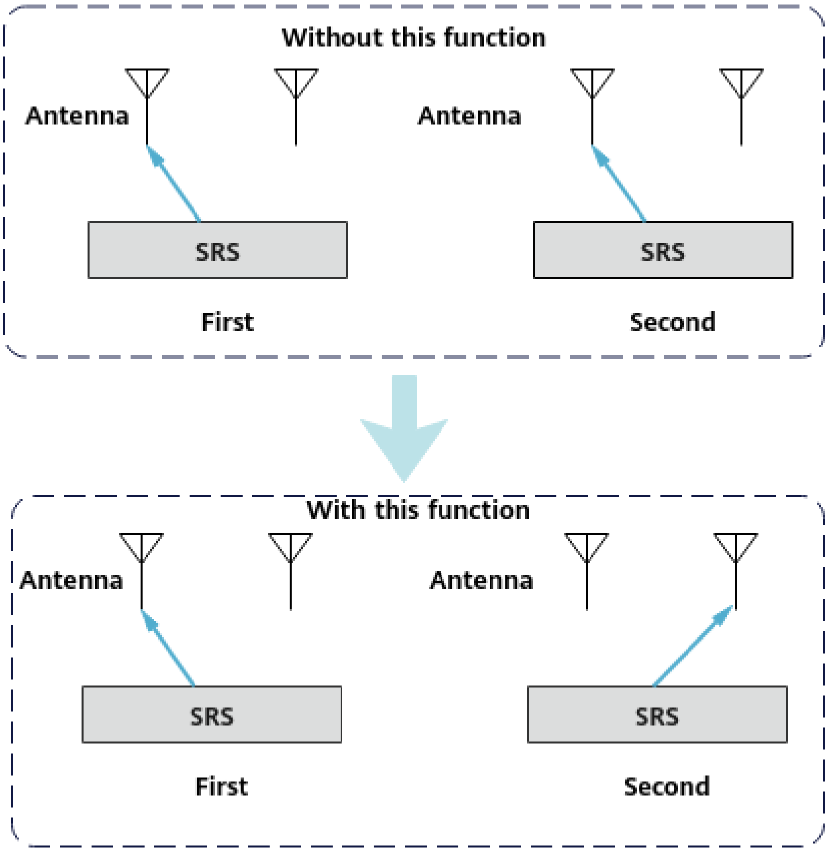
8.1 SRS Antenna Selection for RedCap Users in FWA Scenarios (Low-Frequency TDD)

8.1.1 Principles

In TDD, a base station can obtain downlink channel state information from SRS measurement due to uplink and downlink channel reciprocity. However, as a RedCap CPE always uses the same transmit antenna to send SRSs, the base station cannot obtain complete downlink channel information, which affects downlink user experience. As such, SRS antenna selection for RedCap users in FWA scenarios is introduced. After this function is enabled, **the base station identifies FWA users by 5QI and allocates 1T2R SRS antenna selection resources to 1T2R RedCap CPEs identified as FWA users, so that the base station can receive SRSs from such users over different antennas at different time points and obtain complete channel information**. This improves the accuracy of channel estimation and betters downlink user experience of RedCap CPEs. **Figure 8-1** illustrates this function.

SRS antenna selection for RedCap users in FWA scenarios is controlled by the **REDCAP_SRS_ANT_SWITCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter. This function applies only to SA networking and takes effect only in TDD massive MIMO cells. It is recommended that this function be enabled on FWA networks where RedCap CPE number allocation is available.

Figure 8-1 SRS antenna selection for RedCap users in FWA scenarios



8.1.2 Network Analysis

8.1.2.1 Benefits

After this function is enabled, the average downlink throughput of RedCap FWA users that support 1T2R antenna selection increases.

If RedCap FWA users that support 1T2R antenna selection are located at the cell edge (with the RSRP less than –105 dBm), this function may not deliver any benefits.

In peak rate scenarios, this function does not deliver any benefits because the data rates of RedCap FWA users that support 1T2R antenna selection have reached their peak values.

8.1.2.2 Impacts

The impacts between this function and other functions are described in **8.1.3.2.3 Function Impacts**.

This function has the following network impacts: After this function is enabled, the average uplink or downlink throughput of eMBB users may decrease. In addition, the average uplink/downlink cell throughput (**Cell Uplink Average Throughput (DU)**/**Cell Downlink Average Throughput (DU)**) and average uplink/downlink user throughput (**User Uplink Average Throughput (DU)**/**User Downlink Average Throughput (DU)**) fluctuate.

NOTE

The average uplink/downlink throughput of eMBB users can be calculated using the following 5QI-specific indicators:

- Average uplink throughput of 5QI-specific bearers = (**N.ThpVol.UL.DuCell5QI** – **N.ThpVol.UL.SmallPkt.DuCell5QI**)/**N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- Average downlink throughput of 5QI-specific bearers = (**N.ThpVol.DL.DuCell5QI** – **N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**

8.1.3 Requirements

8.1.3.1 Licenses

None

8.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.1.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
UE-characteristic-based SRS period adaptation	**USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	SRS antenna selection for RedCap users in FWA scenarios depends on UE-characteristic-based SRS period adaptation.
SRS interference coordination based on self-contained slots	**SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	SRS antenna selection for RedCap users in FWA scenarios requires SRS interference coordination based on self-contained slots.
Scenario-specific SRS resource allocation policy	**NRDUCellSrs**.***ScenarioSrsResAllocPol***	*Channel Management*	SRS antenna selection for RedCap users in FWA scenarios requires that **NRDUCellSrs**.***ScenarioSrsResAllocPol*** be set to **CAPC_PRI**.
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	SRS antenna selection for RedCap users in FWA scenarios requires RedCap Introduction.
FWA user identification	**gNBQciBearer**.***ServiceType***	*FWA*	FWA-related service types must be configured for the FWA user identification function. Otherwise, SRS antenna selection for RedCap users in FWA scenarios does not take effect.

8.1.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
UE bandwidth adaptation	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*	SRS antenna selection for RedCap users in FWA scenarios can be enabled only when either of the following conditions is met: ● UE bandwidth adaptation is disabled. ● UE bandwidth adaptation is enabled and **NRDUCellBwp**.***DlMinCarrierBw*** is set to **20M**.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot be enabled together with SRS antenna selection for RedCap users in FWA scenarios.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper Cell cannot be enabled together with SRS antenna selection for RedCap users in FWA scenarios.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell cannot be enabled together with SRS antenna selection for RedCap users in FWA scenarios.

8.1.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Non-overlapping between gap and NR resources	**GAP_AVOID_NR_RES_SW** option of the **gNBMobilityCommParam**.***MeasAlgoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When SRS antenna selection for RedCap users in FWA scenarios is enabled, it is good practice to enable also non-overlapping between gap and NR resources to stagger gap and NR resources and prevent resource conflicts from rendering negative results for cell edge users.

8.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R or 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.1.3.4 Cells

The TX/RX mode of the cell must be 32T32R or 64T64R.

8.1.3.5 Others

The UE capability must meet both of the following conditions:

- The value of the **supportOfRedCap-r17** field in **RedCapParameters** is **supported**.
- The value of the **supportedSRS-TxPortSwitch** field in **BandCombinationList** is **t1r2**.

8.1.4 Operation and Maintenance

8.1.4.1 Data Configuration

8.1.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
RedCap Algorithm Switch	**NRDUCellRedCap**.*RedCapAlgoSwitch*	REDCAP_SRS_ANT_SWITCH_SW	• Select this option when there are RedCap CPEs identified as FWA users and their downlink experience needs to be improved. • It is good practice to deselect this option when the downlink experience of eMBB users needs to be guaranteed.

8.1.4.1.2 Using MML Commands

Before using MML commands, refer to **8.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SRS antenna selection for RedCap users in FWA scenarios
MOD NRDUCELLREDCAP: NrDuCellId=1, RedCapAlgoSwitch=REDCAP_SRS_ANT_SWITCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling SRS antenna selection for RedCap users in FWA scenarios
MOD NRDUCELLREDCAP: NrDuCellId=1, RedCapAlgoSwitch=REDCAP_SRS_ANT_SWITCH_SW-0;
```


8.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.1.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap users to check whether this function has taken effect.

8.1.4.3 Network Monitoring

After this function is enabled, you can monitor the running status of this function by observing the increase in the average downlink throughput of RedCap users and the decrease in the average uplink/downlink throughput of eMBB users.

The following 5QI-related counters can be used to calculate the average downlink throughput of RedCap users and average uplink/downlink throughput of eMBB users:

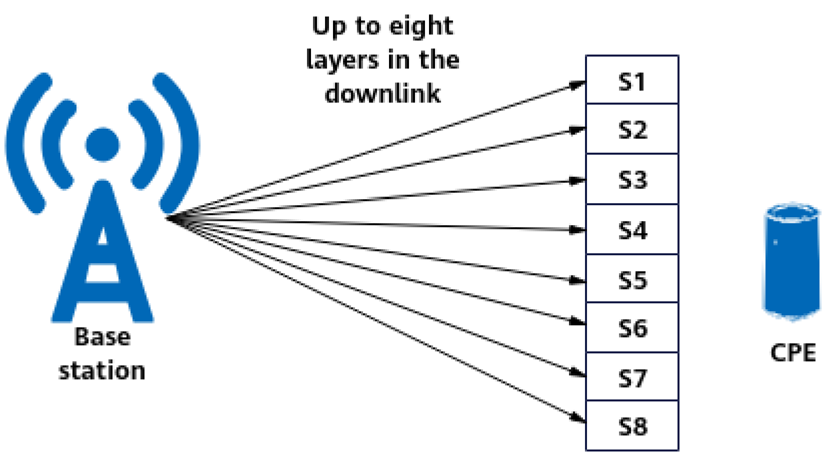
- Average downlink throughput of RedCap users = (**N.ThpVol.DL.RedCap – N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**
- Average uplink throughput of 5QI-specific bearers = (**N.ThpVol.UL.DuCell5QI – N.ThpVol.UL.SmallPkt.DuCell5QI**)/ **N.ThpTime.UL.RmvSmallPkt.DuCell5QI**
- Average downlink throughput of 5QI-specific bearers = (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/ **N.ThpTime.DL.RmvLastSlot.DuCell5QI**

9 Downlink 8-Layer Transmission for a Single FWA User

9.1 Principles

Downlink 8-layer transmission for a single FWA user is designed for UEs capable of downlink eight-layer transmission. When channel conditions permit, **this function uses multi-antenna technologies to enable spatial multiplexing of time-frequency resources on the PDSCH for a single UE. A maximum of eight layers of data can be transmitted in the downlink for the UE, increasing the single-UE peak rate. Figure 9-1 illustrates this function.**

Figure 9-1 Downlink 8-layer transmission for a single FWA user



For low-frequency NR TDD cells with 32 or more TX channels, the maximum number of PDSCH multiplexing layers for a single user is theoretically equal to Min(Number of gNodeB TX antennas, Number of UE RX antennas, Cell downlink maximum MIMO layer number quota specified by **NRDUCellPdsch**.***MaxMimoLayerNum***). The number of gNodeB TX antennas can be configured using the **NRDUCellTrp**.***TxRxMode*** parameter, and the RF module hardware capability needs to be considered during the configuration.

Table 9-1 Maximum number of PDSCH layers for a single user that supports 8-layer transmission

Number of gNodeB TX Antennas	Number of UE RX Antennas	Cell Downlink Maximum MIMO Layer Number Quota	Maximum Number of PDSCH Layers for a Single User That Supports 8-Layer Transmission
64T	8R	LAYER_8	8
32T	8R	LAYER_8	8

Downlink 8-layer transmission for a single FWA user is controlled by the **DL_SU_EIGHT_LAYER_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter. In NSA networking or in CA scenarios with three or more CCs, this function also requires selecting the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter.

In CA scenarios with three or more CCs, the **NRDUCellCarrMgmt**.***DlEightLayerMaxCCNum*** parameter specifies the maximum number of CCs where downlink 8-layer transmission can take effect. The details are as follows:

- When this parameter is set to **1CC**, downlink 8-layer transmission can take effect on at most one CC for a UE with three or more CCs aggregated.
- When this parameter is set to **2CC**, downlink 8-layer transmission can take effect on at most two CCs for a UE with three or more CCs aggregated.
- When this parameter is set to **3CC**, downlink 8-layer transmission can take effect on at most three CCs for a UE with three or more CCs aggregated.
- When this parameter is set to **NOT_CONFIG**, the maximum number of CCs where downlink 8-layer transmission can take effect equals the maximum number of CCs that can be aggregated.

For a UE with three or more CCs aggregated, the UE's capabilities also affect the number of CCs where downlink 8-layer transmission can take effect. Specifically, this CC number equals min(UE's supported number of CCs where downlink 8-layer transmission can take effect, Base station's maximum allowed number of CCs where downlink 8-layer transmission can take effect).

To allocate SRS resources and data channel resources more properly in a cell with downlink 8-layer transmission for a single FWA user enabled, the following functions need to be enabled:

- Resource allocation adjustment for downlink 8-layer transmission: This function enables the base station to properly allocate SRS resources to 8R UEs based on the overall cell performance. This function is controlled by the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoExtSw*** parameter.
- MU pairing adjustment for downlink 8-layer transmission: This function enables 8R UEs to be paired when their channel conditions are good and

there is multipath propagation, so that data channel resources are properly allocated in a cell. This function is controlled by the **DL_EIGHT_LAYER_MU_ADJ_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoExtSw*** parameter and applies only to low-frequency TDD massive MIMO cells.

NOTE

- For details about the UE capabilities required for this function, see the related fields in 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V17.1.0:
  - UE capability of downlink 8-layer transmission: indicated by the value **eightLayers** of the **MIMO-LayersDL** field in **maxNumberMIMO-LayersPDSCH** in **FeatureSetDownlinkPerCC** in the corresponding frequency band.
  - UE capability of downlink 8-layer antenna selection: indicated by the value **t2r8** of the **supportedSRS-TxPortSwitchBeyond4Rx-r17** field in **srs-AntennaSwitchingBeyond4RX-r17**. If a UE has this capability, the base station allocates SRS resources to the UE to increase the downlink throughput of the UE.
- According to section 7.4.1.1.2 "Mapping to physical resources" in 3GPP TS 38.211 V15.7.0, when single-symbol DMRS type 1 is used, a maximum of four antenna ports are supported. In this case, if this function is enabled, a maximum of four layers of downlink transmission is allowed for UEs capable of downlink 8-layer transmission.

9.2 Network Analysis

9.2.1 Benefits

After this function is enabled:

- UEs that use eight RX antennas can obtain diversity gains.
- In locations with many multipath signals and good channel conditions, UEs supporting 8-layer transmission use rank 5 or higher. This brings multi-layer gains.

UEs near the cell center are scheduled with more layers than 2T4R UEs, leading to a higher average downlink rank and a higher **User Downlink Average Throughput (DU)**. In addition, there is a high probability that the **Cell Downlink Average Throughput (DU)** increases because a single UE is scheduled with more layers and has higher spectral efficiency, and the cell capacity and downlink cell spectral efficiency increase. In peak rate scenarios, when channel conditions are good, the downlink peak rate of a single user is higher than that of a baseline 2T4R user.

NOTE

Downlink cell spectral efficiency = **N.ThpVol.DL**/(**N.PRB.DL.DrbUsed.Avg** x **N.DL.PDSCH.Tti.Num**)

In peak rate scenarios, the peak rate of a single user using downlink 8-layer transmission can be reached only when the cell bandwidth meets the corresponding requirements for the line rate over an eCPRI interface. (For details, see the requirements for the line rate over an eCPRI interface in typical NR TDD cell configurations in *eCPRI*).

When downlink 8-layer transmission for a single FWA user is enabled and rank 5 or 6 is used in the downlink, using DMRS type 2 leads to fewer resource overheads and better downlink user performance. Therefore, it is recommended that the **NRDUCellPdsch**.***DlDmrsConfigType*** parameter be set to **TYPE2**. For details about DMRS, see *Downlink Scheduling*.

9.2.2 Impacts

The impacts between this function and other functions are described in **9.3.2.3 Function Impacts**.

After this function is enabled:

- A higher number of layers is mistakenly selected for users capable of downlink 8-layer transmission at a very small number of locations where this number of layers is not permitted by the channel conditions. As a result, the downlink IBLER increases, and a fallback to the original number of layers is triggered for the users. In this case, the downlink user throughput slightly decreases.
- If users capable of downlink 8-layer transmission are located in positions with good channel conditions, some users in the cell cannot participate in downlink MU pairing. In this case, the average downlink cell throughput may slightly decrease.
- In a few cases, certain user behavior may decrease SRS SINR measurement values, leading to an increase in the proportion of SINR measurement values that fall into small-index ranges. The distribution of SRS SINR measurement values is indicated by counters **N.UL.SRS.PreSINR.Index0** through **N.UL.SRS.PreSINR.Index5**.
- On a network with a high proportion of users capable of downlink 8-layer transmission, when these users are scheduled with more downlink layers, these users will pose stronger downlink interference to and may consequently affect the downlink performance of users in neighboring cells in the overlapping area.
- In a lightly or moderately loaded cell with a high proportion of tail packet traffic, when the downlink spectral efficiency of users capable of 8-layer transmission increases, it is possible that the downlink cell spectral efficiency increases but the downlink user-perceived rate decreases.
- For both uplink and downlink CCEs on the PDCCH, the average aggregation level may increase, the CCE allocation success rate may decrease, and the CCE usage may change.
- In NSA networking or in CA scenarios with three or more CCs, if the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is selected, the **User Downlink Average Throughput (DU)** increases and the PUCCH consumes more resources.
- If downlink 8-layer transmission is in effect on a single carrier of a UE with three or more CCs aggregated and the cell load is light, there is an extremely low probability that the UE's downlink throughput decreases after the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is selected.

9.3 Requirements

9.3.1 Licenses

The 8 Layers for a Single FWA User feature requires the following license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-081231	8 Layers for a Single FWA User (Trial)	NR0S08LSFU00	per Cell

This feature depends on the SU-MIMO Multiple Layers feature, which requires the following license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-010020	SU-MIMO Multiple Layers	NR0S0PREUM00	per Cell

In addition, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Downlink SU-MIMO	**DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	Downlink 8-layer transmission for a single FWA user can take effect only when downlink SU-MIMO is enabled.
UE-characteristic-based SRS period adaptation	**USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	Downlink 8-layer transmission for a single FWA user can be enabled only when UE-characteristic-based SRS period adaptation is enabled.
Downlink Maximum MIMO Layer Number Quota	**NRDUCellPdsch**.***MaxMimoLayerNum***	*MIMO (TDD)*	Downlink 8-layer transmission for a single FWA user can be enabled only when **Downlink Maximum MIMO Layer Number Quota** is set to **LAYER_6** or a higher value other than **LAYER_DEFAULT**.

9.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	High-speed Railway Superior Experience cannot work with downlink 8-layer transmission for a single FWA user.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot work with downlink 8-layer transmission for a single FWA user.

Function Name	Function Switch	Reference	Description
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper Cell cannot work with downlink 8-layer transmission for a single FWA user.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell cannot work with downlink 8-layer transmission for a single FWA user.
Dynamic-codebook-based HARQ feedback over the PUCCH	**PUCCH_DYN_HARQ_ACK_CB_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtS-witch*** parameter	*Carrier Aggregation*	Dynamic-codebook-based HARQ feedback over the PUCCH cannot work with downlink 8-layer transmission for a single FWA user.
Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	Single DCI cannot work with downlink 8-layer transmission for a single FWA user.

9.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, Co-MM does not take effect for the user.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	If downlink 8-layer transmission for a single FWA user has taken effect, intra-base-station DL CoMP does not take effect for the user.

Function Name	Function Switch	Reference	Description
Inter-base-station downlink joint transmission	**INTER_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	None	If downlink 8-layer transmission for a single FWA user has taken effect, inter-base-station downlink joint transmission does not take effect for the user.

Function Name	Function Switch	Reference	Description
3CC aggregation	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL3CC**	*CA Experience Improvement*	● When the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is deselected, downlink 8-layer transmission for a single FWA user cannot take effect together with 3CC aggregation for a UE. If 3CC aggregation is enabled, enabling downlink 8-layer transmission for a single FWA user may decrease the success rate of SCell configuration for 3CC aggregation. ● When the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank**.***DlRankSelAlgoSw*** parameter is selected, downlink 8-layer transmission for a single FWA user can take effect for

Function Name	Function Switch	Reference	Description
			CA UEs with three or more CCs aggregated.
Intelligent MCS optimization for downlink SU-MIMO	**DL_SU_MCS_INTEL_OPT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter	*Massive MIMO AHR (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, intelligent MCS optimization for downlink SU-MIMO does not take effect for the user.
Spectral efficiency threshold optimization	**SPCT_EFF_THLD_OPT_SW** option of the **NRDUCellDlRank**.***AiRankParaOptSwitch*** parameter	None	If downlink 8-layer transmission for a single FWA user has taken effect, spectral efficiency threshold optimization does not take effect for the user.
Multi-DCI Retransmission Adapt Switch	**MULTI_DCI_RETRANS_ADAPT_SW** option of the **NrDuCellAlgoSwitch**.***MultiDciAlgoSwitch*** parameter	None	If downlink 8-layer transmission for a single FWA user has taken effect, multi-DCI retransmission adaptation does not take effect for the user.
TTI-level channel shutdown	**NRDUCellPowerSaving**.***ComChnDynShutdownSwitch*** parameter and **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	If downlink 8-layer transmission for a single FWA user has taken effect, the energy saving benefits of TTI-level channel shutdown decrease.

Function Name	Function Switch	Reference	Description
Maximum UE Saturation Power Offset	**NRDUCellChnPwr**.***MaxUeSatPwrOffset***	None	If downlink 8-layer transmission for a single FWA user has taken effect, the **Maximum UE Saturation Power Offset** configuration does not take effect for the user.
DL MU Non-AS Optimal Layer Select Sw	**DL_MU_OPTIMAL_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	None	If downlink 8-layer transmission for a single FWA user has taken effect, the **DL MU Non-AS Optimal Layer Select Sw** configuration does not take effect for the user.
Downlink anti-interference scheduling	**DL_ANTI_INTRF_SCH_SW** option of the **NRDUCELLPDSCH**.***DlPdschAlgoSwitch*** parameter	*MIMO (TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, downlink anti-interference scheduling does not take effect when rank 5 or higher is used for the user.
Port number adaptation for CSI-RS for CM	**NRDUCellCsirs**.***Csirs4P8PAdaptiveSwThld***	*Channel Management*	If downlink 8-layer transmission for a single FWA user has taken effect, port number adaptation for CSI-RS for CM does not take effect for the user.

Function Name	Function Switch	Reference	Description
Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	**DL_MU_NLOS_IR_PRECODING_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*Massive MIMO AHR (Low-Frequency TDD)* *Coordinated Interference Management (Low-Frequency TDD)*	● When the **NRDUCellPdschPrecode**.***DlMuIRPrecodePol*** parameter is set to **MU_ALLUSER_LOWLATENCY** or **MU_ALLUSER** and downlink 8-layer transmission for a single FWA user has taken effect, "interference rejection precoding for downlink MU-MIMO in NLOS scenarios" does not take effect for the user. ● When the **NRDUCellPdschPrecode**.***DlMuIRPrecodePol*** parameter is set to **MU_ALLUSER_POWER_ENH** and downlink 8-layer transmission for a single FWA user has taken effect, "interference rejection precoding for downlink MU-MIMO in NLOS scenarios" can take effect for the user.
DL MU Intrf Reject Precoding Policy	**NRDUCellPdschPrecode**.***DlMuIRPrecodePol*** set to **MU_ALLUSER_LOWLATENCY** or **MU_ALLUSER**	*Massive MIMO AHR (Low-Frequency TDD)*	

Function Name	Function Switch	Reference	Description
Interference rejection precoding in mobile scenarios	**DL_MU_MOBILE_IR_PRECODING_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, interference rejection precoding in mobile scenarios does not take effect for the user.
User CSI-RS Beam Type	**NRDUCellComp**.***UserCsirsBeamType*** set to **Type8**	None	If downlink 8-layer transmission for a single FWA user has taken effect, user-level CSI-RS beam type 8 is not supported for the user.
Closed-loop PUSCH antenna selection	**PUSCH_CLOSED_LOOP_ANT_SEL_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, closed-loop PUSCH antenna selection does not take effect for the user.
SRS beam-specific noise reduction	**BEAM_SPECIFIC_SRS_DENOISE_SW** option of the **NRDUCellSrs**.***SrsDetectionAlgoSwitch*** parameter	*Massive MIMO AHR (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, SRS beam-specific noise reduction does not take effect for the user.
SRS resource set adjustment	**SRS_RES_SET_ADJ_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	If downlink 8-layer transmission for a single FWA user has taken effect, SRS resource set adjustment does not take effect for the user.

Function Name	Function Switch	Reference	Description
SRS Transmit Port	**NRDUCellUeCoop**.***SrsTxPort***	None	If **NRDUCellUeCoop**.***SrsTxPort*** is set to **1T4R** or **2T4R** and downlink 8-layer transmission for a single FWA user has taken effect, the SRS antenna selection capability of 1T4R or 2T4R does not take effect for the user.
PUCCH grouping adaptation for inter-FR CA	**PUCCH_GROUP_ADAPT_SW** option of the **NRDUCellCarrMgmt**.***InterFrCaEnhSw*** parameter	*Carrier Aggregation*	If downlink 8-layer transmission for a single FWA user has taken effect, PUCCH grouping adaptation for inter-FR CA does not take effect for the user.
Robust weights	**DL_ROBUST_WEIGHT_SW** option of the **NRDUCellPdschPrecode**.***PrecodingAlgoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	If downlink 8-layer transmission for a single FWA user has taken effect, robust weights do not take effect for the user in the downlink.

9.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Cells

The TX/RX mode of the cell must be 32T32R or 64T64R.

9.3.5 Others

This function requires that UEs support downlink 8-layer transmission and 8-layer antenna selection. UE capabilities are indicated as follows:

- A UE supports downlink 8-layer transmission if the **MIMO-LayersDL** field in **maxNumberMIMO-LayersPDSCH** in **FeatureSetDownlinkPerCC** is **eightLayers** for the corresponding frequency band.
- A UE supports 8-layer antenna selection if the **supportedSRS-TxPortSwitchBeyond4Rx-r17** field in **srs-AntennaSwitchingBeyond4RX-r17** is **t2r8**.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
SU-MIMO Multiple Layers Switch	**NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw***	DL_SU_MULTI_LAYER_SW	Select this option if downlink 8-layer transmission for a single FWA user is to be enabled.
Downlink Maximum MIMO Layer Number Quota	**NRDUCellPdsch**.***MaxMimoLayerNum***	None	Set this parameter to **LAYER_8** or a higher value if downlink 8-layer transmission for a single FWA user is to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank Selection Algorithm Switch	**NRDUCellDlRank**.***DlRankSelAlgoSw***	DL_SU_EIGHT_LAYER_SW	Select this option if downlink 8-layer transmission for a single FWA user is to be enabled.
PDCCH Aggregation Level Adaptation Policy	**NRDUCellPdcch**.***PdcchAggLvlAdaptPol***	None	It is recommended that this parameter be set to **SEPARATE_ADAPT** if the **NRDUCellPdcch**.***UeSpecificPdcchAggLvl*** parameter is set to **ADAPTIVE**.
DL Rank Selection Algorithm Switch	**NRDUCellDlRank**.***DlRankSelAlgoSw***	DL_EIGHT_LAYER_SCENARIO_EXT_SW	Select this option in NSA networking or in CA scenarios with three or more CCs.
DL Eight Layer Maximum CC Number	**NRDUCellCarrMgmt**.***DlEightLayerMaxCCNum***	None	The value **1CC** is recommended in CA scenarios with three or more CCs.
DL Rank Selection Algorithm Extension Switch	**NRDUCellDlRank**.***DlRankSelAlgoExtSw***	DL_EIGHT_LAYER_RES_ALLO_ADJ_SW	Select this option.
DL Rank Selection Algorithm Extension Switch	**NRDUCellDlRank**.***DlRankSelAlgoExtSw***	DL_EIGHT_LAYER_MU_ADJ_SW	Select this option.

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink SU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_SW-1;
//Setting the maximum number of downlink MIMO layers
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_8;
//Enabling downlink 8-layer transmission for a single FWA user
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SU_EIGHT_LAYER_SW-1;
//Enabling resource allocation adjustment for downlink 8-layer transmission
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoExtSw=DL_EIGHT_LAYER_RES_ALLO_ADJ_SW-1;
//Enabling MU pairing adjustment for downlink 8-layer transmission
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoExtSw=DL_EIGHT_LAYER_MU_ADJ_SW-1;
//(Optional, recommended) Setting the PDCCH Aggregation Level Adaptation Policy parameter to
SEPARATE_ADAPT if the UE-specific PDCCH Aggregation Level parameter is set to ADAPTIVE
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAggLvlAdaptPol=SEPARATE_ADAPT;
//(Optional) Selecting the DL_EIGHT_LAYER_SCENARIO_EXT_SW option in NSA networking or in CA
scenarios with three or more CCs
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_EIGHT_LAYER_SCENARIO_EXT_SW-1;
//(Optional) Setting the DlEightLayerMaxCCNum parameter in CA scenarios with three or more CCs
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlEightLayerMaxCCNum=1CC;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deselecting the DL_EIGHT_LAYER_SCENARIO_EXT_SW option
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_EIGHT_LAYER_SCENARIO_EXT_SW-0;
//Disabling downlink 8-layer transmission for a single FWA user
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SU_EIGHT_LAYER_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Use a monitoring item to check whether downlink 8-layer transmission for a single FWA user has taken effect. Specifically, on the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **User Performance Monitoring** > **User Common Monitoring** to observe the monitoring item **Code1 DL Average Layer Num**. If the value of this monitoring item is not 0, this function has taken effect.

9.4.3 Network Monitoring

Use the following counters to monitor the number of 8R-capable UEs in a cell.

Counter Description	Counter Name
Number of 8R-capable UEs in a cell	**N.User.RRCConn.8R.Max** **N.User.RRCConn.8R.Avg**

Use the following counters to monitor the status of this function. After this function takes effect, the values of one or more of these counters are expected to change from zero to non-zero values.

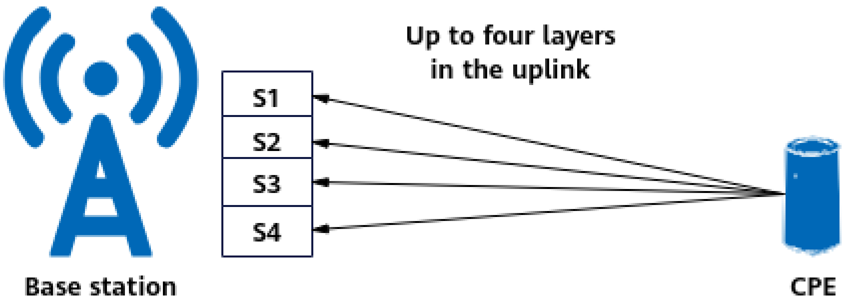
Counter Description	Counter Name
Numbers of RBs for downlink SU-MIMO with rank 5 to rank 8	**N.ChMeas.SU.MIMO.DL.RANK5.RB** **N.ChMeas.SU.MIMO.DL.RANK6.RB** **N.ChMeas.SU.MIMO.DL.RANK7.RB** **N.ChMeas.SU.MIMO.DL.RANK8.RB**
Numbers of downlink TBs transmitted with rank 5 to rank 8	**N.PDSCH.NbrTbDl.Rank5** **N.PDSCH.NbrTbDl.Rank6** **N.PDSCH.NbrTbDl.Rank7** **N.PDSCH.NbrTbDl.Rank8**
Numbers of downlink TBs initially transmitted with rank 5 to rank 8	**N.PDSCH.InitTbDl.Rank5** **N.PDSCH.InitTbDl.Rank6** **N.PDSCH.InitTbDl.Rank7** **N.PDSCH.InitTbDl.Rank8**

10 Uplink 4-Layer Transmission for a Single FWA User

10.1 Principles

Uplink 4-layer transmission for a single FWA user is controlled by the **UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter. This function is designed for UEs capable of uplink four-layer transmission. When channel conditions permit, **this function uses multi-antenna technologies to enable spatial multiplexing of time-frequency resources on the PUSCH for a single UE. A maximum of four layers of data can be transmitted in the uplink for the UE, increasing the single-UE peak rate. Figure 10-1 illustrates this function.**

Figure 10-1 Uplink 4-layer transmission for a single FWA user



For low-frequency NR TDD cells with 32 or more RX channels, the maximum number of PUSCH multiplexing layers for a single UE is theoretically equal to Min(Number of gNodeB RX antennas, Number of UE TX antennas, Cell uplink maximum MIMO layer count quota specified by **NRDUCellPusch**.***MaxMimoLayerCnt***). Table 10-1 lists the details. The number of gNodeB RX antennas is specified by the **NRDUCellTrp**.***TxRxMode*** parameter. The RF module hardware capability needs to be considered during the configuration.

Table 10-1 Maximum number of PUSCH layers for a single UE that supports four-layer transmission

Number of gNodeB RX Antennas	Number of UE TX Antennas	Cell Uplink Maximum MIMO Layer Count Quota	Maximum Number of PUSCH Layers for a Single UE That Supports Four-Layer Transmission
64R	4T	LAYER_4	4
32R	4T	LAYER_4	4

NOTE

This function requires a UE to support uplink four-layer transmission. This capability is indicated by the value **fourLayers** of the **MIMO-LayersUL** field in **maxNumberMIMO-LayersCB-PUSCH** of **FeatureSetUplinkPerCC** for the corresponding frequency band. For details about this capability, see the descriptions of related fields in section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V18.2.0.

10.2 Network Analysis

10.2.1 Benefits

It is recommended that this function be enabled in a cell where there are UEs supporting uplink 4-layer transmission. The number of such UEs can be observed using the following counters.

Counter Description	Counter Name
Average number of UEs capable of uplink four-layer transmission in a cell	**N.User.RRCConn.UL.4Layers.Avg**
Maximum number of UEs capable of uplink four-layer transmission in a cell	**N.User.RRCConn.UL.4Layers.Max**

When this function is enabled:

- UEs that use four TX antennas can obtain transmit diversity gains.
- In locations with many multipath signals and good channel conditions, UEs supporting four-layer transmission use rank 2 or higher. This brings multi-layer gains.
- UEs supporting uplink four-layer transmission near the cell center are scheduled with more layers than 2T UEs, leading to a higher average uplink

rank and a higher **User Uplink Average Throughput (DU)**. In addition, there is a high probability that the **Cell Uplink Average Throughput (DU)** increases because the number of scheduling layers for a single UE, spectral efficiency, and cell capacity increase. In peak rate scenarios, when channel conditions are good, the uplink peak rate of a single UE is higher than that of a 2T UE.

It is recommended that uplink precise channel adaptation for 4T UEs also be enabled to enhance the accuracy of multi-layer channel calculation for 4T UEs and further increase the uplink user-perceived rate in multi-layer scenarios. This function is controlled by the **UL_4T_PREC_CHANNEL_ADAPT_SW** option of the **NRDUCellPusch**.***PuschMeasSw*** parameter.

10.2.2 Impacts

The impacts between this function and other functions are described in **10.3.2.3 Function Impacts**.

After this function is enabled:

- A higher number of layers is mistakenly selected for UEs capable of uplink four-layer transmission at a very small number of locations where this number of layers is not permitted by the channel conditions. As a result, the uplink IBLER increases, and a fallback to the original number of layers is triggered for the UEs. In this case, the uplink UE throughput slightly decreases.
- In medium- and heavy-load scenarios, the uplink MU pairing rate, number of uplink paired layers, and uplink PRB usage of a cell may change, and the average uplink cell throughput may slightly decrease.
- As the number of layers increases in a cell, the average PUSCH interference in the cell may increase. In addition, the average PUSCH interference in neighboring cells may also increase, causing the uplink experience of UEs in neighboring cells to slightly deteriorate.

10.3 Requirements

10.3.1 Licenses

The Uplink 4 Layers for a Single FWA User feature requires the following license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100205	Uplink 4 Layers for a Single FWA User (Trial)	NR0S00ULSF40	per Cell

This feature depends on the SU-MIMO Multiple Layers feature, which requires the following license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-010020	SU-MIMO Multiple Layers	NR0S0PREUM00	per Cell

In addition, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Uplink SU-MIMO	**UL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	Uplink 4-layer transmission for a single FWA user can take effect only when uplink SU-MIMO is enabled.

Function Name	Function Switch	Reference	Description
Uplink Maximum MIMO Layer Count Quota	**NRDUCellPusch**.***MaxMimoLayerCnt***	*MIMO (TDD)*	Uplink 4-layer transmission for a single FWA user can be enabled only when **Uplink Maximum MIMO Layer Count Quota** is set to **LAYER_4** or a higher value.
SRS resource adjustment for 4T UEs	**UL_4T_SRS_RES_ADJ_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	Uplink 4-layer transmission for a single FWA user can take effect only when SRS resource adjustment for 4T UEs is enabled.

10.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	High-speed Railway Superior Experience cannot work with uplink 4-layer transmission for a single FWA user.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot work with uplink 4-layer transmission for a single FWA user.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper Cell cannot work with uplink 4-layer transmission for a single FWA user.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell cannot work with uplink 4-layer transmission for a single FWA user.

10.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, Co-MM does not take effect for the user.
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	If uplink 4-layer transmission for a single FWA user has taken effect, intra-base-station UL CoMP does not take effect for the user.
High-reliability uplink MCS selection	**UL_HIGH_RELIAB_MCS_SEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, high-reliability uplink MCS selection does not take effect for the user.
Uplink PAMC	**UL_PAMC_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	If uplink 4-layer transmission for a single FWA user has taken effect, uplink PAMC does not take effect for the user.
Uplink scheduling on a per RB basis	**UL_1RB_SCH_SW** option of the **NRDUCellRes**.***SchedulingPolicySwitch*** parameter	*Network Slicing*	If uplink 4-layer transmission for a single FWA user has taken effect, whether uplink scheduling on a per RB basis takes effect for the user is affected.

Function Name	Function Switch	Reference	Description
Full-power dual transmission	**UL_RANK1_FULL_POWER_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, full-power dual transmission does not take effect for the user.
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, PUSCH MU-MIMO does not take effect for the user.
Spatial channel estimation under weak uplink coverage	**UL_WEAK_COV_SPACE_CHN_ESTI_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, spatial channel estimation under weak uplink coverage does not take effect for the user.
Closed-loop PUSCH antenna selection	**PUSCH_CLOSED_LOOP_ANT_SEL_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If uplink 4-layer transmission for a single FWA user has taken effect, closed-loop PUSCH antenna selection does not take effect for the user.

10.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and the UBBPg/UBBPh series boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

10.3.4 Cells

The TX/RX mode of the cell must be 32T32R or 64T64R.

10.3.5 Others

UEs must support uplink four-layer transmission. A UE supports uplink four-layer transmission if the **MIMO-LayersUL** field in **maxNumberMIMO-LayersCB-PUSCH** of **FeatureSetUplinkPerCC** is **fourLayers** for the corresponding frequency band.

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Table 10-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 10-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
SU-MIMO Multiple Layers Switch	**NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw***	UL_SU_MULTI_LAYER_SW	Select this option before enabling uplink 4-layer transmission for a single FWA user.
Uplink Maximum MIMO Layer Count Quota	**NRDUCellPusch**.***MaxMimoLayerCnt***	None	Set this parameter to **LAYER_4** or a larger value before enabling uplink 4-layer transmission for a single FWA user.

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Switch	**NRDUCellSrs**.***SrsAlgoSwitch***	UL_4T_SRS_RES_ADJ_SW	Select this option before enabling uplink 4-layer transmission for a single FWA user.
Uplink Rank Algorithm Switch	**NRDUCellUlRank**.***UlRankAlgoSw***	UL_SU_FOUR_LAYER_SW	Select this option if uplink 4-layer transmission for a single FWA user is required.
Uplink Rank Algorithm Switch	**NRDUCellUlRank**.***UlRankAlgoSw***	UL_RANK_FAST_DECREASE_SW	If the inter-layer SRS SINR difference is large, it is recommended that fast uplink rank reduction based on SRS SINR be enabled to increase the uplink UE throughput. For details about fast uplink rank reduction based on SRS SINR, see *MIMO (TDD)*.
PUSCH Measurement Sw	**NRDUCellPusch**.***PuschMeasSw***	UL_4T_PREC_CHANNEL_ADAPT_SW	Select this option if uplink 4-layer transmission for a single FWA user is enabled.

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Performing the following configurations before enabling uplink 4-layer transmission for a single FWA user
//Enabling uplink SU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SuMimoMultipleLayerSw=UL_SU_MULTI_LAYER_SW-1;
//Setting the maximum number of uplink MIMO layers
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_4;
//Enabling SRS resource adjustment for 4T UEs
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=UL_4T_SRS_RES_ADJ_SW-1;

//Enabling uplink 4-layer transmission for a single FWA user
MOD NRDUCELLULRANK: NrDuCellId=0, UlRankAlgoSw=UL_SU_FOUR_LAYER_SW-1;
//Enabling uplink precise channel adaptation for 4T UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-1;
//(Optional) Enabling fast uplink rank reduction based on SRS SINR if the inter-layer SRS SINR difference is
large
MOD NRDUCELLULRANK: NrDuCellId=0, UlRankAlgoSw=UL_RANK_FAST_DECREASE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink 4-layer transmission for a single FWA user
MOD NRDUCELLULRANK: NrDuCellId=0, UlRankAlgoSw=UL_SU_FOUR_LAYER_SW-0;
//(Optional) Disabling fast uplink rank reduction based on SRS SINR
MOD NRDUCELLULRANK: NrDuCellId=0, UlRankAlgoSw=UL_RANK_FAST_DECREASE_SW-0;
//Disabling uplink precise channel adaptation for 4T UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-0;
//Disabling SRS resource adjustment for 4T UEs
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=UL_4T_SRS_RES_ADJ_SW-0;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

Use a monitoring item to check whether uplink 4-layer transmission for a single FWA user has taken effect. Specifically, on the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **User Performance Monitoring** > **User Common Monitoring** to observe the monitoring item **Code0 UL Average Layer Num**. If the value of this monitoring item is greater than two, this function has taken effect.

10.4.3 Network Monitoring

Use the following counters to monitor the status of this function. After this function takes effect, the values of one or more of these counters are expected to change from zero to non-zero values.

Counter Description	Counter Name
Number of RBs for uplink SU-MIMO in rank 3 mode	**N.ChMeas.SU.MIMO.UL.RANK3.RB**
Number of RBs for uplink SU-MIMO in rank 4 mode	**N.ChMeas.SU.MIMO.UL.RANK4.RB**
Number of TBs transmitted with rank 3 in the uplink	**N.PUSCH.TbUl.Rank3**
Number of TBs transmitted with rank 4 in the uplink	**N.PUSCH.TbUl.Rank4**
Number of uplink TBs initially transmitted with rank 3	**N.PUSCH.InitTbUl.Rank3**
Number of uplink TBs initially transmitted with rank 4	**N.PUSCH.InitTbUl.Rank4**

11 FWA mmWave Coverage Extension (High-Frequency TDD)

11.1 Principles

FWA mmWave coverage extension is used to extend the cell coverage and improve the long-distance coverage capability, as shown in **Figure 11-1**. **Theoretically, FWA mmWave coverage extension can support a maximum cell radius of 10 km. Theoretical maximum cell radius can be determined by the PRACH demodulation distance or slot structure, whichever is smaller.** The maximum cell radius can be configured using the **NRDUCell**.***CellRadius*** parameter. This function is enabled when **NRDUCell**.***CellRadius*** is set to a value greater than 2410 m.

Figure 11-1 FWA mmWave coverage extension

PRACH Demodulation

In a random access procedure, the gNodeB calculates the maximum supported cell radius based on the preamble format transmitted from a user. Before this function is enabled, the maximum cell radius is 2.41 km as calculated by the base station based on the format of the preamble transmitted from a user. After this function is enabled, the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter can be set to **192** to improve the preamble-format-based access capability and increase the cell coverage radius.

Slot Configuration and Slot Structure

A user farther away from the base station requires a longer time to transmit signals in advance, and therefore a larger number of GP symbols need to be configured. Therefore, the maximum cell radius can be increased by changing the slot configuration and slot structure. With this function, the slot configuration **NRDUCell**.***SlotAssignment*** can only be set to **4_1_DDDSU**, and the slot structure **NRDUCell**.***SlotStructure*** can only be set to **SS204**, **SS205**, **SS206**, **SS207**, **SS208**, **SS209**, and **SS210**. **Table 11-1** lists the mapping between the GP length and cell coverage capability without considering the preamble format.

Table 11-1 Theoretical maximum cell radius of the 4:1 slot configuration (DDDSU) with different numbers of GP symbols in FWA mmWave coverage extension scenarios

Slot Structure	Number of GP Symbols	Theoretical Maximum Cell Radius (km)
SS204	4	3.25
SS205	5	4.59
SS206	6	5.93
SS207	7	7.27
SS208	8	8.61
SS209	9	9.95
SS210	10	11.29

NOTE

In this document, the maximum cell radius refers to the theoretical maximum value with the FWA mmWave coverage extension function enabled. Generally, the actual cell radius is less than the theoretical maximum value.

11.2 Network Analysis

11.2.1 Benefits

After the FWA mmWave coverage extension function is enabled, the coverage area of the base station is expanded. Theoretically, the maximum cell coverage radius can reach 10 km. In addition, PRACH access resources decrease and uplink data transmission resources increase, which causes the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**), average uplink user throughput (**User Uplink Average Throughput (DU)**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) to increase.

NOTE

Whether the maximum cell radius can reach 10 km depends on the factors responsible for path loss difference, such as the base station's transmit power and antenna altitude as well as antenna gains and transmit power of terminals.

11.2.2 Impacts

The impacts between this function and other functions are described in **11.3.2.3 Function Impacts**.

This function has the following network impacts:

- This function uses the configuration with the PRACH index of 192, which renders fewer PRACH access resources than using the baseline configuration with the PRACH index of 188. When a large number of users perform contention-based access simultaneously, the collision probability increases. As a result, the contention-based random access success rate (**N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att** x 100%) decreases.
- This function uses the slot structure configuration of SS204, SS205, SS206, SS207, SS208, SS209, or SS210. The corresponding number of GP symbols is 4 to 10, which renders fewer downlink data transmission resources than the baseline configuration of 3 GP symbols. This may decrease the average downlink user throughput (**User Downlink Average Throughput (DU)**) and increase the downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**).
- After this function is enabled, users far away from the base station can access the network and perform data transmission. Given that the RSRP or SINR of these users is lower than that of users relatively close to the base station, the average uplink user throughput (**User Uplink Average Throughput (DU)**) and average downlink user throughput (**User Downlink Average Throughput (DU)**) decrease.
- When the cell radius configured in FWA mmWave coverage extension is greater than 5000 m, the wide beam scenario is required. Using wide beams will alter the cell coverage, which results in changes in related performance indicators, such as those related to the number of users in a cell, user throughput, access, handover, and service drop.

11.3 Requirements

11.3.1 Licenses

The following license is needed to enable FWA mmWave coverage extension for common cells (for which the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **NORMAL_CELL**).

Feature ID	Feature Name	Model	Sales Unit
FOFD-070301	mmWave Coverage Extension	NR0S00MMCE00	per Cell

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td colspan="4">For the license control item NR0S00MMCE00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

NOTE

The number of licenses for a sector must be the same as the number of carriers. Otherwise, this function cannot take effect.

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	The FWA mmWave coverage extension function requires that the slot configuration be **4_1_DDDSU**.

Function Name	Function Switch	Reference	Description
Slot structure	**NRDUCell**.***SlotStructure***	*Standards Compliance*	● When the cell radius configured in FWA mmWave coverage extension is greater than 2410 m and less than or equal to 5000 m, the slot structure must be set to **SS204**, **SS205**, or **SS206**. ● When the cell radius configured in FWA mmWave coverage extension is greater than 5000 m, the slot structure must be set to **SS206**, **SS207**, **SS208**, **SS209**, or **SS210**.
Wide beam scenario	**NRDUCellTrpMmwavBeam**.***CoverageScenario*** set to **SCENARIO_104**	*mmWave Beam Management (High-Frequency TDD)*	When the cell radius configured in FWA mmWave coverage extension is greater than 5000 m and less than or equal to 10000 m, FWA mmWave coverage extension requires that the wide beam scenario be enabled.

11.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
PRACH coverage enhancement	**PRACH_COV_ENH_SW** option of the **NRDUCellPrach**.***RachAlgoOptSwitch*** parameter	*Random Access Control*	PRACH coverage enhancement cannot work with FWA mmWave coverage extension.

11.3.2.3 Function Impacts

None

11.3.3 Hardware

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD-capable mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only the HAAU5323 supports this function. For details, see the AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Cells

All activated cells in a sector must have the same radius.

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

Table 11-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
PRACH Configuration Index	**NRDUCellPrach**.***PrachConfigurationIndex***	Set this parameter to **192**.
Slot Assignment	**NRDUCell**.***SlotAssignment***	Set this parameter to **4_1_DDDSU**.

Parameter Name	Parameter ID	Setting Notes
Slot Structure	**NRDUCell**.***SlotStructure***	Set this parameter to **SS204**, **SS205**, **SS206**, **SS207**, **SS208**, **SS209**, or **SS210**. You are advised to set this parameter based on the number of GP symbols corresponding to the required cell radius.
Cell Radius	**NRDUCell**.***CellRadius***	Set this parameter based on the network plan. If this parameter is set to a value greater than **2410** and less than or equal to **10000** (in units of meters), the FWA mmWave coverage extension function takes effect.

11.4.1.2 Using MML Commands

Before using MML commands, refer to **11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Deactivating all cells in a sector
DEA NRCELL: NrCellId=0;
//Setting the slot configuration and slot structure
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, FrequencyBand=N258,
SlotAssignment=4_1_DDDSU, SlotStructure=SS206;
//Configuring the PRACH index
MOD NRDUCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=192;
//Configuring the cell radius, which must be the same for all activated cells in a sector
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellRadius=5000;
//Activating the corresponding cell
ACT NRCELL: NrCellId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating the corresponding cell
DEA NRCELL: NrCellId=0;
//Configuring the cell radius
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellRadius=2410;
//Activating the corresponding cell
ACT NRCELL: NrCellId=0;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

After FWA mmWave coverage extension is activated, perform the following operations to check whether it has taken effect:

1. Run the **LST NRDUCELL** command. Verify that the **NRDUCell**.***CellRadius*** parameter value is greater than **2410**.
2. Run the **DSP NRDUCELL** command to check the cell status and verify that the cell has been activated.

11.4.3 Network Monitoring

After this function is enabled, if number allocation is available for users who are more than 2.41 km away from the base station, you can check the increases in **N.RA.TA.UE.Index8**, **N.RA.TA.UE.Index9**, and **N.RA.TA.UE.Index10** to monitor the running status of this function.

NOTE

- **N.RA.TA.UE.Index8**: measures the number of random access times when the TA values are in the range of [186, 385] (which corresponds to a distance of 1.7 km to 3.7 km).
- **N.RA.TA.UE.Index9**: measures the number of random access times when the TA values are in the range of [386, 785] (which corresponds to a distance of 3.7 km to 7.6 km).
- **N.RA.TA.UE.Index10**: measures the number of random access times when the TA values are in the range of [786, 1485] (which corresponds to a distance of 7.6 km to 14.5 km).

12 FWA Ultra-Large Coverage

12.1 Principles

FWA Ultra-Large Coverage has been introduced to improve coverage for FWA users at the cell edge. This feature includes the following functions:

12.1.1 FWA Device-Pipe-based Uplink Dense Codebooks

12.1.2 FWA Device-Pipe-based Uplink High Power

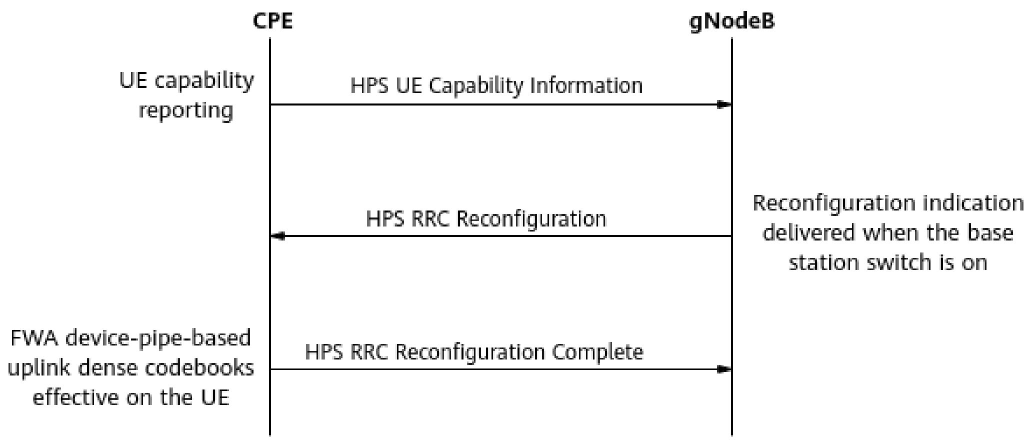
12.1.1 FWA Device-Pipe-based Uplink Dense Codebooks

The "FWA device-pipe-based uplink dense codebooks" function enables the base station to instruct UEs to expand their codebook sets for the PUSCH beyond standard codebooks. This function applies to UEs that support uplink dense codebooks when channel conditions permit. It makes uplink data transmission more accurate and improves coverage for UEs at the cell edge.

This function is controlled by the **UL_DENSER_CODEBOOK_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter. This function requires that the base station and UEs complete device-pipe identification. For details about this procedure, see *Device-Pipe Synergy*. This section describes only processing exclusive to FWA device-pipe-based uplink dense codebooks in comparison to basic device-pipe identification. **Figure 12-1** illustrates the related message exchange.

1. UE capability reporting: A UE reports its capability of device-pipe-based uplink dense codebooks to the gNodeB through the **support-DenserCodebook** field in an HPS UE Capability Information message.
2. Function status notification: The gNodeB sends an HPS RRC Reconfiguration message to notify the UE of the function status and related configurations. If the FWA device-pipe-based uplink dense codebooks function is enabled, this function takes effect after UEs supporting this function receive this message.
3. Function activation response: If the FWA device-pipe-based uplink dense codebooks function successfully takes effect on the UE, the UE sends an HPS RRC Reconfiguration Complete message as a response to the gNodeB.

Figure 12-1 Message exchange for device-pipe synergy



This function applies to low-frequency TDD massive MIMO scenarios.

12.1.2 FWA Device-Pipe-based Uplink High Power

The FWA device-pipe-based uplink high power function increases the transmit power of 4T UEs that support uplink high power. It achieves this by enabling the gNodeB to instruct such UEs to use high power. This function improves coverage for UEs at the cell edge.

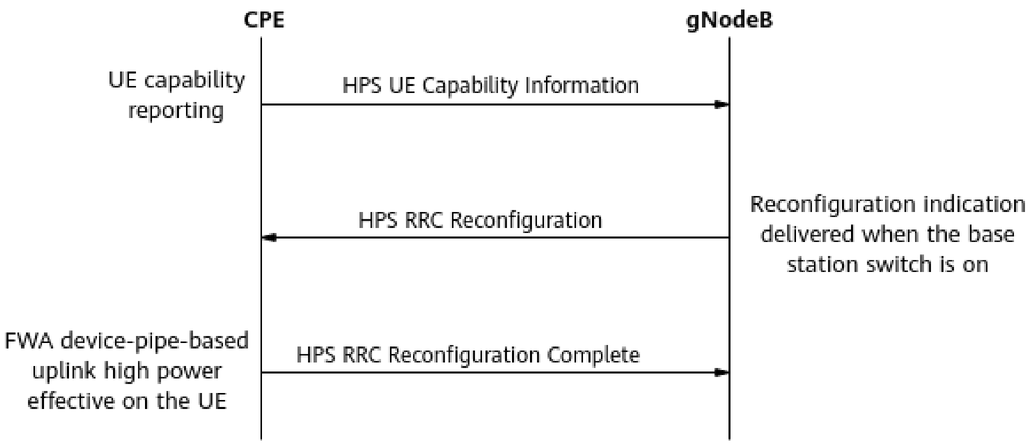
This function is enabled when parameters are set as follows:

- The **UL_HIGH_POWER_UE_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter is selected.
- The **NRDUCellSelConfig**.***MaxUeTransmitPower*** parameter is set to **254**.

This function requires that the base station and UEs complete device-pipe identification. For details about this procedure, see *Device-Pipe Synergy*. This section describes only processing exclusive to FWA device-pipe-based uplink high power in comparison to basic device-pipe identification. **Figure 12-2** illustrates the related message exchange.

1. UE capability reporting: A UE reports its capability of device-pipe-based uplink high power to the gNodeB through the **hps-ue-PowerClass** field in an HPS UE Capability Information message.
2. Function status notification: The gNodeB sends an HPS RRC Reconfiguration message to notify the UE of the function status and related configurations. If the FWA device-pipe-based uplink high power function is enabled, this function takes effect after UEs supporting this function receive this message.
3. Function activation response: If the FWA device-pipe-based uplink high power function successfully takes effect on the UE, the UE sends an HPS RRC Reconfiguration Complete message as a response to the gNodeB.

Figure 12-2 Message exchange for device-pipe synergy



This function applies to low-frequency TDD massive MIMO scenarios.

12.2 Network Analysis

12.2.1 Benefits

FWA Ultra-Large Coverage improves uplink coverage for 4T UEs supporting this feature and increases the **User Uplink Average Throughput (DU)** for such UEs at the cell edge compared to 2T UEs.

It is recommended that uplink precise channel adaptation for 4T UEs also be enabled to enhance the accuracy of multi-layer channel calculation for 4T UEs and further increase the uplink user-perceived rate in multi-layer scenarios. This function is controlled by the **UL_4T_PREC_CHANNEL_ADAPT_SW** option of the **NRDUCellPusch**.***PuschMeasSw*** parameter.

12.2.2 Impacts

The impacts between this function and other functions are described in **7.1.3.2.3 Function Impacts**.

When FWA Ultra-Large Coverage is enabled in a cell, the following impacts occur:

- In poor channel conditions, incorrect beam directions may be selected for UEs at a few positions. As a result, the uplink IBLER increases and the uplink UE throughput slightly decreases for these UEs.
- The average PUSCH interference in the cell may increase. In addition, the average PUSCH interference in neighboring cells may also increase, causing the uplink experience of UEs in neighboring cells to slightly deteriorate.

12.3 Requirements

12.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-100206	FWA Ultra-Large Coverage (Trial)	NR0S00FULC30	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	**SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter	*Device-Pipe Synergy*	Device-pipe identification must be enabled to enable FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power.
SRS resource adjustment for 4T UEs	**UL_4T_SRS_RES_ADJ_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	SRS resource adjustment for 4T UEs must be enabled for FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power to take effect.

<table>
<tr><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>RRC-reconfiguration-based dynamic waveform switching</td><td>UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter</td><td>Turbo Uplink (Low-Frequency TDD)</td><td rowspan="2">Either RRC-reconfiguration-based dynamic waveform switching or DCI-based dynamic waveform switching must be enabled for FWA device-pipe-based uplink dense codebooks to take effect. DCI-based dynamic waveform switching has the following UE capability requirements:
● If only one carrier is configured in a band, UEs must have the capability indicated by the dynamicWaveformSwitch-r18 field.
● If multiple carriers are configured in a band, the value of the dynamicWaveformSwitchIntraCA-r18 field in the reported UE capability information must be greater than or equal to the number of carriers configured in the band.</td></tr>
<tr><td>DCI-based dynamic waveform switching</td><td>DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter</td><td>Uplink Scheduling</td></tr>
</table>

12.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	Fusion Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.

12.3.2.3 Function Impacts

None

12.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and the UBBPg/UBBPh series boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

12.3.4 Cells

The TX/RX mode of the cell must be 32T32R or 64T64R.

12.3.5 Others

Uplink 4T UEs are required and they must have the following capabilities:

- Capability of device-pipe-based uplink dense codebooks for the FWA device-pipe-based uplink dense codebooks function. This capability is indicated by the **support-DenserCodebook** field in the RRC_HPS_UE_CAP_INFO message. If the value of this field from a UE is **supported**, the UE has this capability.
- Capability of device-pipe-based uplink high power for the FWA device-pipe-based uplink high power function. This capability is indicated by the **hps-ue-PowerClass** field in the RRC_HPS_UE_CAP_INFO message. If the value of this field from a UE is **pc1-32dbm-ext**, the UE has this capability.

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

Table 12-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 12-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Specified UE Cooperation Switch	**NRDUCellUeCoop**.***SpecUeCooperationSwitch***	• Select the **UL_DENSER_CODEBOOK_SW** option if the FWA device-pipe-based uplink dense codebooks function is required. • Select the **UL_HIGH_POWER_UE_SW** option if the FWA device-pipe-based uplink high power function is required.
Max UE Transmit Power	**NRDUCellSelConfig**.***MaxUeTransmitPower***	Set this parameter to **254**.

Parameter Name	Parameter ID	Setting Notes
PUSCH Measurement Sw	**NRDUCellPusch**.***PuschMeasSw***	Select the **UL_4T_PREC_CHANNEL_ADAPT_SW** option if FWA Ultra-Large Coverage is enabled.

12.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- FWA device-pipe-based uplink dense codebooks

```
//Enabling uplink precise channel adaptation for 4T UEs if FWA Ultra-Large Coverage is enabled
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-1;
//Enabling FWA device-pipe-based uplink dense codebooks
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=UL_DENSER_CODEBOOK_SW-1;
```

- FWA device-pipe-based uplink high power

```
//Enabling uplink precise channel adaptation for 4T UEs if FWA Ultra-Large Coverage is enabled
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-1;
//Enabling FWA device-pipe-based uplink high power
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=254;
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=UL_HIGH_POWER_UE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- FWA device-pipe-based uplink dense codebooks

```
//Disabling FWA device-pipe-based uplink dense codebooks
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=UL_DENSER_CODEBOOK_SW-0;
//Disabling uplink precise channel adaptation for 4T UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-0;
```

- FWA device-pipe-based uplink high power

```
//Disabling FWA device-pipe-based uplink high power
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=UL_HIGH_POWER_UE_SW-0;
//Disabling uplink precise channel adaptation for 4T UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=UL_4T_PREC_CHANNEL_ADAPT_SW-0;
```

12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

On the MAE-Access, perform as follows to check whether FWA Ultra-Large Coverage has taken effect:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left of the displayed **Signaling Trace Management** tab, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.

Step 3 Enable a UE capable of device-pipe synergy to initially access the network.

Step 4 On the displayed **Uu Interface Trace** page, check for the RRC_HPS_UE_CAP_INFO message. If this message is present, check its contained fields as follows:

- Check the value of the **support-DenserCodebook** field. If the value is **supported**, the UE supports the FWA device-pipe-based uplink dense codebooks function. Otherwise, the UE does not support this function.
- Check the value of the **hps-ue-PowerClass** field. If the value is **pc1-32dbm-ext**, the UE supports the FWA device-pipe-based uplink high power function. Otherwise, the UE does not support this function.

Step 5 If the UE supports both of the functions, check for the RRC_HPS_CONN_RECFG message on the **Uu Interface Trace** page. If this message is present, check its contained fields as follows:

- Check the value of the **hps-DenserCodebook** field. If the value is **enabled**, the FWA device-pipe-based uplink dense codebooks function has taken effect.
- Check the value of the **hps-Pc132dbmExtTransmitPower** field. If the value is **enabled** and SIB1 does not contain the *p-Max* IE, the FWA device-pipe-based uplink high power function has taken effect.

----End

12.4.3 Network Monitoring

On the MAE-Access, choose **User Performance Monitoring** > **User Common Monitoring** > **Uplink MAC Throughput**. Check if the single-UE uplink MAC throughput increases after function activation.

13 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

14 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

15 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

16 Reference Documents

- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *QoS Management*
- *Network Slicing*
- *Admission Control*
- *URLLC*
- *Hyper Cell*
- *Cell Combination*
- *Multi-Operator Sharing*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *CoMP*
- *UE Power Saving*
- *MIMO (TDD)*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Resource Control in NSA Networking*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Random Access Control*
- *EPC-based NSA Basics*
- *EPC-based NSA Experience Improvement*
- *SA Mobility Management in Idle Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *Flexible User Steering*
- *Mobility Load Balancing in Idle Mode*

- *Mobility Load Balancing in Connected Mode*
- *Ultimate Video Experience*
- *CA Experience Improvement*
- *RedCap*
- *Standards Compliance*
- *Channel Management*
- *Device-Pipe Synergy*
- *DRX*
- *Turbo Uplink (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *SA Mobility Management in Connected Mode*
- *Co-MM (Low-Frequency TDD)*
- *High Speed Mobility*
- *eCPRI*
- *mmWave Beam Management (High-Frequency TDD)*
- *Scalable Bandwidth*
- *Fusion Cell (Low-Frequency TDD)*
- *Multi-Frequency Smart Aggregation*
- *License Control Item Lists*
- *Uplink Flexible Spectrum Scheduling*
- *QCI-based Resource Control*
- *Service-differentiated Experience Guarantee*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *Performance Management*
- *Adaptive Managed Latency*